

Can Statistical Models Provide Evidence for Election Frauds? Does the eforensics Election-Fraud Model Demonstrate South Korean Election Frauds Across Two Decades?

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Abstract

Walter Mebane’s eforensics model, and in particular his analysis of the 2020 South Korean National Assembly election, was one of the flash points for South Korean early-vote election fraud claims. In political methodology, this is a special case of a general question: can statistical models really provide evidence of election frauds? Despite errors in the initial Mebane analysis, the corrected eforensics analysis still suggests widespread election frauds across two decades of South Korean elections, with the worst cases actually reported for elections that have never attracted mainstream fraud claims and that predate the introduction of early voting. Using three synthetic fraud-free null datasets that reflect the observed voter turnout and vote-share behavior, we show that these results can be attributed to estimation bias. While this does not directly prove that election fraud models are useless, it shows that they need significant improvements before they can be relied upon.

Keywords: election forensics; Bayesian finite mixture models; null calibration; eforensics; early voting; South Korea; election fraud claims

1 Introduction

In an era of electronic election administration and processing, and in some cases electronic voting, the conventional physical methods of verifying election integrity have recently come under question. Absentee votes, mail-in votes, and early votes have complicated election integrity checks, which has coincided with the recent rise of election fraud claims regarding these types of votes in the United States and other parts of the world.

In light of this background, the potential role that a statistical model can play in detecting election frauds has become more critical. One of the most well-known statistical election-fraud models is the eforensics model, and Walter Mebane, the creator of this model, has used it to argue for a high probability of election frauds involving early votes in several elections, including the 21st South Korean general (National Assembly) election [Mebane, 2020]. More recently, the same model has been used to raise questions about the 2024 presidential election results in Pennsylvania [Election Truth Alliance, 2025, Walker and Huseman, 2026]. His analysis of the 21st South Korean general election was an early catalyst for the subsequent election fraud claims in South Korea, especially those surrounding early voting. Min Kyung-wook, the sitting conservative lawmaker in April 2020 who lost his seat in the 21st general election (year 2020) and filed an election-invalidity lawsuit (eventually dismissed by the Supreme Court in 2022; Lawtimes [Beobryul Sinmun], 2022), is reported to have cited Mebane’s paper as grounds for his claim [Newsis, 2020].

While recent events have shifted the focus of election fraud claims in South Korea from early votes to general voting-administration irregularities and the alleged incompetence of the National Election Commission that oversees South Korean elections [News1, 2026, Kyunghyang Shinmun [Khan], 2026b, New

Daily, 2026], election fraud theorists and even the leader (Jang Dong-hyeok) of the main South Korean conservative party (People Power Party) are still pushing for the elimination of early votes [Seoul Shinmun, 2026a,b, MBC News, 2026], and it cannot be denied that the Mebane model had historical importance in the development of this sentiment. It is therefore important to clarify how much these models can actually do for us.

This paper argues that South Korean elections cleanly illustrate why the eforensics model, or at least its estimation process (as specified by its creators), may be flawed. We provide a series of validation tests that partially resemble the South Korean elections but should not be classified as election frauds by the design of (and the theory behind) the eforensics model. It turns out that eforensics model estimation suggests a high fraud probability even for these synthetic fraud-free data.

Furthermore, while the original Mebane analysis covered only the 21st South Korean general election, and was later corrected by Min and Yoo [2021] toward a much lower probability of election frauds once the flaws in the original analysis were fixed, the eforensics model actually suggests a much higher probability of election frauds in the elections held before early voting was introduced in South Korea, especially in those won by the main conservative party rather than the main liberal (or ‘democratic’) party that election fraud claims usually target.

The eforensics model also only detects election fraud favoring a single party in a single estimation. We therefore evaluate election frauds favoring each of the two mainstream parties (conservative and democratic/liberal) separately. It turns out that fraud shares do not change much between them, suggesting an internal model estimation bias.

We briefly review the history behind the South Korean early-voting system and early fraud claims. Nationwide in-person early voting (*sajeon tupyoo*) came into effect in 2013 and was first used in the 2014 local elections. The first general and presidential elections held under this early-vote system were the 20th general election (2016) and the 19th presidential election (2017). Before 2014, no mainstream election-fraud claim was raised about a Korean national election.¹ Relatively mainstream fraud allegations first emerged after the 21st general election (April 2020), focusing on early votes [East Asia Research Center, 2020]. Initially considered a fringe conspiracy theory, the fraud claim was given more media attention after the Mebane analysis. In the report, whose full text circulated in Korean media [Finance Today, 2020], roughly 28.7% of early-vote units (geographic election districts) were flagged as fraudulent [Mebane, 2020], and the fact that this was the work of a University of Michigan professor with proper expertise in statistical election analysis, supported by his prior work on Iran, Russia, and other contested elections, gave Korean election-fraud activists a justification far more reputable than YouTube analyses. The episode has recently been revisited by U.S. journalists in the context of Mebane’s Pennsylvania analysis. Mebane has since removed the Korea working paper from his website, acknowledging that it contained errors [Walker, 2026].

While Min and Yoo [2021] (which, like the original Mebane analysis, dealt with the 21st general election only) pointed out data-processing flaws in the original Mebane analysis due to a misunderstanding of the South Korean election system, the belief that early votes are fraudulent for statistical reasons still lives on, even if the focus has largely shifted away from them. For example, Jang Dong-hyeok, the leader of the People Power Party, pointed to in-precinct early-vote returns in the June 2026 local elections in which the top two candidates each received exactly the same number of votes in two different *dong* of the same metropolitan race. He asserted that such a coincidence had a probability of “1 in 590 million” [Kyunghyang Shinmun [Khan], 2026a, Seoul Shinmun, 2026b], to which some Korean statisticians voiced objections [Ohmynews, 2026]. Because it is easy to commit statistical fallacies (or at least such fallacy-based critiques can easily be stated), the importance of properly validated election forensics models is hard to overstate.

¹Partial qualification: the 2017 documentary *The Plan* alleged ballot-sorting-machine misclassification in the 2012 presidential election, but this never became a popular movement comparable to the post-2020 controversy.

2 The eforensics model

The eforensics R package [Ferrari et al., 2019] implements the Bayesian finite mixture model of election frauds developed in Mebane [2016] and Ferrari et al. [2018], which generalizes Klimek et al. [2012]. We use its quasi-binomial-logistic (qbl) model. This section explains the general logic of the model. Readers interested in the full details should consult the appendix.

The model analyzes a single election (which means we are not aggregating different elections for fraud analysis) at the level of an electoral *unit*, which is a geographic election district such as a neighborhood (*dong* in Korean) or a parliamentary constituency (*seongeogu*). In this model, the *leader* is not the party that actually won but the party chosen by the analyst to gain from fraud.

Each unit has a latent *population-wise* (or ‘true’, which is used interchangeably) turnout and a latent *population-wise* leader share, which form the fraud-free reference.

Unit i belongs to one of three classes: no fraud, incremental fraud or extreme fraud. In a fraudulent unit, part of the would-be abstainers are counted as votes for the leader (*manufactured* votes), and part of the votes cast for other parties are counted for the leader (*stolen* votes). Under incremental fraud these proportions are expected to be below 70%, and under extreme fraud above 70% (the default threshold in the official eforensics package).

The latent values cannot be left unrestricted. If every unit had its own freely estimated true turnout and leader share, the model could reproduce any observed result without fraud by setting those values equal to the observed ones, and fraud could never be detected. Therefore, the model needs to group these units, which is done at the province level. The provinces are Korea’s first-level administrative divisions, roughly comparable to small states in the United States, with Seoul further split into its three Gangnam-area districts (Gangnam, Seocho and Songpa) and the rest of the city, because the two differ markedly.

Each unit’s latent turnout and leader share on the logit scale are drawn at random from a common distribution whose mean depends on the unit’s province through province fixed effects. Fraud is then inferred from units that depart from this distribution. This feature can be the model’s weak point: genuine differences among units within a province that an assumed distribution cannot capture can look like fraud to the model. The fraud-free null datasets of Section 3.5 test whether this weak point explains the results for the Korean data. It does not, because the false positives persist even when such differences are removed.

The mixture weights $\pi = (\pi_1, \pi_2, \pi_3)$ are the probabilities that a unit belongs to each class, and the reported *fraud share* is $\pi_2 + \pi_3$.

Estimation is by MCMC via JAGS, as in the official package.

One detail of the official package is relevant to robustness. The package builds the leader-vote probability with the observed-turnout form rather than the latent-turnout form of the published model. In the qbl model the two forms differ only by sampling noise, and Ferrari et al. [2018] show that they are identical in a simplified version of the model. A plausible reason for the package’s choice, not discussed in the eforensics papers, is that the latent-turnout form far more often generates units with more leader votes than votes cast, which the package rejects as invalid (see the appendix). Section 4.5 confirms that our results are qualitatively the same under either form.

3 Data and experimental design

3.1 Elections and levels of aggregation

We assembled official National Election Commission results, obtained from the Korean government’s open-data portal,² for eleven Korean national elections: five general (18th [2008], 19th [2012], 20th [2016], 21st

²General (legislative, National Assembly) election results: <https://www.data.go.kr/data/15025527/fileData.do>. Presidential election results: <https://www.data.go.kr/data/15025528/fileData.do>.

[2020], 22nd [2024]) and six presidential (16th [2002], 17th [2007], 18th [2012], 19th [2017], 20th [2022], 21st [2025]). Five of these (the 2008 and 2012 general elections and the 2002, 2007, and 2012 presidential elections) predate early voting, and the remaining six were held under the early-voting system.

Each election was separately analyzed at two aggregation levels: *dong* (Korean administrative neighborhood, *beopjeong eupmyeondong*), with 2,700–3,600 units after filtering, and *constituency* (*seongeogu*), with 198–253 units.

For early-voting-era elections, analysis was also done separately for each vote channel. At the *dong* level, the vote channels are: 1) in-precinct early votes, 2) same-day votes, 3) in-precinct + same-day votes (the pooled channel, which cannot include out-of-precinct votes because they cannot be attributed to any *dong*). At the constituency level, they are: 1) in-precinct early votes, 2) out-of-precinct early votes (in the South Korean system, these are attributable to each constituency), together with the few other ballots reported only per constituency, 3) total early votes (in-precinct + out-of-precinct), 4) same-day votes, 5) total votes.

Pre-early-voting elections use the single ‘total-no-early’ channel. Because ‘pooled’, ‘total’ and ‘total-no-early’ cover different sets of ballots, Section 3.2 and the vote-channel table in the appendix define each channel precisely.

Units with 50 or fewer votes in any analyzed channel, or with zero leader votes across both early and same-day channels, were dropped. Because eforensics models turnout and leader share on the logit scale, units with either quantity closer than 0.001 to 0 or 1 were also excluded, since the transform is undefined at the boundary.

3.2 Terminology: cells, channels, leader coding, fraud share, and flags

Several terms used throughout the paper need clarification.

Unit. A *unit* is a geographic election district, the level at which the model is fit: a *dong* or a constituency, depending on the aggregation level. All model quantities, including the latent true turnout and leader share, are defined per unit. The finest level at which Korean results are published is the *polling district* (*tupyogu*), several of which make up a *dong*. We use it only for the Klimek comparison of Section 5.1.

Observed- and latent-turnout forms. The model builds each unit’s leader-vote probability, the chance that an eligible voter’s ballot is counted for the leader, from turnout, leader share and any fraud. The *observed-turnout form*, which the package uses, takes the turnout actually observed in the unit. The *latent-turnout form* takes the unit’s latent true turnout (Section 2).

Sweep. In the Klimek model (Section 5.1), a *sweep*, the term used in Klimek et al. [2018], is one search over all candidate parameter values, fitted to one simulated set of units whose turnout and vote shares are drawn from Normal distributions centered on the empirical modes (the most common values) of turnout and vote share. Each sweep therefore uses a single simulation.

Cell and run. A *cell* is one (election, aggregation level, vote channel) triple, for example ‘21st general election, *dong* level, in-precinct early channel’. Each cell is fit separately for each data source (real data, marginal null, joint null and self-null) and each leader coding, and a single such fit is a *run*.

Vote channel. A *channel* is the set of ballots whose turnout and leader share are modeled. In-precinct early and same-day ballots are reported per *dong*, while out-of-precinct early ballots and a few smaller categories (home- or ship-bound, overseas and misfiled ballots, and absentee ballots before early voting) are reported only per constituency. Three names are easily confused. *pooled* (*dong* level) is in-precinct early plus same-day ballots and omits all constituency-only ballots, 8–13% of valid votes. *total* (constituency level) is the complete vote. *total-no-early* (pre-early-voting elections, both levels) is every ballot cast at a polling station on election day. Despite its name, it excludes the 3–4% of absentee and overseas ballots, so that both levels cover the same ballots. A table in the appendix lists all channels and their composition.

Leader. The *leader* is the party (or presidential candidate) selected in advance to benefit from fraud, which need not be the party that actually won. ‘Democratic-coded’ (dem) runs select the main liberal party lineage (currently the Democratic Party of Korea) or its presidential candidate, and ‘conservative-coded’ (con) runs select the main conservative party lineage (currently the People Power Party) or its candidate.

Fraud share. The *fraud share* of a run is the estimated share of units in the two fraud classes, $\pi_2 + \pi_3$, reported as the sum of the posterior medians of π_2 and π_3 with an exact joint 95% highest posterior density (HPD) interval. π is a single mixture vector for the whole cell, so there is no unit-specific mixture weight. Its posterior mean is approximately the unweighted average of the units’ posterior probabilities of being fraudulent. Two consequences follow. First, the fraud share is a share of *units*, not of voters or votes: a dong of 500 eligible voters and one of 50,000 count equally. Second, it adds up fraud probability spread across all units, so it can be large even when no single unit is more likely fraudulent than not.

Fraudulent votes. F_w is the estimated total number of fraudulent (manufactured plus stolen) votes and F_t its manufactured-vote component, both defined in the appendix. Unlike the fraud share, these totals weight each unit by its electorate. Their intervals are exact 95% HPD intervals, computed from the posterior draws.

Units flagged. A unit is flagged when its most probable class, given its posterior class probabilities averaged across chains, is incremental or extreme fraud. A unit with a fraud probability of 0.45 is therefore not flagged, although it contributes 0.45 to the fraud share.

Convergence and multimodality flags. A run *fails convergence* when the largest Gelman–Rubin potential scale reduction factor $\hat{R}$ [Gelman and Rubin, 1992] of π_1 , π_2 and π_3 across the four chains exceeds 1.05. Following Mebane [2023], a multimodality flag is raised when, for some component π_j , the chain-specific posterior means differ by more than 0.01 and the Hartigan dip test [Hartigan and Hartigan, 1985] rejects unimodality at the 5% level.

3.3 MCMC configuration and chain length

Model estimation used JAGS with 4 chains throughout the analysis, each with 1,000 adaptation iterations, 300 burn-in iterations and 1,000 sampling iterations. We call this the *short-chain configuration*. (By comparison, Min and Yoo [2021] used even shorter chains.) Most runs fail the $\hat{R} \leq 1.05$ criterion, so one might ask whether this non-convergence is an artifact of insufficient sampling. We checked this with the package’s optional PSRF stopping rule (Section 3.4), which keeps extending the chains until $\hat{R} < 1.05$ for every mixture weight. We reran all 35 constituency-level cells with this rule, for the real data and both nulls under both leader codings (210 runs), starting from the short-chain configuration and allowing up to ten extensions, the package’s default limit. A run that uses all ten extensions draws 15,300 iterations per chain after adaptation, 11.8 times the 1,300 of a short-chain run.

Longer chains do not produce convergence. Of the 210 runs, 173 used all ten extensions without meeting the rule. Of the 37 that met it, 36 did so at the first check before any extension (median maximum $\hat{R}$ of 1.02, against 1.59 for the other runs). Across all 210 runs, the median maximum $\hat{R}$ is 1.42 after extension and 1.47 in the short-chain runs, the share of runs with $\hat{R} \leq 1.05$ is 16.7% and 17.1%, and extension lowers $\hat{R}$ in only 100 of the 210 runs. The fraud-share estimates barely move (Figure 1). They correlate with the short-chain estimates at $r = 0.96$, with a median absolute difference of 0.4 percentage points and no systematic shift (mean difference 0.0 points). Running the chains about twelve times longer thus neither resolves the non-convergence nor changes the estimates, so the short-chain configuration is close to the best achievable. The complete PSRF results are listed in the appendix.

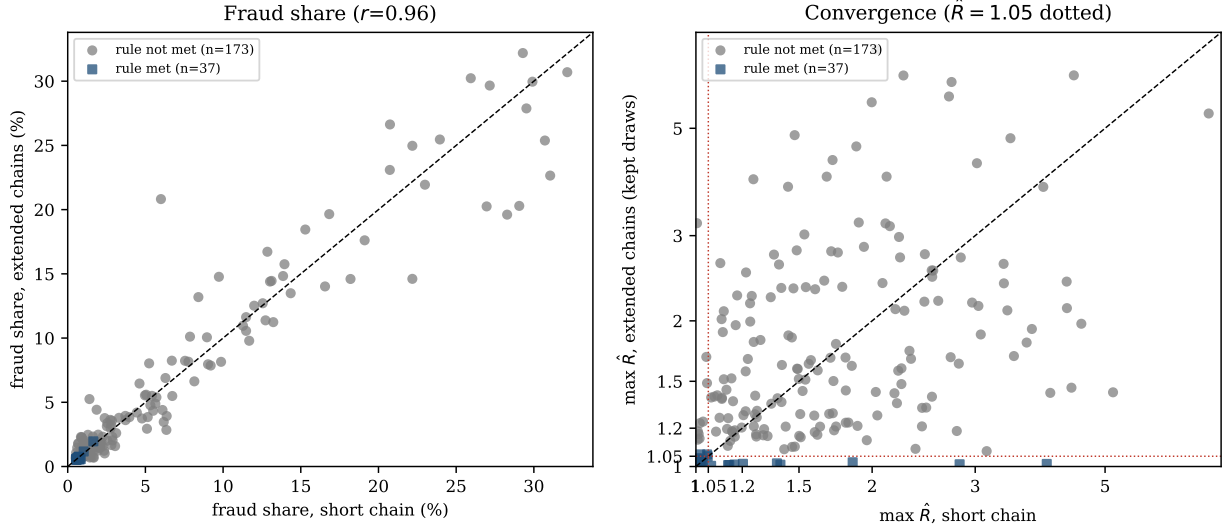


Figure 1: Chain-length check: the 210 constituency-level runs rerun with the package’s PSRF stopping rule (chains extended up to ten times, about twelve times as many iterations) against the same runs in the short-chain configuration. Left: fraud share. Right: maximum $\hat{R}$ on the mixture weights, on a log scale, with the 1.05 criterion dotted. Blue: runs that met the PSRF rule.

3.4 The eforensics stopping rule and the Gelman-Rubin diagnostic

The eforensics package does not use Gelman-Rubin $\hat{R}$ by default for the estimation stopping rule, though an option requiring $\hat{R} < 1.05$ is available. Ferrari et al. [2018] required both a potential scale reduction factor (PSRF, that is, $\hat{R}$) of at most 1.1 and a small Monte Carlo standard error (MCSE). By default, the current package instead requires each chain’s MCSE for π to be below 0.05 and otherwise extends the chains up to ten times. Except for the PSRF check (Section 3.3) and the latent-turnout robustness check (Section 4.5), all eforensics runs in this paper use the package’s default model, sampler and stopping-rule settings.

Per-chain MCSE cannot detect cross-chain problems: chains that are individually stable but settled in different regions of the posterior. $\hat{R}$ detects this divergence between chains.

No published statement explains why the official R eforensics package drops the PSRF stopping rule by default, but two reasons are plausible. First, the rule is rarely satisfied, and running the chains longer does not change that. In our PSRF check, only 37 of 210 runs met it. The runs that did converge had essentially no estimated fraud: 35 of the 37 have fraud shares of 1% or less, and none exceeds 2.0%. Second, Mebane [2023] treats disagreement between chains in the mixture weights not as a convergence failure but as a substantive signal of lost votes or other misspecification, and this may have been part of the reason why the official R package does not use PSRF by default.

3.5 Three fraud-free controls (validation datasets for the model)

The central methodological innovation of this study is a set of synthetic datasets that match each real dataset’s turnout and vote-share distributions while containing, from the model’s point of view, no fraud. Each keeps every unit’s number of eligible voters and province and replaces only its vote counts. The appendix gives the exact constructions.

Marginal null. Each unit’s turnout and leader share are redrawn independently around their observed values, with small Beta-distributed noise. This treats each unit’s real turnout and leader share as its latent true

values and checks whether the model has difficulty with the real differences between units within a province and is therefore biased toward reporting fraud. This dataset has the weakest claim to being fraud-free, but that turns out not to matter, given the results for the other two.

Joint null. Each unit’s turnout and leader share are drawn on the logit scale from a bivariate Normal distribution with its province’s mean and the covariance of the real data around the province means. This discards all unit-specific information and asks whether the model has difficulty with data generated by province fixed effects and the empirical covariance alone. If it does, the model (or its estimation) carries an implicit theory of which covariance structures are consistent with a fraud-free election.

Model-consistent self-null. We fit the model to the real data, take the estimated parameters of its fraud-free class, and simulate a new dataset from the model’s fraud-free process with every unit forced into that class ($\pi = (1, 0, 0)$). This dataset is then fit again in the same way. Because it comes from the estimated model’s no-fraud process, a substantial fraud estimate here is a very strong indicator of problems in the model’s estimation, since the refit should recover $\pi_1 \approx 1$.

4 Results

4.1 False-positive fraud on no-fraud null data

Across the 232 marginal-null and joint-null runs under both leader codings (listed in the appendix), 77 (33.2%) report a fraud share above 5%. In 54 of them, even the lower bound of the 95% HPD interval exceeds 5%. The most extreme false positive, the joint null of the 2025 presidential election at the constituency level (total channel, Democratic-coded), has an estimated fraud share of 31.1% (95% HPD 21.6–39.6%) and 3.7 million estimated fraudulent votes (95% HPD 2.9–4.7 million). A table in the appendix lists selected extreme cases, showing that objections to synthetic null datasets cannot target just one type. The model-consistent null dataset does not change these false-positive issues either (see the appendix).

Real-data fraud share correlates with the marginal-null fraud share at $r = 0.992$ and with the joint-null fraud share at $r = 0.942$ across the 58 Democratic-coded cells, and at $r = 0.981$ and $r = 0.985$ across the 58 conservative-coded cells (Figure 2). The model-consistent self-null dataset, discussed in Section 4.2, leads to essentially the same conclusion.

This suggests that the eforensics model cannot distinguish the real Korean election data from its synthetic fraud-free twin datasets for the purpose of fraud detection.

4.2 Confirmation with the model’s own no-fraud process (the self-null dataset)

Even for the model-consistent self-null dataset, the eforensics model still reports substantial fraud. The marginal and joint nulls of Section 3.5 use noise processes chosen outside the model, and therefore there is still room for objections even to the joint-null dataset. The self-null dataset rules this out.

We ran self-null estimation for all 58 cells under both leader codings with the short-chain configuration. With the Democratic-coded leader, the median fraud share is 1.8% and the mean 4.6%, with 17 cells (29.3%) exceeding 5% and 5 (8.6%) exceeding 15%. With the conservative-coded leader, the median is 1.3% and the mean 4.2%, with 14 cells (24.1%) exceeding 5% and 5 (8.6%) exceeding 15% (see the appendix).

Paired with its real-data counterpart, the self-null fraud share correlates with the real-data fraud share at $r = 0.948$ with the Democratic-coded leader and $r = 0.978$ with the conservative-coded leader (58 cells each). Figure 3 shows the comparison, with the complete listing in the appendix.

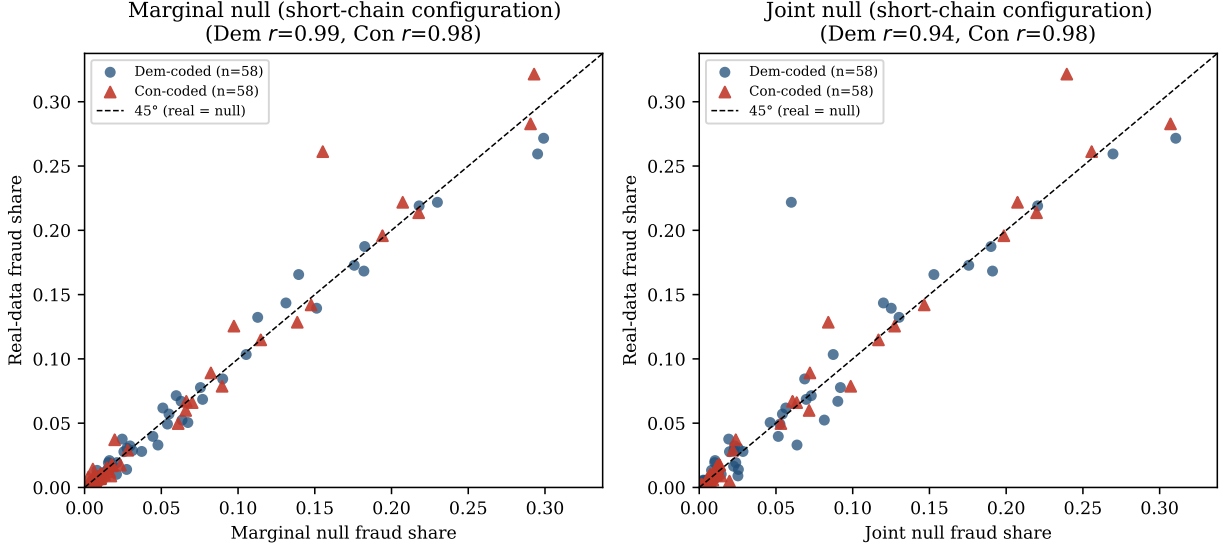


Figure 2: Real-data fraud share vs. fraud-free null fraud share, short-chain configuration ($n_{\text{iter}}=1,000$, $\text{burn.in}=300$), for all 58 Dem(ocratic)-coded cells (blue circles) and 58 con(servative)-coded cells (red triangles).

4.3 The pre-early-voting natural experiment

The model assigns large fraud shares to elections that predate early voting, where no mainstream fraud claim has ever existed. A table in the appendix reports real-data results for the five pre-early-voting elections. With the Democratic-coded leader, the presidential elections of 2002, 2007 and 2012 receive fraud shares of 7–17%, or 0.3–1.6 million estimated fraudulent votes. At the dong level, the 2012 and 2002 presidential elections (17.3% and 13.9%) are exceeded only by the pooled channels of the 2025 and 2022 presidential elections. When the conservative party is coded as the leader (appendix table), the real-data fraud share for the 2012 presidential election reaches 22.2% at the constituency level and 26.1% at the dong level, the largest dong-level estimate of any election, with up to 1.9 million estimated fraudulent votes.

Figure 4 places these estimates on a timeline that reflects the introduction of early voting and the onset of fraud claims. Note that the estimated fraud shares in elections with no mainstream fraud claim are as large as those in elections with such claims.

4.4 Bidirectional leader-coded confirmation

The eforensics model’s fraud mechanisms are unidirectional, so a single run can only test for fraud favoring either the Democratic (liberal) party or the conservative party. Testing both leader codings reproduces the same pathology for either side (appendix table; Figures 5 and 6).

4.4.1 The 2012 presidential election

The eforensics model does not always implicate the South Korean Democratic side more. In the 2012 presidential election, the model reports more fraud favoring the conservative winner ($\sim 26\%$ dong, $\sim 22\%$ constituency) than favoring the Democratic loser ($\sim 17\%$ at both levels; appendix table and Figure 5). Neither estimate, however, should count as evidence of fraud in either direction. This election is discussed separately because it is one for which a significant fraud estimate is reported even though no mainstream

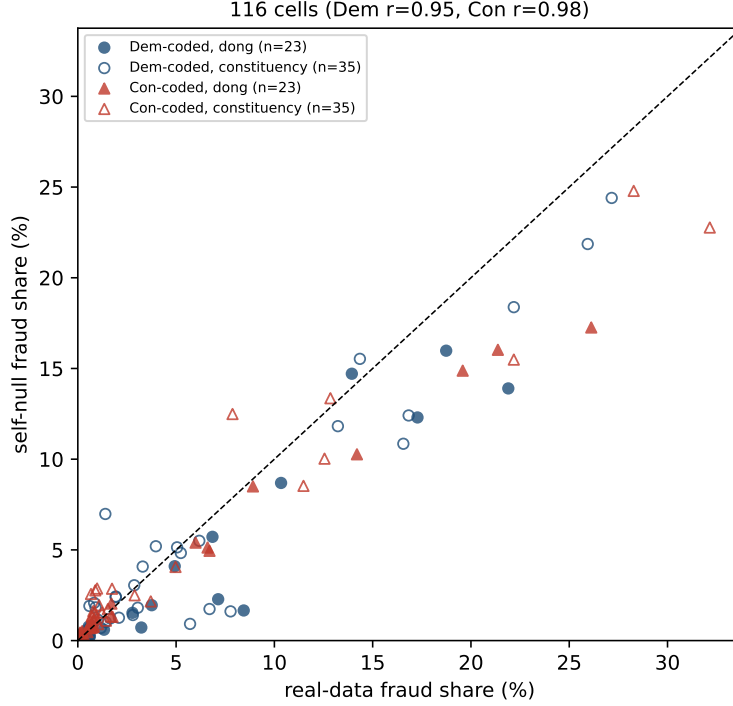


Figure 3: Real-data fraud share vs. model-consistent self-null fraud share (π forced to $(1, 0, 0)$), short-chain configuration, all 58 cells under each leader coding.

election fraud claim has ever existed, and because it was held before the introduction of early voting.

4.5 Robustness to the latent-turnout form

All results above use the package’s observed-turnout form, in which the leader’s vote probability is built from each unit’s observed turnout. To check whether the false positives come from using the observed-turnout instead of the latent-turnout form, we re-estimated every run with the latent-turnout form of the leader-vote probability. Both forms lead to qualitatively the same results. As before, the appendix provides the full results.

The latent-turnout form lowers the estimates: the median fraud share falls from 1.4% to 0.8%, and the number of fraud-free null runs reporting more than 5% fraud falls from 77 to 46 of 232. It also converges somewhat more often, although most runs still fail the $\hat{R}$ criterion (78% against 89%). But it does not remove the false positives, either on the fraud-free nulls or in the elections without fraud claims, and it does not separate real data from fraud-free data (Figure 7). The largest fraud-free estimate rises to 40.0% (95% HPD 8.2–50.1%) for the joint null of the 2025 presidential election at the constituency level (conservative-coded), with 3.5 million estimated fraudulent votes. Real-data estimates still track the null estimates at $r = 0.97$ with both joint and marginal nulls under the Democratic-coded leader and at $r = 0.86$ (joint) and 0.96 (marginal) under the conservative-coded leader. In 60 of the 116 cells, the joint-null estimate is at least as large as the real-data estimate. The three presidential elections held before early voting still receive fraud shares of 4.0–17.1% (Democratic-coded) and 3.1–18.4% (conservative-coded), with up to 1.7 million estimated fraudulent votes.

The latent-turnout self-null gives the same picture. Fraud shares exceed 5% in 11 (Democratic-coded) and 10 (conservative-coded) of the 58 fraud-free cells and correlate with the real-data estimates at $r = 0.95$

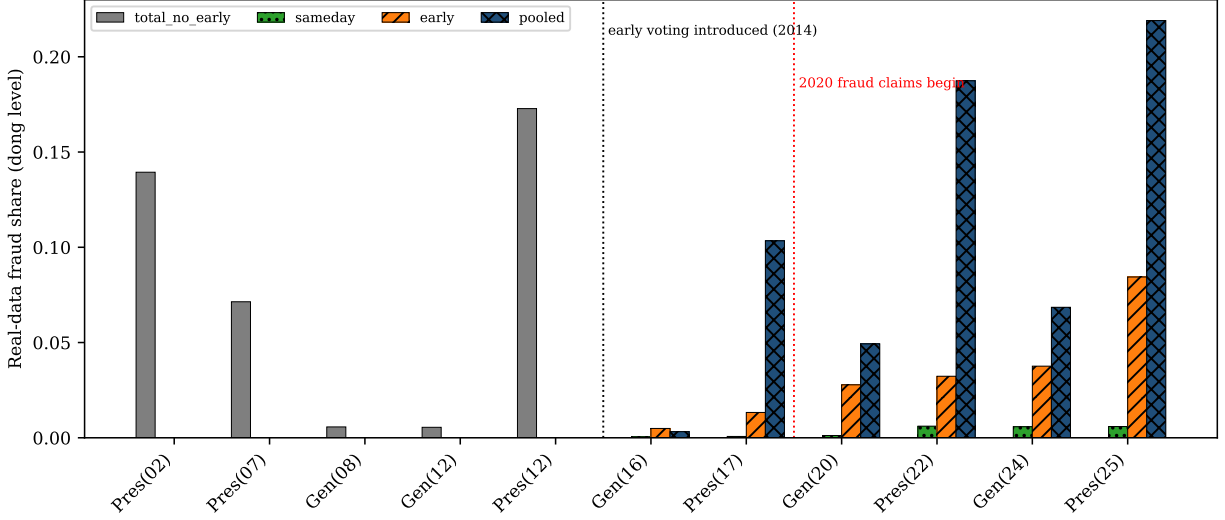


Figure 4: Real-data fraud share by election and channel (dong level, short-chain configuration, Democratic-coded leader), in chronological order. Dashed lines mark the introduction of early voting (2014) and the onset of organized fraud claims (2020). Estimates in the era with no fraud claims are as large as those in the era with fraud claims, and same-day-only channels remain near zero throughout.

and 0.97.

5 Discussion: is eforensics credible as a fraud detector?

To summarize, we have noted that the eforensics model has three problems.

First, the estimated model assigns fraud shares to datasets that are supposed to be fraud-free by construction. The three null datasets address different potential criticisms of such datasets, with the strongest test coming from the model-consistent self-null, in which we take the eforensics model’s estimated fraud-free process from the real data and generate a synthetic dataset. Even for this dataset, eforensics model estimation results in significant fraud detection that is incompatible with the theoretical design of the dataset. These failures persist when the leader’s vote probability is built from latent instead of observed turnout (Section 4.5).

Second, the estimated model fails historical validation. Unless one is claiming that the eforensics model uncovered election frauds preceding both existing mainstream election fraud claims and early votes, significant fraud detection for these pre-2016 elections suggests some issue with eforensics model estimation.

Lastly, the estimated model suggests similar scales of fraud regardless of which party is coded as the artificially favored leader. If taken at face value, this would suggest that in South Korean elections, every side is committing some level of election fraud.

5.1 Does this extend to other election forensics methods?

The misidentification of ordinary demographic and regional heterogeneity as a fraud signature in aggregate (turnout, vote-share) data is not necessarily confined to the eforensics model. It is a plausible concern for any method that infers manipulation from just the joint turnout/vote-share distribution, including the widely used “election-fingerprint cluster-detection” approach of Klimek et al. [2012].

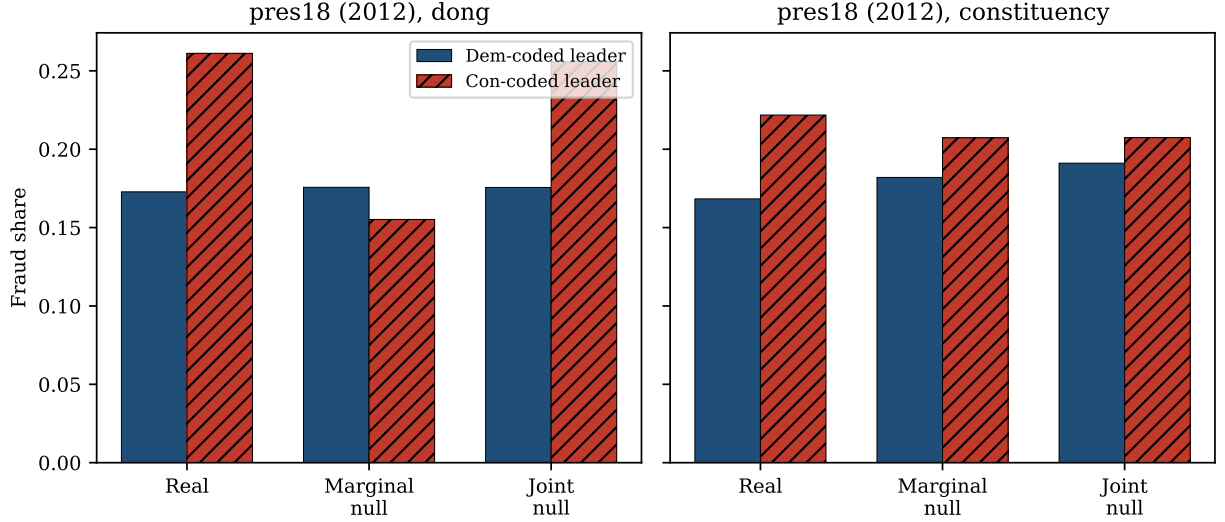


Figure 5: 2012 presidential election fraud share under Democratic-coded (blue) and conservative-coded (red) leader, real data and both fraud-free nulls, at dong and constituency level, short-chain configuration.

To check whether this concern applies, we fit the fraud model of Klimek et al. [2012], updated in Klimek et al. [2018] for the Turkish election analysis, to the real data and to the joint-null data. The model assumes that without fraud, every unit’s turnout and leader share come from one nationwide pair of Normal distributions centered on the modes of the empirical turnout and leader-share distributions. These Normal distributions provide simulated fraud-free data for each *sweep* (Section 3.2). The model estimation then chooses the shares of units with incremental and extreme fraud on top of the simulated fraud-free data so that the simulated joint distribution of turnout and leader share matches the observed one. This is repeated 200 times (*sweeps*). We run this analysis only for the five elections held before early voting, at the dong and polling-district levels and under both leader codings. The appendix describes the procedure and reports all results.

The Klimek model fails the same historical and fraud-free validation tests as eforensics (Figure 8). On the raw dong-level fingerprints, none of the ten real-data fits (five elections, both leader codings) falls below 5%. The fraud-free joint null is qualitatively similar: none of the ten fits falls below 5%, and the real-data and joint-null shares correlate at $r = 0.80$.

At the polling-district level, the real-data and joint-null fraud shares are weakly negatively correlated ($r = -0.42$), but some real-data and joint-null fits still show large fraud shares, and some null shares clearly exceed the corresponding real-data shares. Klimek et al. [2012] present insensitivity to the level of aggregation as a key property of their method, but this does not hold for the Korean elections: the same election’s estimates differ between the two levels by up to 80 percentage points.

Neither removing province means (Figure 8) nor excluding the regions with the strongest partisan leanings (see the appendix) resolves these problems.

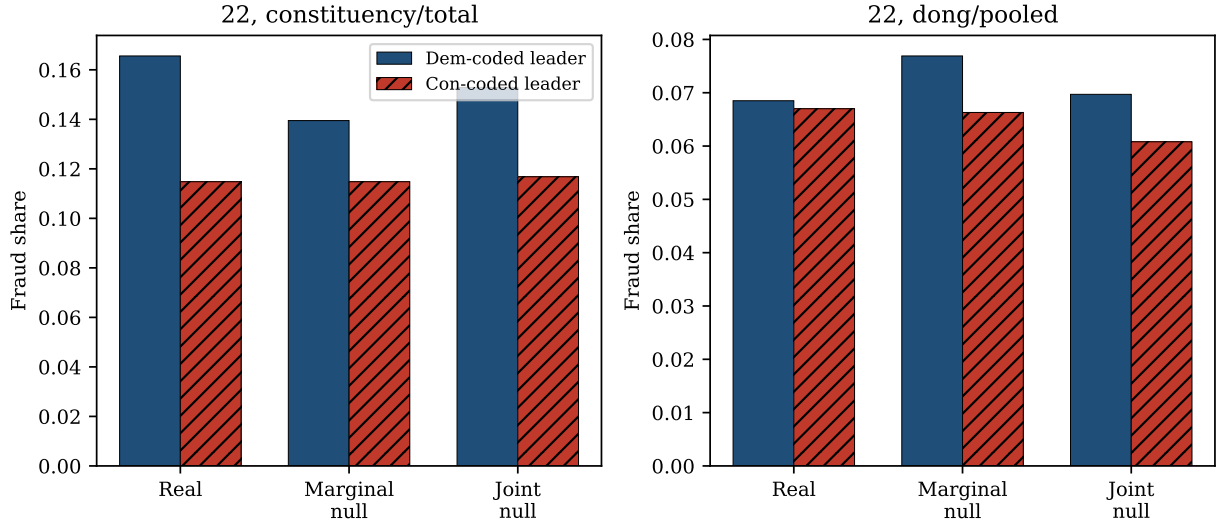


Figure 6: 2024 general election fraud share under Democratic-coded (blue) and conservative-coded (red) leader, real data and both fraud-free nulls, for the two largest-signal channels, short-chain configuration.

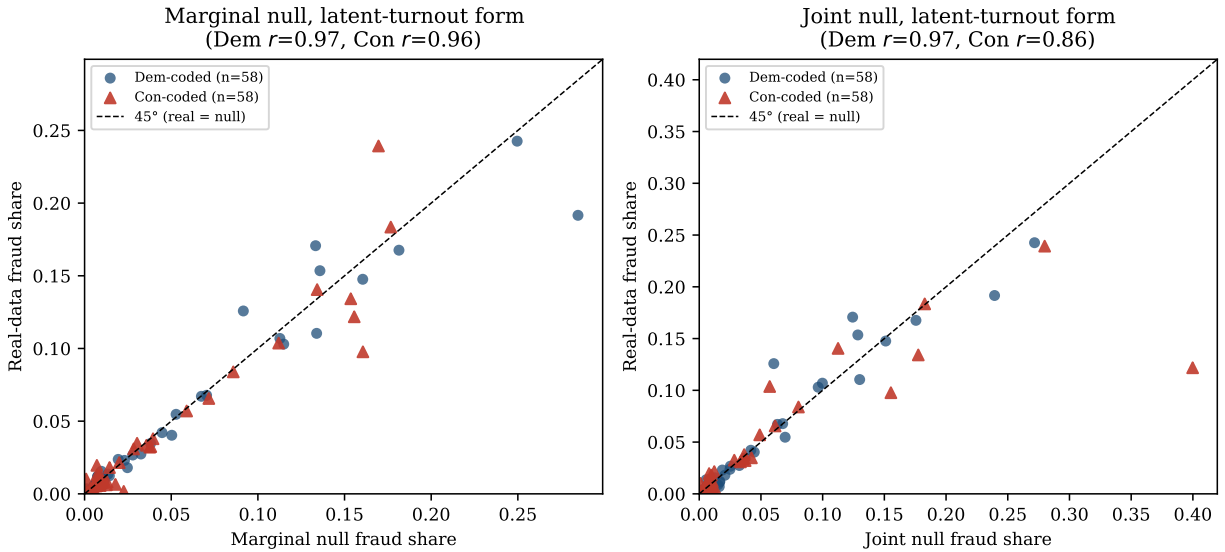


Figure 7: Real-data fraud share against marginal-null (left) and joint-null (right) fraud share under the latent-turnout form, both leader codings, short-chain configuration. Compare Figure 2 for the observed-turnout form.

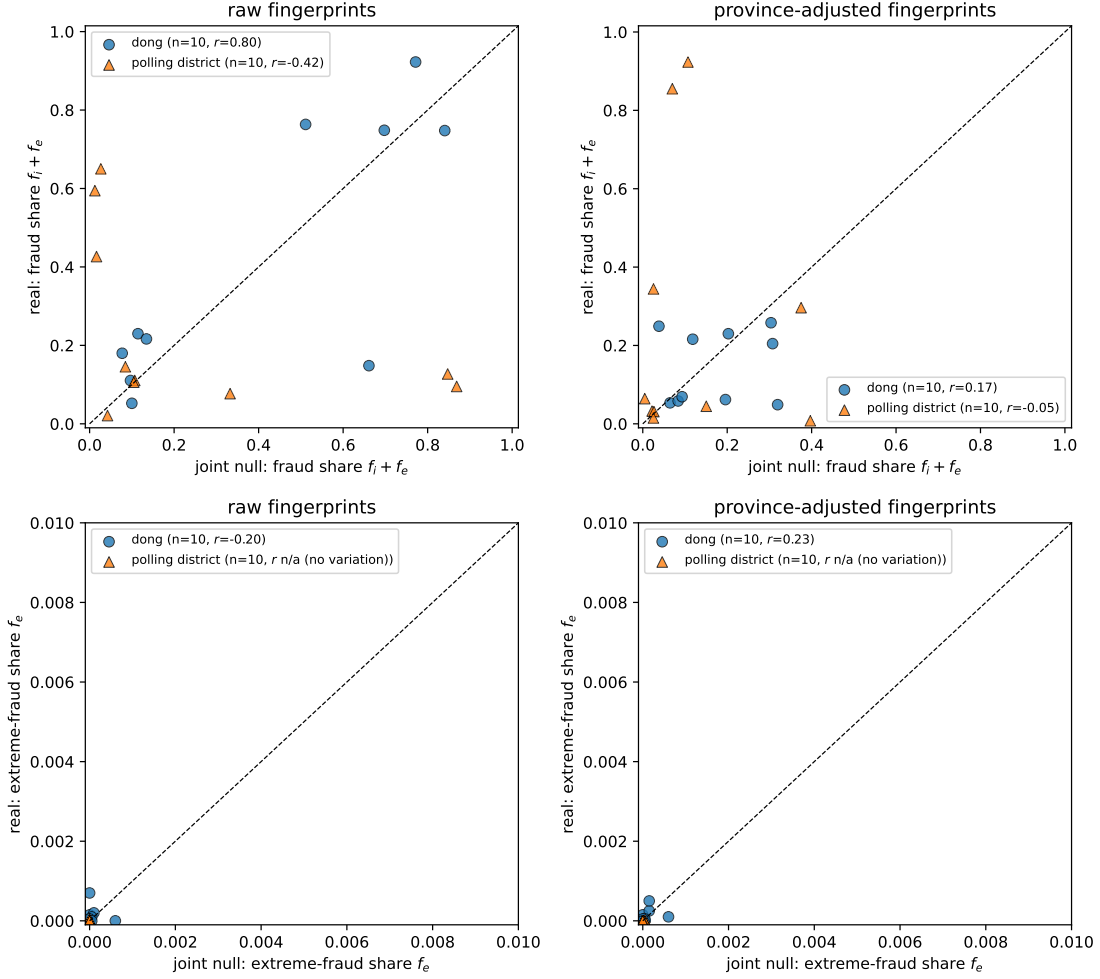


Figure 8: Klimek et al. fraud-model estimates for the real dataset (vertical axis) against its joint-null dataset (horizontal axis): the total fraud share $f_i + f_e$ (top) and the extreme-fraud share f_e (bottom), on the raw fingerprints (left) and after removing province means (right). The five elections held before early voting, under both leader codings, at the dong (circles) and polling-district (triangles) levels (10 runs per level in each panel). r is computed within each level and is undefined for the polling-district f_e , which is essentially zero in every fit. Each estimate is the mean over 200 sweeps.

6 Conclusion

In this paper, we reported eforensics model estimation results on South Korean national election data (both legislative and presidential) from 2002 to 2025, covering six presidential elections and five general (legislative) elections. Some of these elections predate the introduction of early voting as well as mainstream election fraud claims.

The eforensics model estimation failed historical validations and fraud-free null tests, which means that its widespread fraud detection across elections cannot be trusted. Early voting turned out to be irrelevant to the model’s South Korean election fraud estimates. Furthermore, the fitted model suggested fraud regardless of which side was designated as favored by fraud. The broad incremental-fraud component of the Klimek et al. [2012] fingerprint model shows the same validation failures.

This suggests limits on how far a statistical model can go in detecting election frauds. If the problem were confined to the tests not involving fraud-free synthetic-null validation, one could say that improving the theoretical model might somehow change the outcome. However, the fraud-free synthetic-null validation suggests that the problem can be deeper: it involves problems with Bayesian estimation, because estimation on theoretically fraud-free datasets, especially the model-consistent self-null dataset, is somehow biased toward detecting fraud. This is not directly a pre-estimation theory problem.

We also note that this paper is not targeted against those wishing for more election transparency in South Korea. The opposite is true. Even for those critical of the National Election Commission in South Korea, we believe it would be helpful to clarify how much a statistical model can do to settle election fraud issues today. The eforensics model case suggests that there is still a long way to go for statistical election forensics, and that non-statistical and non-mathematical work may be required to either prove or disprove election frauds and to establish the necessary transparency in South Korean elections and, more generally, in elections in the United States and in other parts of the world.

Data availability

The raw election-result CSV files used throughout this study are official National Election Commission returns published by the Korean government’s open-data portal, <https://www.data.go.kr>. General election results are available at <https://www.data.go.kr/data/15025527/fileData.do>, and presidential election results at <https://www.data.go.kr/data/15025528/fileData.do>.

The processed unit-level tables, all model outputs, and the full analysis and figure-generation code are provided at https://mkimacad.github.io/south_korean_election_forensics. Given its length, the codebase is not reproduced in this manuscript. The appendix documents every script’s purpose, software requirements, and the commands needed to reproduce each stage of the analysis.

A Model specification

This appendix gives the full specification of the `qbl` model summarized in the main text’s model section (“The eforensics model”).

The `eforensics` R package [Ferrari et al., 2019] implements the Bayesian finite mixture model of election frauds developed in Mebane [2016] and Ferrari et al. [2018], which is a generalization of Klimek et al. [2012]. Its `qbl` (quasi-binomial-logistic) variant assigns each unit i , with eligible voters N_i , leader votes W_i , valid votes V_i , and abstentions $A_i = N_i - V_i$, to one of three latent classes $Z_i \in \{1, 2, 3\}$ (no fraud, incremental fraud, extreme fraud) with population mixture weights $\pi = (\pi_1, \pi_2, \pi_3)$.

Which party counts as the ‘leader’ is selected in advance by the analyst. The term does not refer to the party that actually led the vote in unit i . Fraud detection works unidirectionally: a single fit can only detect fraud favoring the designated party. W_i above is that party’s vote count.

The prior on the mixture weights is specified as:

$$Z_i \sim \text{Categorical}(\pi), \quad \pi_k = \frac{u_k}{u_1 + u_2 + u_3}, \quad u_1 \sim U(0, 1), \quad u_2, u_3 \mid u_1 \stackrel{\text{iid}}{\sim} U(0, u_1). \quad (\text{A.1})$$

This prior imposes $\pi_2 \leq \pi_1$ and $\pi_3 \leq \pi_1$. Each unit has a latent *true* turnout τ_i and a latent *true* leader share ν_i (the values that would have been observed without fraud), modeled as follows:

$$\text{logit } \tau_i = x_i^\top \beta_\tau + \theta_i, \quad \theta_i \sim N(\alpha_\tau, \sigma_\tau^2); \quad \text{logit } \nu_i = x_i^\top \beta_\nu + \eta_i, \quad \eta_i \sim N(\alpha_\nu, \sigma_\nu^2), \quad (\text{A.2})$$

We can think of the above equation as follows: $x_i^\top \beta_\tau$ and $x_i^\top \beta_\nu$ provide the province-level fixed effects (where ‘province’ would be something like states in the United States). The terms θ_i and η_i then add unit-level noise, and together these determine the latent true values before any fraud effect is applied.

We use the full set of provinces, which currently comprises Seoul, the six metropolitan cities, Sejong, and the nine provinces. Seoul is further split into two groups: its three Gangnam-area districts (Gangnam, Seocho and Songpa) and the rest of the city, so the model has 18 province groups from 2012 on and 17 in the earlier elections. The exact set varies slightly across the earlier elections in our data, because Sejong and the special self-governing provinces were created later.³

The hyperpriors are $\alpha_\tau, \alpha_\nu \sim N(0, 1)$, $\sigma_\tau^2, \sigma_\nu^2 \sim \text{Exponential}(5)$, and independent $N(0, 1)$ priors on the province coefficients. The intercept entries of β_τ and β_ν are given a prior with precision 10^4 , which is close to fixing them at zero, so that the overall intercept is carried by α_τ and α_ν instead.

We now discuss how fraud is introduced into the latent τ_i and ν_i . In a fraudulent unit, a proportion of would-be abstainers is “manufactured” into leader votes and a proportion of would-be opposition votes is “stolen” for the leader. Writing (ι_i^M, ι_i^S) for the manufactured and stolen proportions under incremental fraud and (χ_i^M, χ_i^S) for the manufactured and stolen proportions under extreme fraud, the expected observed abstention proportion a_i^* and leader-vote proportion w_i^* (both relative to N_i) are

$$a_i^* = \begin{cases} 1 - \tau_i, & Z_i = 1, \\ (1 - \tau_i)(1 - \iota_i^M), & Z_i = 2, \\ (1 - \tau_i)(1 - \chi_i^M), & Z_i = 3, \end{cases} \quad (\text{A.3})$$

$$w_i^* = \begin{cases} \tau_i \nu_i, & Z_i = 1, \\ \tau_i \nu_i + \iota_i^M (1 - \tau_i) + \iota_i^S \tau_i (1 - \nu_i), & Z_i = 2, \\ \tau_i \nu_i + \chi_i^M (1 - \tau_i) + \chi_i^S \tau_i (1 - \nu_i), & Z_i = 3, \end{cases} \quad (\text{A.4})$$

³The code also supports a coarser four-megaregion grouping (Capital, Gangwon-Chungcheong, Yeongnam, and Honam-Jeju, as in Min and Yoo, 2021). The results reported in this paper use the full province set.

Then A_i (the realized abstention count) and W_i (the realized leader-vote count) are drawn as:

$$A_i \sim \text{Binomial}(N_i, a_i^*), \quad W_i \sim \text{Binomial}(N_i, w_i^*) \quad (\text{A.5})$$

Equation (A.4) builds the leader-vote probability w_i^* from the unit's latent true turnout τ_i . We call this the *latent-turnout form*. The official R package instead builds it from the turnout actually observed in the unit, replacing τ_i in equation (A.4) by the turnout implied by the observed abstention count A_i :

$$\tilde{\tau}_i = \begin{cases} 1 - A_i/N_i, & Z_i = 1, \\ 1 - (A_i/N_i)/(1 - \iota_i^M), & Z_i = 2, \\ 1 - (A_i/N_i)/(1 - \chi_i^M), & Z_i = 3, \end{cases} \quad (\text{A.6})$$

$$w_i^* = \begin{cases} \tilde{\tau}_i \nu_i, & Z_i = 1, \\ \tilde{\tau}_i \nu_i + \iota_i^M(1 - \tilde{\tau}_i) + \iota_i^S \tilde{\tau}_i(1 - \nu_i), & Z_i = 2, \\ \tilde{\tau}_i \nu_i + \chi_i^M(1 - \tilde{\tau}_i) + \chi_i^S \tilde{\tau}_i(1 - \nu_i), & Z_i = 3, \end{cases} \quad (\text{A.7})$$

We call this the *observed-turnout form*. It is the package default and is used for every result in the paper except the robustness check described in the main text (“Robustness to the latent-turnout form”). The two forms differ only in which turnout enters w_i^* . Everything else, including the binomial draws of equation (A.5), is the same.

The package follows Ferrari et al. [2018]. In the version of the model they analyze first, observed abstention and leader share are exact, noise-free functions of latent turnout, latent leader share and the fraud proportions (their equations 1–2): in a fraud-free unit, for example, observed abstention is exactly $1 - \tau_i$. Latent turnout can then be recovered from observed abstention, so writing the leader-vote probability with observed turnout (their equation 3) describes the same model. They prefer to write it that way because the MCMC sampler can then handle the unknown fraud proportions without integrating over them analytically (their equation 9). The package's `qbl` model, however, draws the counts A_i and W_i binomially around a_i^* and w_i^* (equation (A.5)). Observed turnout then differs from latent turnout by sampling noise, and the two forms are no longer the same model. Ferrari et al. [2018] do not discuss this difference.

The package's documentation does not say why it uses the observed-turnout form, but simulating from the fitted model suggests a practical reason. Under the latent-turnout form, A_i and W_i in equation (A.5) are drawn independently, so nothing ties the leader's votes to the number of votes actually cast. The model therefore assigns positive probability to units in which the leader receives more votes than were cast ($W_i > N_i - A_i$). When we simulated data from the fitted latent-turnout model, such units did occur, in places where the leader wins almost every vote, and the package rejects these data as invalid. The observed-turnout form ties the leader-vote probability to the observed turnout, so the expected number of leader votes, $\nu_i(N_i - A_i)$ in a fraud-free unit, can never exceed the votes cast, and the draw varies less because the abstention count is fixed. This makes invalid draws much rarer, though not impossible, since W_i remains binomial in N_i rather than in the number of votes cast. For a fraud-free unit with 5,000 eligible voters, turnout of 30% and a leader share of 95%, an invalid draw occurs with probability of roughly 5% under the latent-turnout form and under 1% under the observed-turnout form. The observed-turnout form therefore gives up some fidelity to the model's latent structure in exchange for far fewer impossible datasets.

The fraud proportions are generated as follows (M referring to “manufactured” and S to “stolen”):

$$\begin{aligned} \iota_i^M &= N_i^{\iota M}/N_i, & N_i^{\iota M} &\sim \text{Binomial}(N_i, \mu_i^{\iota M}) \\ \iota_i^S &= N_i^{\iota S}/N_i, & N_i^{\iota S} &\sim \text{Binomial}(N_i, \mu_i^{\iota S}) \\ \chi_i^M &= N_i^{\chi M}/N_i, & N_i^{\chi M} &\sim \text{Binomial}(N_i, \mu_i^{\chi M}) \\ \chi_i^S &= N_i^{\chi S}/N_i, & N_i^{\chi S} &\sim \text{Binomial}(N_i, \mu_i^{\chi S}) \end{aligned} \quad (\text{A.8})$$

$$\begin{aligned}
\mu_i^{\iota M} &= k \operatorname{logit}^{-1}(\theta_i^{\iota M}), & \mu_i^{\chi M} &= k + (1 - k) \operatorname{logit}^{-1}(\theta_i^{\chi M}), \\
\mu_i^{\iota S} &= k \operatorname{logit}^{-1}(\theta_i^{\iota S}), & \mu_i^{\chi S} &= k + (1 - k) \operatorname{logit}^{-1}(\theta_i^{\chi S}), \\
\theta_i^l &\sim N(\alpha_l, \sigma_l^2), & \alpha_l &\sim N(0, 1), & \sigma_l^2 &\sim \text{Exponential}(5), & l &\in \{\iota M, \iota S, \chi M, \chi S\},
\end{aligned} \tag{A.9}$$

with $k = 0.7$ in the package. Unlike turnout and leader share, the fraud proportions carry no province effects in our runs, although the R package allows such effects.

Each unit belongs to exactly one class, so the manufactured and stolen vote proportions realized in unit i are

$$m_i = \begin{cases} 0, & Z_i = 1, \\ \iota_i^M (1 - \tau_i), & Z_i = 2, \\ \chi_i^M (1 - \tau_i), & Z_i = 3, \end{cases} \quad s_i = \begin{cases} 0, & Z_i = 1, \\ \iota_i^S \tau_i (1 - \nu_i), & Z_i = 2, \\ \chi_i^S \tau_i (1 - \nu_i), & Z_i = 3. \end{cases} \tag{A.10}$$

The implied number of fraudulent votes in unit i is then $N_i(m_i + s_i)$, and the election-level totals reported throughout this paper are obtained by summing the posterior means of these per-unit quantities:

$$F_t = \sum_i N_i \mathbb{E}[m_i \mid \text{data}], \quad F_w = \sum_i N_i \mathbb{E}[m_i + s_i \mid \text{data}]. \tag{A.11}$$

F_t therefore counts only manufactured votes (would-be abstentions converted into leader votes), while F_w counts manufactured and stolen votes together. We compute both from the posterior draws. For each draw, the fraudulent votes in each unit follow from its drawn class, fraud proportions, latent turnout and latent leader share, and summing over units gives the totals for that draw. This also yields exact 95% HPD intervals for the totals. The per-unit fraud summaries that the package itself returns cannot be used for this, because they summarize the fraud proportions ι and χ , which are rates relative to would-be abstainers and would-be opposition voters rather than shares of the electorate. Multiplying them by N_i , as the package’s local-fraud summary does, overstates the number of fraudulent votes that the model implies.

Fraud is thus identified purely from the distribution of turnout/vote-share pairs, with no information other than the vote returns and the province of each unit. Estimation is by MCMC via JAGS [Plummer, 2003].

Every eforensics estimation in this paper calls the official `eforensics` R package [Ferrari et al., 2019] directly, installed from Mebane’s repository (`UMeforensics/eforensics_public`, https://github.com/UMeforensics/eforensics_public). No part of the fraud model, its priors or its sampler was re-implemented. Our code contribution is limited to data preparation, the three null-data (synthetic data) generators (described in the main text’s section on fraud-free controls), post-processing of posterior output, and the separate implementation of the Klimek et al. [2012] model (described in the main text’s section on other election forensics methods).

A.1 Construction of the fraud-free null datasets

This subsection gives the exact construction of the three fraud-free datasets described in the main text’s section on fraud-free controls.

For a *real-data* cell, unit i has eligible voters N_i , observed turnout $\tau_i = V_i/N_i$, and observed leader share $\nu_i = W_i/V_i$ with associated province $p(i)$. Every null dataset keeps N_i and $p(i)$ at their real values and replaces only the vote counts (V_i, W_i) by synthetic counts (V_i^*, W_i^*) .

Marginal null. For each unit i , independently draw

$$\tau_i^* \sim \text{Beta}(\max(\kappa\tau_i, \tfrac{1}{2}), \max(\kappa(1 - \tau_i), \tfrac{1}{2})), \quad \nu_i^* \sim \text{Beta}(\max(\kappa\nu_i, \tfrac{1}{2}), \max(\kappa(1 - \nu_i), \tfrac{1}{2})), \tag{A.12}$$

with concentration $\kappa = 200$ (distinct from the fraud threshold $k = 0.7$ of equation (A.9)), so that $E[\tau_i^*] \approx \tau_i$, $E[\nu_i^*] \approx \nu_i$, and $\text{Var}(\tau_i^*) \approx \tau_i(1 - \tau_i)/(\kappa + 1)$ (the floor of $\frac{1}{2}$ on the shape parameters only prevents

degenerate draws for units extremely close to the boundary). The draws are clipped to $[0.001, 0.999]$ and the counts are reconstructed as

$$V_i^* = \max(1, \text{round}(N_i \tau_i^*)), \quad W_i^* = \text{round}(V_i^* \nu_i^*). \quad (\text{A.13})$$

The point of this dataset is to see what happens if we take each unit’s real turnout and leader vote share as the latent true unit-level population parameters and inject Beta noise independently into τ_i^* and ν_i^* . This mainly checks whether the model (or its estimation) has difficulty modeling the empirical differences between units within a province, and is thereby biased toward reporting fraud. Admittedly, this dataset has the weakest claim to being fraud-free, but this turns out to be irrelevant given the results from the other synthetic datasets, which have much stronger fraud-free justifications.

Joint null. Let $y_i = (\text{logit } \tau_i, \text{logit } \nu_i)^\top$ and let $\bar{y}_p$ be the mean of y_i over the units of province p . Compute the residuals $r_i = y_i - \bar{y}_{p(i)}$ and their pooled 2×2 sample covariance $\Sigma = \text{Cov}(r) + 10^{-6} I_2$ across all units in the cell, and simulate

$$y_i^* = \bar{y}_{p(i)} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N_2(0, \Sigma), \quad \tau_i^* = \text{logit}^{-1}(y_{i1}^*), \quad \nu_i^* = \text{logit}^{-1}(y_{i2}^*), \quad (\text{A.14})$$

followed by the same clipping and count reconstruction as in equation (A.13).

Unlike the marginal null, this synthetic dataset discards all unit-specific information, retaining only the province means and the pooled covariance Σ . The joint-null validation therefore asks whether the model (or its estimation) has difficulty modeling data generated by province fixed effects and the empirical covariance Σ alone. If it does, the implication is that the model carries an implicit electoral theory about which covariance structures are consistent with a fraud-free election, and that the empirically observed Σ is not among them.

Model-consistent self-null. For a given cell (and leader coding), we first fit the qbl eforensics model to the real data and obtain the posterior medians of the estimated model’s class-1 (“no fraud”) hyperparameters in equation (A.2): the province coefficient vectors $\hat{\beta}_\tau, \hat{\beta}_\nu$, the random-intercept means $\hat{\alpha}_\tau, \hat{\alpha}_\nu$, and the random-intercept variances $\hat{\sigma}_\tau^2, \hat{\sigma}_\nu^2$. We then simulate a synthetic dataset from the model’s class-1 (fraud-free) data-generating equations with every unit forced into class 1 (population mixture weights fixed at $\pi = (1, 0, 0)$ by construction):

$$\begin{aligned} \theta_i &\sim N(\hat{\alpha}_\tau, \hat{\sigma}_\tau^2), \quad \mu_{\tau i} = \text{logit}^{-1}(x_i^\top \hat{\beta}_\tau + \theta_i), \quad A_i^* \sim \text{Binomial}(N_i, 1 - \mu_{\tau i}), \\ \eta_i &\sim N(\hat{\alpha}_\nu, \hat{\sigma}_\nu^2), \quad \mu_{\nu i} = \text{logit}^{-1}(x_i^\top \hat{\beta}_\nu + \eta_i), \quad W_i^* \sim \text{Binomial}(N_i, \mu_{\nu i}(1 - A_i^*/N_i)), \end{aligned} \quad (\text{A.15})$$

with $V_i^* = N_i - A_i^*$. The resulting synthetic data are then fit again through the same eforensics estimation process. Because this dataset uses the *estimated* model’s fraud-free-class process, an estimated heavy fraud probability for this dataset is a very strong indicator of estimation problems in the eforensics model, since the refit should ideally recover $\pi_1 \approx 1$.

B Additional results

This section collects the tables summarized in the main text: the vote channels (terminology section), the largest false positives on the fraud-free nulls (“False-positive fraud on no-fraud null data”), the largest self-null results (“Confirmation with the model’s own no-fraud process”), the elections held before early voting (“The pre-early-voting natural experiment”), the 2012 presidential election under both leader codings (“Bidirectional leader-coded confirmation”), and the latent-turnout comparison (“Robustness to the latent-turnout form”).

Table B.1: Vote channels. “Constituency-only” ballots are those the National Election Commission reports only per constituency: out-of-precinct early ballots, home- or ship-bound ballots, overseas ballots and misfiled ballots (early-voting era), or absentee ballots and, in 2012, overseas ballots (pre-early-voting elections), plus a negligible number of misfiled ballots in both eras. Shares are the range across the elections of each era, as a percentage of all valid votes in the election.

Channel	Level	Ballots included	Share of votes
<i>Early-voting era (2016–2025)</i>			
early	dong, constituency	in-precinct early	14–37%
sameday	dong, constituency	same-day	51–78%
pooled	dong	in-precinct early + same-day	87–92%
early-out	constituency	constituency-only (92–95% out-of-precinct early)	8–13%
early-total	constituency	early + early-out	22–49%
total	constituency	early-total + same-day (all valid votes)	100%
<i>Pre-early-voting elections (2002–2012)</i>			
total-no-early	dong, constituency	election-day polling-station ballots	96–97%

Table B.2: Selected fraud-free null runs with large false-positive fraud shares (short-chain configuration, Democratic-coded).

Cell (fraud-free null)	Fraud share	Fraudulent votes (F_w)	Units flagged
2025 pres., constituency, total, joint	31.1%	3,725,654	37/242
2025 pres., constituency, total, marginal	29.9%	3,456,141	31/242
2022 pres., constituency, total, marginal	29.5%	3,220,048	33/248
2022 pres., constituency, total, joint	27.0%	2,991,506	28/248
2017 pres., constituency, total, marginal	23.0%	2,323,658	15/250
2025 pres., dong, pooled, joint	22.1%	1,417,322	445/3553

Table B.3: The three largest model-consistent self-null results under each leader coding, with π forced to $(1, 0, 0)$ (short-chain configuration).

Leader	Cell (self-null, π forced to $(1, 0, 0)$)	Fraud share	Units flagged
Dem	2025 pres., constituency, total	24.4%	19/242
Dem	2022 pres., constituency, total	21.9%	19/248
Dem	2017 pres., constituency, total	18.4%	9/250
Con	2025 pres., constituency, total	24.8%	19/242
Con	2022 pres., constituency, total	22.8%	22/248
Con	2012 pres., dong, total-no-early	17.3%	271/3479

Table B.4: Real-data results for elections held before early voting existed (short-chain configuration, Democratic-coded).

Election	Level	Fraud share	Fraudulent votes (F_w)	Units flagged
Pres. 2002	Constituency	14.3%	810,037	17/244
Pres. 2002	Dong	13.9%	595,592	304/3515
Pres. 2007	Constituency	6.7%	266,127	8/248
Pres. 2007	Dong	7.1%	269,010	154/3576
Pres. 2012	Constituency	16.8%	1,602,947	17/251
Pres. 2012	Dong	17.3%	1,158,472	275/3479
Gen. 2008	Constituency	0.7%	0	0/198
Gen. 2008	Dong	0.6%	10,602	4/2733
Gen. 2012	Constituency	0.7%	0	0/209
Gen. 2012	Dong	0.5%	11,457	4/2923

Table B.5: 2012 presidential election, Democratic- vs. conservative-coded leader (short-chain configuration).

Level	Leader coding	Source	Fraud share	$\hat{R}$
Dong	Dem	Real	17.3%	9.90
Dong	Dem	Marginal null	17.6%	8.33
Dong	Dem	Joint null	17.6%	3.47
Dong	Con	Real	26.1%	15.44
Dong	Con	Marginal null	15.5%	10.82
Dong	Con	Joint null	25.6%	12.64
Constituency	Dem	Real	16.8%	2.43
Constituency	Dem	Marginal null	18.2%	1.22
Constituency	Dem	Joint null	19.1%	1.66
Constituency	Con	Real	22.2%	3.40
Constituency	Con	Marginal null	20.7%	2.46
Constituency	Con	Joint null	20.7%	3.67

Table B.6: Latent-turnout versus observed-turnout form: the same cells under both leader codings, short-chain configuration. “Not converged” is the share of runs with $\max \hat{R} > 1.05$. r is the correlation between the two forms’ fraud shares across the runs in each row.

Data	Runs	Runs with fraud share > 5%		Median fraud share		Not converged		r
		Observed	Latent	Observed	Latent	Observed	Latent	
Real data	116	36 (31%)	23 (20%)	1.4%	0.8%	87%	75%	0.96
Marginal null	116	38 (33%)	24 (21%)	1.6%	0.9%	89%	82%	0.96
Joint null	116	39 (34%)	22 (19%)	1.2%	0.8%	91%	77%	0.92
Both nulls	232	77 (33%)	46 (20%)	1.4%	0.8%	90%	79%	0.94

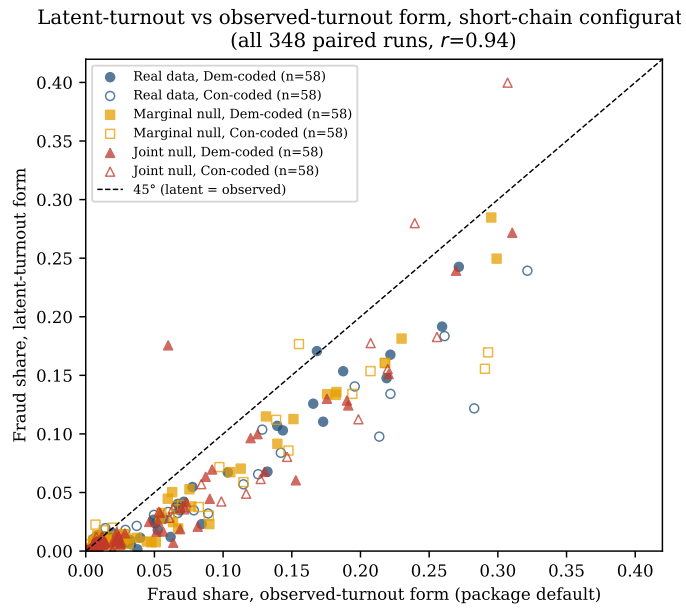


Figure B.1: Fraud share under the latent-turnout form against the observed-turnout (package-default) form, for every paired run: real data and both nulls, both leader codings, short-chain configuration.

C Complete results listing

Tables C.1–C.2 list every completed model run referenced in the main text: the short-chain configuration under the Democratic- and conservative-coded leaders, 348 rows in total. Each row also reports the number and percentage of units flagged, that is, units whose maximum a posteriori class is fraudulent. The 95% interval reported alongside each fraud-share estimate is the exact joint HPD interval of the fraud share $\pi_2 + \pi_3$, computed from the per-draw sum of the monitored π draws. F_w is the posterior mean of the estimated number of fraudulent (manufactured plus stolen) votes, computed from the posterior draws as described in Appendix A (equation (A.11)). Rows with $\hat{R}$ marked * fail the $\hat{R} \leq 1.05$ convergence criterion. The marker is applied to the unrounded value, so a row printed as exactly 1.05 may still carry it. Election codes follow National Election Commission numbering: Pres02/07/12/17/22/25 are the 16th/17th/18th/19th/20th/21st presidential elections, and Gen08/12/16/20/24 are the 18th/19th/20th/21st/22nd general elections.

Table C.1: Short-chain configuration, Democratic-coded leader (n.iter=1000, burn.in=300, n.adapt=1000). Full coverage, all 11 elections. N is the unit count and “Flag.” the number (and percentage) of units receiving a non-clean MAP classification. The 95% HPD interval is the exact joint interval for the fraud share $\pi_2 + \pi_3$, computed from the per-draw sum. F_w is the posterior mean of the estimated number of fraudulent (manufactured plus stolen) votes implied by the model.

Elec.	Level	Channel	Source	N	Flag. (%)	Fraud %	95% HPD	F_w	$\hat{R}$
Gen08	const.	total-no-early	real	198	0 (0.0)	0.71	[0.0,2.4]	0	1.00
Gen08	const.	total-no-early	marg.null	198	0 (0.0)	0.67	[0.0,2.2]	0	1.01
Gen08	const.	total-no-early	joint.null	198	0 (0.0)	0.69	[0.0,2.4]	0	1.02
Gen08	dong	total-no-early	real	2733	4 (0.1)	0.57	[0.3,0.9]	10,602	1.32*
Gen08	dong	total-no-early	marg.null	2733	5 (0.2)	0.75	[0.4,1.3]	13,975	1.68*
Gen08	dong	total-no-early	joint.null	2733	0 (0.0)	0.26	[0.1,0.5]	11,832	1.50*
Gen12	const.	total-no-early	real	209	0 (0.0)	0.65	[0.0,2.3]	0	1.02
Gen12	const.	total-no-early	marg.null	209	0 (0.0)	0.67	[0.0,2.3]	0	1.00
Gen12	const.	total-no-early	joint.null	209	0 (0.0)	0.66	[0.0,2.4]	0	1.06*
Gen12	dong	total-no-early	real	2923	4 (0.1)	0.55	[0.2,1.0]	11,457	1.84*
Gen12	dong	total-no-early	marg.null	2923	4 (0.1)	0.58	[0.3,0.9]	15,718	1.12*
Gen12	dong	total-no-early	joint.null	2923	0 (0.0)	0.34	[0.1,0.6]	12,389	1.41*
Pres02	const.	total-no-early	real	244	17 (7.0)	14.35	[8.8,21.0]	810,037	1.89*
Pres02	const.	total-no-early	marg.null	244	20 (8.2)	13.12	[8.6,18.6]	722,392	1.46*
Pres02	const.	total-no-early	joint.null	244	16 (6.6)	12.00	[7.5,17.6]	558,194	1.60*
Pres02	dong	total-no-early	real	3515	304 (8.6)	13.94	[11.3,15.7]	595,592	2.95*
Pres02	dong	total-no-early	marg.null	3515	316 (9.0)	15.12	[10.9,16.7]	697,266	4.38*
Pres02	dong	total-no-early	joint.null	3515	218 (6.2)	12.51	[9.3,15.9]	627,247	6.27*
Pres07	const.	total-no-early	real	248	8 (3.2)	6.70	[3.1,11.1]	266,127	1.61*
Pres07	const.	total-no-early	marg.null	248	8 (3.2)	6.29	[3.0,9.9]	274,189	1.30*
Pres07	const.	total-no-early	joint.null	248	8 (3.2)	9.03	[5.4,14.7]	402,848	1.69*
Pres07	dong	total-no-early	real	3576	154 (4.3)	7.14	[5.9,8.3]	269,010	1.97*
Pres07	dong	total-no-early	marg.null	3576	114 (3.2)	5.97	[4.2,8.1]	245,375	4.78*
Pres07	dong	total-no-early	joint.null	3576	136 (3.8)	7.30	[6.3,8.4]	299,489	1.52*
Pres12	const.	total-no-early	real	251	17 (6.8)	16.83	[10.2,24.6]	1,602,947	2.43*
Pres12	const.	total-no-early	marg.null	251	18 (7.2)	18.20	[13.4,24.1]	1,675,172	1.22*
Pres12	const.	total-no-early	joint.null	251	14 (5.6)	19.11	[13.0,25.6]	1,502,692	1.66*
Pres12	dong	total-no-early	real	3479	275 (7.9)	17.28	[8.4,19.4]	1,158,472	9.90*
Pres12	dong	total-no-early	marg.null	3479	264 (7.6)	17.57	[9.7,19.5]	1,186,553	8.33*
Pres12	dong	total-no-early	joint.null	3479	325 (9.3)	17.56	[14.2,19.3]	1,062,781	3.47*
Gen16	const.	early	real	234	0 (0.0)	1.93	[0.1,5.3]	20,217	2.32*
Gen16	const.	early	marg.null	234	0 (0.0)	1.55	[0.3,3.5]	17,213	1.04
Gen16	const.	early	joint.null	234	0 (0.0)	1.01	[0.0,3.0]	6,734	1.29*
Gen16	const.	early-out	real	234	0 (0.0)	0.59	[0.0,2.0]	0	1.01

Table C.1 – continued

Elec.	Level	Channel	Source	N	Flag. (%)	Fraud %	95% HPD	F_w	$\hat{R}$
Gen16	const.	early-out	marg.null	234	0 (0.0)	0.77	[0.0,2.9]	4,023	1.85*
Gen16	const.	early-out	joint.null	234	0 (0.0)	0.59	[0.0,1.9]	0	1.02
Gen16	const.	early-total	real	234	0 (0.0)	2.80	[0.1,5.4]	35,501	2.04*
Gen16	const.	early-total	marg.null	234	0 (0.0)	3.72	[1.5,6.8]	56,130	1.21*
Gen16	const.	early-total	joint.null	234	0 (0.0)	2.85	[1.0,5.7]	38,858	1.26*
Gen16	const.	sameday	real	234	0 (0.0)	0.62	[0.0,2.1]	0	1.01
Gen16	const.	sameday	marg.null	234	0 (0.0)	0.60	[0.0,2.2]	0	1.01
Gen16	const.	sameday	joint.null	234	0 (0.0)	0.59	[0.0,2.0]	0	1.02
Gen16	const.	total	real	234	0 (0.0)	1.02	[0.0,2.9]	56,676	1.54*
Gen16	const.	total	marg.null	234	1 (0.4)	2.08	[0.5,4.7]	153,186	1.26*
Gen16	const.	total	joint.null	234	1 (0.4)	1.43	[0.1,4.0]	78,475	1.82*
Gen16	dong	early	real	3135	0 (0.0)	0.49	[0.1,0.9]	7,282	3.13*
Gen16	dong	early	marg.null	3135	0 (0.0)	0.94	[0.2,1.4]	14,170	3.35*
Gen16	dong	early	joint.null	3135	0 (0.0)	0.23	[0.0,1.8]	10,076	7.68*
Gen16	dong	pooled	real	3135	2 (0.1)	0.32	[0.1,0.6]	12,961	1.34*
Gen16	dong	pooled	marg.null	3135	2 (0.1)	0.64	[0.3,1.0]	43,165	1.64*
Gen16	dong	pooled	joint.null	3135	3 (0.1)	0.30	[0.1,0.5]	15,685	1.26*
Gen16	dong	sameday	real	3135	0 (0.0)	0.05	[0.0,0.2]	68	1.12*
Gen16	dong	sameday	marg.null	3135	0 (0.0)	0.04	[0.0,0.1]	0	1.01
Gen16	dong	sameday	joint.null	3135	0 (0.0)	0.04	[0.0,0.1]	0	1.01
Pres17	const.	early	real	250	0 (0.0)	3.05	[1.0,6.9]	111,908	2.03*
Pres17	const.	early	marg.null	250	0 (0.0)	2.76	[0.7,5.5]	75,342	1.47*
Pres17	const.	early	joint.null	250	0 (0.0)	2.54	[0.9,5.0]	40,605	1.24*
Pres17	const.	early-out	real	250	0 (0.0)	0.90	[0.0,6.5]	15,260	3.92*
Pres17	const.	early-out	marg.null	250	0 (0.0)	0.85	[0.0,7.5]	26,352	4.30*
Pres17	const.	early-out	joint.null	250	0 (0.0)	2.52	[0.1,5.6]	26,295	2.43*
Pres17	const.	early-total	real	250	0 (0.0)	0.73	[0.0,6.0]	21,048	3.98*
Pres17	const.	early-total	marg.null	250	0 (0.0)	0.84	[0.0,2.7]	9,875	1.38*
Pres17	const.	early-total	joint.null	250	0 (0.0)	0.75	[0.0,4.1]	11,312	2.82*
Pres17	const.	sameday	real	250	0 (0.0)	0.56	[0.0,1.8]	0	1.01
Pres17	const.	sameday	marg.null	250	0 (0.0)	0.54	[0.0,1.8]	0	1.01
Pres17	const.	sameday	joint.null	250	0 (0.0)	0.56	[0.0,1.9]	0	1.00
Pres17	const.	total	real	250	5 (2.0)	22.18	[15.1,29.4]	2,380,072	2.03*
Pres17	const.	total	marg.null	250	15 (6.0)	22.99	[16.9,29.3]	2,323,658	1.36*
Pres17	const.	total	joint.null	250	2 (0.8)	6.00	[1.6,27.0]	924,350	7.52*
Pres17	dong	early	real	3491	0 (0.0)	1.33	[0.5,3.0]	47,017	5.42*
Pres17	dong	early	marg.null	3491	0 (0.0)	0.81	[0.4,3.0]	37,082	6.00*
Pres17	dong	early	joint.null	3491	0 (0.0)	0.80	[0.3,3.0]	36,000	6.97*
Pres17	dong	pooled	real	3491	56 (1.6)	10.34	[5.2,12.0]	769,178	7.09*
Pres17	dong	pooled	marg.null	3491	67 (1.9)	10.53	[6.3,12.0]	818,482	5.69*
Pres17	dong	pooled	joint.null	3491	53 (1.5)	8.73	[5.8,12.0]	573,592	7.17*
Pres17	dong	sameday	real	3491	0 (0.0)	0.07	[0.0,0.2]	7,905	1.27*
Pres17	dong	sameday	marg.null	3491	0 (0.0)	0.12	[0.0,0.3]	16,718	1.54*
Pres17	dong	sameday	joint.null	3491	0 (0.0)	0.07	[0.0,0.2]	2,666	1.25*
Gen20	const.	early	real	253	0 (0.0)	6.18	[3.3,9.8]	207,346	1.25*
Gen20	const.	early	marg.null	253	0 (0.0)	5.10	[2.5,8.8]	175,069	1.49*
Gen20	const.	early	joint.null	253	0 (0.0)	5.64	[2.5,11.4]	205,836	2.23*
Gen20	const.	early-out	real	253	0 (0.0)	1.93	[0.3,5.0]	21,249	2.14*
Gen20	const.	early-out	marg.null	253	0 (0.0)	2.10	[0.1,5.0]	28,211	2.22*
Gen20	const.	early-out	joint.null	253	0 (0.0)	2.38	[0.1,5.9]	28,261	2.46*
Gen20	const.	early-total	real	253	0 (0.0)	2.87	[0.7,7.4]	132,378	2.33*
Gen20	const.	early-total	marg.null	253	0 (0.0)	3.12	[1.2,5.7]	133,218	1.12*
Gen20	const.	early-total	joint.null	253	0 (0.0)	2.26	[0.6,4.9]	94,003	1.40*
Gen20	const.	sameday	real	253	0 (0.0)	0.52	[0.0,1.8]	0	1.01
Gen20	const.	sameday	marg.null	253	0 (0.0)	0.68	[0.0,2.3]	9,401	1.50*
Gen20	const.	sameday	joint.null	253	0 (0.0)	0.63	[0.0,2.1]	5,711	1.15*

Table C.1 – continued

Elec.	Level	Channel	Source	N	Flag. (%)	Fraud %	95% HPD	F_w	$\hat{R}$
Gen20	const.	total	real	253	7 (2.8)	13.23	[8.9,18.2]	1,309,189	1.26*
Gen20	const.	total	marg.null	253	4 (1.6)	11.28	[3.6,15.8]	1,122,282	2.53*
Gen20	const.	total	joint.null	253	9 (3.6)	13.02	[8.5,18.7]	1,249,384	1.49*
Gen20	dong	early	real	3480	8 (0.2)	2.78	[2.0,3.6]	128,778	1.83*
Gen20	dong	early	marg.null	3480	3 (0.1)	2.56	[2.0,3.2]	120,591	1.18*
Gen20	dong	early	joint.null	3480	2 (0.1)	1.97	[1.5,2.5]	85,859	1.29*
Gen20	dong	pooled	real	3480	40 (1.1)	4.93	[4.1,5.8]	366,990	1.45*
Gen20	dong	pooled	marg.null	3480	53 (1.5)	5.40	[4.6,6.3]	452,370	1.21*
Gen20	dong	pooled	joint.null	3480	47 (1.4)	5.26	[4.4,6.2]	368,088	1.34*
Gen20	dong	sameday	real	3480	0 (0.0)	0.12	[0.0,0.3]	4,759	1.92*
Gen20	dong	sameday	marg.null	3480	0 (0.0)	0.20	[0.1,0.4]	8,143	1.62*
Gen20	dong	sameday	joint.null	3480	1 (0.0)	0.15	[0.0,0.3]	7,174	1.56*
Pres22	const.	early	real	250	1 (0.4)	1.65	[0.5,3.7]	44,546	1.05
Pres22	const.	early	marg.null	250	0 (0.0)	1.86	[0.3,4.1]	63,402	1.47*
Pres22	const.	early	joint.null	250	0 (0.0)	2.24	[0.5,4.9]	133,325	1.49*
Pres22	const.	early-out	real	250	1 (0.4)	3.97	[1.0,8.0]	67,344	2.11*
Pres22	const.	early-out	marg.null	250	0 (0.0)	4.45	[0.3,7.6]	76,885	2.47*
Pres22	const.	early-out	joint.null	250	1 (0.4)	5.15	[0.2,10.2]	89,934	3.36*
Pres22	const.	early-total	real	250	9 (3.6)	5.71	[3.1,9.1]	171,214	1.11*
Pres22	const.	early-total	marg.null	250	7 (2.8)	5.50	[1.9,9.8]	158,258	1.96*
Pres22	const.	early-total	joint.null	250	4 (1.6)	5.42	[2.7,8.8]	205,014	1.22*
Pres22	const.	sameday	real	250	0 (0.0)	2.09	[0.6,4.6]	24,765	1.33*
Pres22	const.	sameday	marg.null	250	0 (0.0)	1.62	[0.1,3.9]	28,305	1.80*
Pres22	const.	sameday	joint.null	250	0 (0.0)	1.03	[0.0,3.4]	27,607	1.80*
Pres22	const.	total	real	248	23 (9.3)	25.94	[13.5,37.7]	2,651,938	4.30*
Pres22	const.	total	marg.null	248	33 (13.3)	29.52	[23.9,36.5]	3,220,048	1.25*
Pres22	const.	total	joint.null	248	28 (11.3)	26.96	[15.5,33.6]	2,991,506	2.71*
Pres22	dong	early	real	3509	63 (1.8)	3.23	[2.4,13.2]	219,086	17.34*
Pres22	dong	early	marg.null	3509	39 (1.1)	2.96	[2.2,9.4]	172,144	12.10*
Pres22	dong	early	joint.null	3509	16 (0.5)	2.31	[1.8,2.9]	108,233	1.29*
Pres22	dong	pooled	real	3509	414 (11.8)	18.74	[14.1,21.2]	1,188,954	11.93*
Pres22	dong	pooled	marg.null	3509	329 (9.4)	18.25	[9.4,20.3]	1,091,262	9.31*
Pres22	dong	pooled	joint.null	3509	405 (11.5)	19.01	[17.7,20.6]	1,156,066	2.24*
Pres22	dong	sameday	real	3509	0 (0.0)	0.61	[0.3,1.0]	20,389	1.73*
Pres22	dong	sameday	marg.null	3509	0 (0.0)	0.69	[0.3,1.0]	19,338	1.65*
Pres22	dong	sameday	joint.null	3509	0 (0.0)	0.63	[0.2,1.0]	24,696	1.90*
Gen24	const.	early	real	246	0 (0.0)	3.30	[0.6,8.4]	174,660	2.59*
Gen24	const.	early	marg.null	246	1 (0.4)	4.78	[1.9,8.3]	198,807	1.56*
Gen24	const.	early	joint.null	246	0 (0.0)	6.37	[0.9,13.9]	316,187	4.38*
Gen24	const.	early-out	real	246	0 (0.0)	1.40	[0.1,3.9]	14,657	1.90*
Gen24	const.	early-out	marg.null	246	0 (0.0)	2.74	[0.1,7.0]	40,878	2.83*
Gen24	const.	early-out	joint.null	246	0 (0.0)	2.55	[0.1,5.2]	31,672	2.12*
Gen24	const.	early-total	real	246	0 (0.0)	0.83	[0.0,2.6]	23,942	1.40*
Gen24	const.	early-total	marg.null	246	0 (0.0)	1.06	[0.1,2.7]	30,869	1.10*
Gen24	const.	early-total	joint.null	246	0 (0.0)	1.18	[0.1,3.5]	60,085	1.72*
Gen24	const.	sameday	real	246	0 (0.0)	1.17	[0.2,3.0]	28,514	1.10*
Gen24	const.	sameday	marg.null	246	0 (0.0)	1.28	[0.1,3.3]	34,718	1.42*
Gen24	const.	sameday	joint.null	246	0 (0.0)	1.25	[0.2,3.1]	34,888	1.27*
Gen24	const.	total	real	246	10 (4.1)	16.56	[11.8,21.8]	1,514,162	1.13*
Gen24	const.	total	marg.null	246	9 (3.7)	13.95	[6.2,20.6]	1,252,077	2.72*
Gen24	const.	total	joint.null	246	8 (3.3)	15.29	[9.4,23.8]	1,505,591	2.37*
Gen24	dong	early	real	2739	5 (0.2)	3.76	[1.7,6.6]	151,697	7.32*
Gen24	dong	early	marg.null	2739	3 (0.1)	2.45	[1.6,3.5]	96,703	2.30*
Gen24	dong	early	joint.null	2739	0 (0.0)	1.92	[1.2,5.1]	119,320	6.55*
Gen24	dong	pooled	real	2739	60 (2.2)	6.85	[5.2,8.3]	449,682	2.38*
Gen24	dong	pooled	marg.null	2739	78 (2.8)	7.69	[6.6,8.9]	505,896	1.24*

Table C.1 – continued

Elec.	Level	Channel	Source	N	Flag. (%)	Fraud %	95% HPD	F_w	$\hat{R}$
Gen24	dong	pooled	joint.null	2739	61 (2.2)	6.97	[5.9,8.1]	459,324	1.26*
Gen24	dong	sameday	real	2739	0 (0.0)	0.58	[0.2,1.0]	15,513	1.75*
Gen24	dong	sameday	marg.null	2739	0 (0.0)	0.74	[0.4,1.1]	21,721	1.28*
Gen24	dong	sameday	joint.null	2739	0 (0.0)	0.68	[0.4,1.1]	24,264	1.19*
Pres25	const.	early	real	253	1 (0.4)	5.05	[2.2,9.4]	331,248	1.71*
Pres25	const.	early	marg.null	253	1 (0.4)	6.73	[3.0,10.9]	356,682	1.70*
Pres25	const.	early	joint.null	253	0 (0.0)	4.62	[1.5,11.7]	302,212	3.14*
Pres25	const.	early-out	real	253	0 (0.0)	5.24	[1.8,10.6]	120,028	2.53*
Pres25	const.	early-out	marg.null	253	0 (0.0)	6.35	[0.1,10.0]	127,584	3.35*
Pres25	const.	early-out	joint.null	253	0 (0.0)	8.15	[1.0,12.6]	160,532	3.07*
Pres25	const.	early-total	real	253	7 (2.8)	7.77	[3.5,12.1]	249,099	1.70*
Pres25	const.	early-total	marg.null	253	10 (4.0)	7.55	[4.5,11.5]	286,978	1.15*
Pres25	const.	early-total	joint.null	253	14 (5.5)	9.20	[3.3,13.8]	402,708	2.22*
Pres25	const.	sameday	real	253	0 (0.0)	1.42	[0.3,3.2]	19,865	1.04
Pres25	const.	sameday	marg.null	253	0 (0.0)	1.51	[0.2,3.9]	19,180	1.60*
Pres25	const.	sameday	joint.null	253	0 (0.0)	1.19	[0.1,3.4]	9,641	1.48*
Pres25	const.	total	real	242	29 (12.0)	27.16	[14.4,34.4]	3,014,376	2.79*
Pres25	const.	total	marg.null	242	31 (12.8)	29.91	[18.6,38.5]	3,456,141	2.73*
Pres25	const.	total	joint.null	242	37 (15.3)	31.05	[21.6,39.6]	3,725,654	2.26*
Pres25	dong	early	real	3553	179 (5.0)	8.44	[4.1,10.1]	278,696	5.98*
Pres25	dong	early	marg.null	3553	210 (5.9)	9.00	[5.2,15.6]	372,668	9.92*
Pres25	dong	early	joint.null	3553	145 (4.1)	6.87	[5.0,15.5]	427,885	11.87*
Pres25	dong	pooled	real	3553	463 (13.0)	21.90	[19.4,23.8]	1,512,267	2.21*
Pres25	dong	pooled	marg.null	3553	350 (9.9)	21.80	[7.9,23.7]	1,364,426	14.12*
Pres25	dong	pooled	joint.null	3553	445 (12.5)	22.05	[19.9,23.8]	1,417,322	1.93*
Pres25	dong	sameday	real	3553	0 (0.0)	0.59	[0.3,1.0]	15,479	1.73*
Pres25	dong	sameday	marg.null	3553	1 (0.0)	0.52	[0.3,0.8]	14,779	1.45*
Pres25	dong	sameday	joint.null	3553	1 (0.0)	0.69	[0.4,1.1]	26,375	1.56*

Table C.2: Short-chain configuration, conservative-coded leader. Full coverage, all 11 elections. N is the unit count and “Flag.” the number (and percentage) of units receiving a non-clean MAP classification. The 95% HPD interval is the exact joint interval for the fraud share $\pi_2 + \pi_3$, computed from the per-draw sum. F_w is the posterior mean of the estimated number of fraudulent (manufactured plus stolen) votes implied by the model.

Elec.	Level	Channel	Source	N	Flag. (%)	Fraud %	95% HPD	F_w	$\hat{R}$
Gen08	const.	total-no-early	real	198	0 (0.0)	0.71	[0.0,2.4]	0	1.01
Gen08	const.	total-no-early	marg.null	198	0 (0.0)	0.68	[0.0,2.2]	0	1.01
Gen08	const.	total-no-early	joint.null	198	0 (0.0)	0.73	[0.0,2.6]	0	1.03
Gen08	dong	total-no-early	real	2733	3 (0.1)	0.25	[0.1,0.5]	2,245	1.24*
Gen08	dong	total-no-early	marg.null	2733	1 (0.0)	0.34	[0.0,0.7]	6,738	2.04*
Gen08	dong	total-no-early	joint.null	2733	0 (0.0)	0.13	[0.0,0.4]	5,146	1.79*
Gen12	const.	total-no-early	real	194	0 (0.0)	0.71	[0.0,2.4]	0	1.02
Gen12	const.	total-no-early	marg.null	194	0 (0.0)	1.03	[0.1,3.2]	22,145	1.39*
Gen12	const.	total-no-early	joint.null	194	0 (0.0)	0.71	[0.0,2.4]	0	1.01
Gen12	dong	total-no-early	real	2656	18 (0.7)	1.77	[1.1,2.4]	37,646	1.64*
Gen12	dong	total-no-early	marg.null	2656	14 (0.5)	1.76	[1.0,2.9]	41,937	2.69*
Gen12	dong	total-no-early	joint.null	2656	10 (0.4)	1.30	[0.9,1.8]	59,816	1.06*
Pres02	const.	total-no-early	real	244	9 (3.7)	12.55	[7.7,17.5]	924,900	1.40*
Pres02	const.	total-no-early	marg.null	244	4 (1.6)	9.73	[4.8,15.9]	766,575	2.24*
Pres02	const.	total-no-early	joint.null	244	9 (3.7)	12.73	[8.4,17.7]	1,017,384	1.39*
Pres02	dong	total-no-early	real	3515	217 (6.2)	14.20	[13.0,15.3]	884,306	1.02
Pres02	dong	total-no-early	marg.null	3515	238 (6.8)	14.76	[13.4,16.2]	1,016,814	1.34*
Pres02	dong	total-no-early	joint.null	3515	222 (6.3)	14.66	[13.5,16.2]	998,896	1.42*
Pres07	const.	total-no-early	real	248	6 (2.4)	4.97	[2.0,8.5]	249,017	1.50*
Pres07	const.	total-no-early	marg.null	248	3 (1.2)	6.10	[3.3,9.5]	464,935	1.11*
Pres07	const.	total-no-early	joint.null	248	2 (0.8)	5.32	[1.7,10.0]	349,953	2.19*
Pres07	dong	total-no-early	real	3576	77 (2.2)	6.60	[5.6,7.6]	418,134	1.42*
Pres07	dong	total-no-early	marg.null	3576	85 (2.4)	7.01	[6.2,8.0]	474,774	1.13*
Pres07	dong	total-no-early	joint.null	3576	86 (2.4)	6.35	[5.5,7.3]	443,338	1.36*
Pres12	const.	total-no-early	real	251	19 (7.6)	22.18	[11.9,31.2]	1,892,942	3.40*
Pres12	const.	total-no-early	marg.null	251	19 (7.6)	20.73	[12.5,28.9]	1,918,551	2.46*
Pres12	const.	total-no-early	joint.null	251	22 (8.8)	20.74	[12.1,32.1]	1,800,095	3.67*
Pres12	dong	total-no-early	real	3479	350 (10.1)	26.12	[10.2,27.9]	1,591,918	15.44*
Pres12	dong	total-no-early	marg.null	3479	240 (6.9)	15.52	[13.2,26.2]	1,156,420	10.82*
Pres12	dong	total-no-early	joint.null	3479	325 (9.3)	25.57	[12.0,27.6]	1,788,721	12.64*
Gen16	const.	early	real	229	0 (0.0)	1.74	[0.2,4.4]	21,009	1.73*
Gen16	const.	early	marg.null	229	0 (0.0)	2.34	[0.2,5.3]	29,437	1.80*
Gen16	const.	early	joint.null	229	0 (0.0)	1.34	[0.2,3.4]	9,940	1.27*
Gen16	const.	early-out	real	229	0 (0.0)	0.62	[0.0,2.2]	0	1.01
Gen16	const.	early-out	marg.null	229	0 (0.0)	0.60	[0.0,2.0]	0	1.01
Gen16	const.	early-out	joint.null	229	0 (0.0)	0.60	[0.0,2.0]	0	1.00
Gen16	const.	early-total	real	229	0 (0.0)	2.89	[0.8,5.8]	41,119	1.35*
Gen16	const.	early-total	marg.null	229	0 (0.0)	2.81	[0.8,5.6]	44,176	1.34*
Gen16	const.	early-total	joint.null	229	0 (0.0)	2.19	[0.3,6.6]	34,851	2.53*
Gen16	const.	sameday	real	229	0 (0.0)	0.60	[0.0,2.1]	0	1.02
Gen16	const.	sameday	marg.null	229	0 (0.0)	0.60	[0.0,2.0]	0	1.02
Gen16	const.	sameday	joint.null	229	0 (0.0)	0.62	[0.0,2.1]	0	1.01
Gen16	const.	total	real	229	0 (0.0)	0.87	[0.0,2.7]	29,066	1.42*
Gen16	const.	total	marg.null	229	0 (0.0)	1.10	[0.1,4.0]	78,979	2.21*
Gen16	const.	total	joint.null	229	0 (0.0)	0.68	[0.0,2.4]	14,017	1.22*
Gen16	dong	early	real	3058	0 (0.0)	0.81	[0.1,2.0]	12,544	5.13*
Gen16	dong	early	marg.null	3058	0 (0.0)	0.25	[0.0,3.5]	12,461	10.44*
Gen16	dong	early	joint.null	3058	0 (0.0)	0.67	[0.2,1.2]	17,272	3.02*
Gen16	dong	pooled	real	3058	19 (0.6)	1.71	[1.2,2.3]	27,042	1.34*
Gen16	dong	pooled	marg.null	3058	13 (0.4)	1.82	[1.0,2.4]	39,228	2.25*

Table C.2 – continued

Elec.	Level	Channel	Source	N	Flag. (%)	Fraud %	95% HPD	F_w	$\hat{R}$
Gen16	dong	pooled	joint.null	3058	9 (0.3)	1.24	[0.8,1.8]	63,299	1.55*
Gen16	dong	sameday	real	3058	0 (0.0)	0.11	[0.0,0.3]	1,221	1.74*
Gen16	dong	sameday	marg.null	3058	1 (0.0)	0.16	[0.0,0.4]	1,916	1.38*
Gen16	dong	sameday	joint.null	3058	0 (0.0)	0.09	[0.0,0.3]	4,250	2.03*
Pres17	const.	early	real	250	0 (0.0)	1.68	[0.0,7.6]	28,255	3.75*
Pres17	const.	early	marg.null	250	0 (0.0)	1.76	[0.0,4.9]	17,810	2.54*
Pres17	const.	early	joint.null	250	0 (0.0)	1.17	[0.1,3.0]	25,265	1.11*
Pres17	const.	early-out	real	250	0 (0.0)	0.72	[0.0,2.3]	1,528	1.20*
Pres17	const.	early-out	marg.null	250	0 (0.0)	0.63	[0.0,2.1]	1,386	1.16*
Pres17	const.	early-out	joint.null	250	0 (0.0)	0.74	[0.0,2.3]	985	1.17*
Pres17	const.	early-total	real	250	0 (0.0)	0.79	[0.1,2.5]	5,295	1.44*
Pres17	const.	early-total	marg.null	250	0 (0.0)	0.56	[0.0,1.9]	0	1.01
Pres17	const.	early-total	joint.null	250	0 (0.0)	0.69	[0.0,2.8]	3,182	1.94*
Pres17	const.	sameday	real	250	0 (0.0)	0.67	[0.0,2.3]	10,028	1.44*
Pres17	const.	sameday	marg.null	250	0 (0.0)	1.00	[0.0,3.3]	33,465	1.88*
Pres17	const.	sameday	joint.null	250	0 (0.0)	0.53	[0.0,1.9]	0	1.00
Pres17	const.	total	real	250	2 (0.8)	12.85	[1.7,18.0]	758,495	4.03*
Pres17	const.	total	marg.null	250	3 (1.2)	13.86	[9.6,18.9]	1,020,581	1.16*
Pres17	const.	total	joint.null	250	2 (0.8)	8.41	[2.4,19.3]	582,590	4.55*
Pres17	dong	early	real	3491	1 (0.0)	0.34	[0.1,2.5]	12,964	9.87*
Pres17	dong	early	marg.null	3491	0 (0.0)	0.27	[0.0,1.1]	6,159	5.01*
Pres17	dong	early	joint.null	3491	0 (0.0)	0.77	[0.0,2.2]	26,011	7.30*
Pres17	dong	pooled	real	3491	132 (3.8)	8.91	[6.4,15.6]	590,076	10.21*
Pres17	dong	pooled	marg.null	3491	75 (2.1)	8.23	[6.8,11.7]	496,571	4.66*
Pres17	dong	pooled	joint.null	3491	51 (1.5)	7.21	[5.4,11.8]	626,806	7.10*
Pres17	dong	sameday	real	3491	0 (0.0)	0.11	[0.0,3.0]	52,554	12.71*
Pres17	dong	sameday	marg.null	3491	1 (0.0)	0.21	[0.1,0.4]	11,271	1.18*
Pres17	dong	sameday	joint.null	3491	0 (0.0)	0.09	[0.0,0.2]	7,294	1.46*
Gen20	const.	early	real	253	0 (0.0)	0.98	[0.1,2.9]	16,702	1.52*
Gen20	const.	early	marg.null	253	0 (0.0)	1.30	[0.1,3.5]	32,389	1.74*
Gen20	const.	early	joint.null	253	0 (0.0)	1.12	[0.1,2.9]	29,039	1.27*
Gen20	const.	early-out	real	253	0 (0.0)	0.89	[0.1,2.7]	4,043	1.51*
Gen20	const.	early-out	marg.null	253	0 (0.0)	0.67	[0.0,2.1]	2,660	1.12*
Gen20	const.	early-out	joint.null	253	0 (0.0)	0.66	[0.0,2.4]	2,826	1.57*
Gen20	const.	early-total	real	253	0 (0.0)	0.71	[0.0,3.1]	16,326	2.18*
Gen20	const.	early-total	marg.null	253	0 (0.0)	1.05	[0.1,3.8]	40,560	2.08*
Gen20	const.	early-total	joint.null	253	0 (0.0)	0.61	[0.0,2.0]	4,780	1.17*
Gen20	const.	sameday	real	253	0 (0.0)	0.58	[0.0,1.9]	0	1.01
Gen20	const.	sameday	marg.null	253	0 (0.0)	0.57	[0.0,2.1]	0	1.04
Gen20	const.	sameday	joint.null	253	0 (0.0)	0.53	[0.0,1.8]	0	1.01
Gen20	const.	total	real	253	2 (0.8)	7.87	[0.4,12.0]	691,155	3.49*
Gen20	const.	total	marg.null	253	3 (1.2)	8.96	[3.9,13.3]	911,978	1.85*
Gen20	const.	total	joint.null	253	8 (3.2)	9.87	[6.2,14.5]	976,766	1.38*
Gen20	dong	early	real	3480	1 (0.0)	0.57	[0.3,1.0]	21,789	2.06*
Gen20	dong	early	marg.null	3480	0 (0.0)	0.51	[0.2,1.5]	24,362	4.47*
Gen20	dong	early	joint.null	3480	0 (0.0)	0.89	[0.5,1.3]	51,057	1.37*
Gen20	dong	pooled	real	3480	71 (2.0)	5.99	[5.2,6.9]	357,963	1.12*
Gen20	dong	pooled	marg.null	3480	78 (2.2)	6.59	[5.8,7.5]	390,560	1.18*
Gen20	dong	pooled	joint.null	3480	92 (2.6)	7.16	[6.2,8.2]	606,694	1.35*
Gen20	dong	sameday	real	3480	0 (0.0)	0.12	[0.0,0.3]	4,028	1.69*
Gen20	dong	sameday	marg.null	3480	0 (0.0)	0.16	[0.1,0.3]	5,420	1.37*
Gen20	dong	sameday	joint.null	3480	0 (0.0)	0.24	[0.1,0.5]	21,181	1.53*
Pres22	const.	early	real	250	0 (0.0)	1.08	[0.1,2.8]	55,280	1.06*
Pres22	const.	early	marg.null	250	0 (0.0)	1.05	[0.2,2.8]	41,920	1.11*
Pres22	const.	early	joint.null	250	0 (0.0)	0.90	[0.1,2.6]	55,485	1.26*
Pres22	const.	early-out	real	250	0 (0.0)	3.71	[0.5,10.0]	68,262	3.04*

Table C.2 – continued

Elec.	Level	Channel	Source	N	Flag. (%)	Fraud %	95% HPD	F_w	$\hat{R}$
Pres22	const.	early-out	marg.null	250	0 (0.0)	1.96	[0.0,5.8]	27,469	2.81*
Pres22	const.	early-out	joint.null	250	0 (0.0)	2.38	[0.4,5.1]	33,816	1.78*
Pres22	const.	early-total	real	250	0 (0.0)	1.03	[0.1,2.8]	9,321	1.09*
Pres22	const.	early-total	marg.null	250	0 (0.0)	1.04	[0.1,2.7]	5,482	1.13*
Pres22	const.	early-total	joint.null	250	0 (0.0)	0.73	[0.0,2.5]	2,767	1.51*
Pres22	const.	sameday	real	250	0 (0.0)	0.65	[0.0,2.1]	3,140	1.17*
Pres22	const.	sameday	marg.null	250	0 (0.0)	0.84	[0.1,2.3]	12,197	1.08*
Pres22	const.	sameday	joint.null	250	0 (0.0)	0.66	[0.0,2.1]	2,917	1.14*
Pres22	const.	total	real	248	30 (12.1)	32.15	[25.9,38.7]	3,249,354	1.24*
Pres22	const.	total	marg.null	248	27 (10.9)	29.29	[12.0,38.4]	2,986,080	5.16*
Pres22	const.	total	joint.null	248	19 (7.7)	23.95	[14.7,34.9]	2,212,030	3.45*
Pres22	dong	early	real	3509	0 (0.0)	0.49	[0.2,0.8]	25,289	1.54*
Pres22	dong	early	marg.null	3509	5 (0.1)	0.71	[0.4,7.3]	83,936	17.23*
Pres22	dong	early	joint.null	3509	0 (0.0)	1.97	[0.4,6.0]	173,002	12.72*
Pres22	dong	pooled	real	3509	272 (7.8)	21.37	[12.3,23.3]	1,381,036	8.97*
Pres22	dong	pooled	marg.null	3509	302 (8.6)	21.77	[12.7,23.9]	1,416,720	9.10*
Pres22	dong	pooled	joint.null	3509	248 (7.1)	21.98	[8.6,24.4]	1,531,157	13.52*
Pres22	dong	sameday	real	3509	0 (0.0)	0.18	[0.1,0.4]	11,927	1.38*
Pres22	dong	sameday	marg.null	3509	0 (0.0)	0.26	[0.1,0.5]	14,084	1.24*
Pres22	dong	sameday	joint.null	3509	1 (0.0)	0.35	[0.1,0.6]	24,200	1.56*
Gen24	const.	early	real	246	0 (0.0)	0.89	[0.0,3.7]	36,725	2.27*
Gen24	const.	early	marg.null	246	0 (0.0)	1.71	[0.1,4.0]	47,608	1.61*
Gen24	const.	early	joint.null	246	0 (0.0)	1.43	[0.1,6.0]	68,137	2.96*
Gen24	const.	early-out	real	246	0 (0.0)	1.23	[0.1,3.5]	7,986	1.54*
Gen24	const.	early-out	marg.null	246	0 (0.0)	1.64	[0.2,4.2]	18,461	1.67*
Gen24	const.	early-out	joint.null	246	0 (0.0)	1.14	[0.0,3.3]	8,830	1.46*
Gen24	const.	early-total	real	246	0 (0.0)	0.80	[0.1,2.5]	13,653	1.13*
Gen24	const.	early-total	marg.null	246	0 (0.0)	0.63	[0.0,2.2]	6,728	1.31*
Gen24	const.	early-total	joint.null	246	0 (0.0)	0.65	[0.0,2.1]	6,531	1.14*
Gen24	const.	sameday	real	246	0 (0.0)	0.58	[0.0,1.9]	0	1.01
Gen24	const.	sameday	marg.null	246	0 (0.0)	0.73	[0.0,2.2]	13,004	1.20*
Gen24	const.	sameday	joint.null	246	0 (0.0)	0.65	[0.0,2.2]	5,123	1.14*
Gen24	const.	total	real	246	6 (2.4)	11.48	[7.3,17.5]	1,128,863	1.59*
Gen24	const.	total	marg.null	246	1 (0.4)	11.48	[6.6,17.2]	1,170,972	1.75*
Gen24	const.	total	joint.null	246	6 (2.4)	11.68	[7.8,16.2]	1,179,027	1.18*
Gen24	dong	early	real	2739	0 (0.0)	0.55	[0.2,6.5]	60,701	14.76*
Gen24	dong	early	marg.null	2739	1 (0.0)	0.66	[0.3,1.1]	28,798	1.43*
Gen24	dong	early	joint.null	2739	0 (0.0)	0.66	[0.4,1.0]	30,922	1.18*
Gen24	dong	pooled	real	2739	61 (2.2)	6.70	[5.4,8.1]	399,395	2.02*
Gen24	dong	pooled	marg.null	2739	48 (1.8)	6.63	[5.3,7.8]	476,771	1.88*
Gen24	dong	pooled	joint.null	2739	45 (1.6)	6.08	[5.0,7.3]	489,054	1.72*
Gen24	dong	sameday	real	2739	0 (0.0)	0.11	[0.0,0.3]	2,048	1.06*
Gen24	dong	sameday	marg.null	2739	0 (0.0)	0.23	[0.1,0.5]	7,558	1.47*
Gen24	dong	sameday	joint.null	2739	0 (0.0)	0.28	[0.1,1.7]	24,872	6.35*
Pres25	const.	early	real	253	0 (0.0)	1.43	[0.2,4.0]	27,552	1.71*
Pres25	const.	early	marg.null	253	0 (0.0)	0.54	[0.0,1.8]	0	1.03
Pres25	const.	early	joint.null	253	0 (0.0)	1.08	[0.1,3.2]	32,696	1.63*
Pres25	const.	early-out	real	253	0 (0.0)	1.53	[0.1,3.7]	10,406	1.72*
Pres25	const.	early-out	marg.null	253	0 (0.0)	1.45	[0.1,3.5]	17,022	1.53*
Pres25	const.	early-out	joint.null	253	0 (0.0)	1.18	[0.0,4.1]	15,274	2.24*
Pres25	const.	early-total	real	253	0 (0.0)	0.64	[0.0,2.4]	258	1.14*
Pres25	const.	early-total	marg.null	253	0 (0.0)	0.56	[0.0,1.8]	0	1.01
Pres25	const.	early-total	joint.null	253	0 (0.0)	0.63	[0.0,2.1]	2,069	1.16*
Pres25	const.	sameday	real	253	0 (0.0)	0.80	[0.0,2.5]	46,336	1.27*
Pres25	const.	sameday	marg.null	253	0 (0.0)	0.95	[0.1,2.7]	68,339	1.23*
Pres25	const.	sameday	joint.null	253	0 (0.0)	0.82	[0.1,2.4]	53,861	1.08*

Table C.2 – continued

Elec.	Level	Channel	Source	N	Flag. (%)	Fraud %	95% HPD	F_w	$\hat{R}$
Pres25	const.	total	real	242	19 (7.9)	28.28	[10.4,36.2]	2,385,712	4.42*
Pres25	const.	total	marg.null	242	25 (10.3)	29.06	[21.7,37.1]	2,823,151	2.00*
Pres25	const.	total	joint.null	242	25 (10.3)	30.71	[17.7,38.4]	2,824,050	3.02*
Pres25	dong	early	real	3553	0 (0.0)	0.22	[0.1,0.4]	8,547	1.46*
Pres25	dong	early	marg.null	3553	0 (0.0)	0.82	[0.1,2.5]	24,900	8.80*
Pres25	dong	early	joint.null	3553	0 (0.0)	0.21	[0.1,0.4]	8,547	1.43*
Pres25	dong	pooled	real	3553	266 (7.5)	19.58	[11.8,22.2]	1,231,744	8.29*
Pres25	dong	pooled	marg.null	3553	210 (5.9)	19.41	[9.2,21.1]	1,269,303	10.52*
Pres25	dong	pooled	joint.null	3553	264 (7.4)	19.85	[18.2,21.6]	1,596,062	1.57*
Pres25	dong	sameday	real	3553	2 (0.1)	0.84	[0.5,1.2]	101,179	1.17*
Pres25	dong	sameday	marg.null	3553	3 (0.1)	0.94	[0.6,1.3]	95,688	1.41*
Pres25	dong	sameday	joint.null	3553	8 (0.2)	0.86	[0.5,1.2]	81,400	1.47*

C.1 Mixture-component probabilities

Tables C.3–C.4 report the three underlying mixture-component probabilities, no-fraud (π_1), incremental fraud (π_2), and extreme fraud (π_3), with their individual 95% highest posterior density intervals, for the same 348 runs listed in Tables C.1–C.2. Because these are marginal posterior medians rather than a single posterior draw, the three need not sum exactly to 1. Across the 348 runs the sum ranges from 0.992 to 1.022, and deviates from 1 by more than 0.01 in only one row. The fraud share reported throughout the main text is $\pi_2 + \pi_3$. Its exact joint interval, given in Tables C.1–C.2, is narrower in every run than the interval obtained by adding the π_2 and π_3 interval bounds shown in Tables C.3–C.4 (the component tables), because adding the two component intervals ignores the posterior dependence between them.

Table C.3: Mixture-component probabilities, short-chain configuration, Democratic-coded leader. Full coverage, all 11 elections.

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen08	const.	total-no-early	real	99.15 [97.6,100.0]	0.34 [0.0,1.4]	0.37 [0.0,1.6]
Gen08	const.	total-no-early	marg.null	99.19 [97.8,100.0]	0.34 [0.0,1.3]	0.33 [0.0,1.4]
Gen08	const.	total-no-early	joint.null	99.16 [97.7,100.0]	0.34 [0.0,1.5]	0.35 [0.0,1.6]
Gen08	dong	total-no-early	real	99.42 [99.1,99.7]	0.55 [0.3,0.9]	0.03 [0.0,0.1]
Gen08	dong	total-no-early	marg.null	99.24 [98.7,99.6]	0.73 [0.4,1.2]	0.03 [0.0,0.1]
Gen08	dong	total-no-early	joint.null	99.73 [99.5,99.9]	0.23 [0.0,0.5]	0.02 [0.0,0.1]
Gen12	const.	total-no-early	real	99.20 [97.7,100.0]	0.32 [0.0,1.5]	0.33 [0.0,1.5]
Gen12	const.	total-no-early	marg.null	99.19 [97.7,100.0]	0.34 [0.0,1.4]	0.33 [0.0,1.4]
Gen12	const.	total-no-early	joint.null	99.19 [97.6,100.0]	0.32 [0.0,1.4]	0.33 [0.0,1.7]
Gen12	dong	total-no-early	real	99.43 [99.0,99.8]	0.53 [0.2,1.0]	0.03 [0.0,0.1]
Gen12	dong	total-no-early	marg.null	99.41 [99.1,99.7]	0.56 [0.3,0.9]	0.03 [0.0,0.1]
Gen12	dong	total-no-early	joint.null	99.65 [99.4,99.9]	0.31 [0.1,0.6]	0.02 [0.0,0.1]

Table C.3 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres02	const.	total-no-early	real	85.53 [79.0,91.2]	14.05 [8.6,20.6]	0.30 [0.0,1.2]
Pres02	const.	total-no-early	marg.null	86.78 [81.4,91.5]	12.67 [7.9,17.9]	0.46 [0.0,1.7]
Pres02	const.	total-no-early	joint.null	87.79 [82.4,92.5]	11.61 [6.6,16.9]	0.39 [0.0,1.8]
Pres02	dong	total-no-early	real	85.98 [84.3,88.7]	13.82 [11.3,15.6]	0.13 [0.0,0.3]
Pres02	dong	total-no-early	marg.null	84.85 [83.4,89.1]	15.00 [10.8,16.5]	0.12 [0.0,0.2]
Pres02	dong	total-no-early	joint.null	87.46 [84.1,90.7]	12.46 [9.3,15.8]	0.05 [0.0,0.2]
Pres07	const.	total-no-early	real	93.18 [88.9,96.9]	6.44 [2.9,10.8]	0.26 [0.0,1.1]
Pres07	const.	total-no-early	marg.null	93.59 [90.1,97.0]	6.00 [2.7,9.6]	0.29 [0.0,1.1]
Pres07	const.	total-no-early	joint.null	90.84 [85.3,94.6]	8.74 [4.8,13.9]	0.29 [0.0,1.3]
Pres07	dong	total-no-early	real	92.84 [91.7,94.1]	7.12 [5.9,8.3]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	marg.null	94.02 [91.9,95.9]	5.95 [4.2,8.1]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	joint.null	92.69 [91.6,93.7]	7.28 [6.3,8.4]	0.02 [0.0,0.1]
Pres12	const.	total-no-early	real	83.04 [75.4,89.8]	16.36 [9.2,23.9]	0.47 [0.0,1.6]
Pres12	const.	total-no-early	marg.null	81.61 [75.9,86.6]	17.84 [12.8,23.1]	0.36 [0.0,1.6]
Pres12	const.	total-no-early	joint.null	80.75 [74.4,87.0]	18.76 [12.3,24.7]	0.36 [0.0,1.6]
Pres12	dong	total-no-early	real	82.68 [80.6,91.6]	17.16 [8.4,19.3]	0.12 [0.0,0.3]
Pres12	dong	total-no-early	marg.null	82.39 [80.5,90.3]	17.43 [9.6,19.2]	0.14 [0.0,0.3]
Pres12	dong	total-no-early	joint.null	82.44 [80.7,85.8]	17.46 [14.0,19.2]	0.10 [0.0,0.3]
Gen16	const.	early	real	97.92 [94.7,99.9]	1.60 [0.0,4.8]	0.33 [0.0,1.3]
Gen16	const.	early	marg.null	98.33 [96.5,99.7]	1.26 [0.2,2.9]	0.29 [0.0,1.2]
Gen16	const.	early	joint.null	98.85 [97.0,100.0]	0.69 [0.0,2.2]	0.32 [0.0,1.4]
Gen16	const.	early-out	real	99.29 [98.0,100.0]	0.29 [0.0,1.3]	0.30 [0.0,1.2]
Gen16	const.	early-out	marg.null	99.03 [97.1,100.0]	0.48 [0.0,2.4]	0.29 [0.0,1.3]
Gen16	const.	early-out	joint.null	99.29 [98.1,100.0]	0.31 [0.0,1.3]	0.29 [0.0,1.1]
Gen16	const.	early-total	real	97.09 [94.6,100.0]	2.53 [0.0,4.9]	0.28 [0.0,1.2]
Gen16	const.	early-total	marg.null	96.14 [93.2,98.5]	3.40 [1.2,6.2]	0.32 [0.0,1.4]
Gen16	const.	early-total	joint.null	97.03 [94.3,99.0]	2.53 [0.5,5.0]	0.32 [0.0,1.3]
Gen16	const.	sameday	real	99.25 [97.9,100.0]	0.31 [0.0,1.4]	0.31 [0.0,1.3]

Table C.3 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen16	const.	sameday	marg.null	99.28 [97.8,100.0]	0.32 [0.0,1.4]	0.28 [0.0,1.3]
Gen16	const.	sameday	joint.null	99.29 [98.0,100.0]	0.29 [0.0,1.3]	0.30 [0.0,1.2]
Gen16	const.	total	real	98.84 [97.1,100.0]	0.75 [0.0,2.4]	0.27 [0.0,1.1]
Gen16	const.	total	marg.null	97.78 [95.3,99.5]	1.77 [0.2,3.9]	0.31 [0.0,1.4]
Gen16	const.	total	joint.null	98.40 [96.0,99.9]	1.10 [0.0,3.5]	0.32 [0.0,1.4]
Gen16	dong	early	real	99.51 [99.1,99.9]	0.46 [0.1,0.9]	0.02 [0.0,0.1]
Gen16	dong	early	marg.null	99.04 [98.6,99.8]	0.92 [0.1,1.4]	0.02 [0.0,0.1]
Gen16	dong	early	joint.null	99.76 [98.2,100.0]	0.21 [0.0,1.8]	0.02 [0.0,0.1]
Gen16	dong	pooled	real	99.67 [99.4,99.9]	0.30 [0.1,0.5]	0.02 [0.0,0.1]
Gen16	dong	pooled	marg.null	99.35 [99.0,99.7]	0.61 [0.3,1.0]	0.02 [0.0,0.1]
Gen16	dong	pooled	joint.null	99.69 [99.5,99.9]	0.27 [0.1,0.5]	0.03 [0.0,0.1]
Gen16	dong	sameday	real	99.94 [99.8,100.0]	0.03 [0.0,0.1]	0.02 [0.0,0.1]
Gen16	dong	sameday	marg.null	99.95 [99.9,100.0]	0.02 [0.0,0.1]	0.02 [0.0,0.1]
Gen16	dong	sameday	joint.null	99.95 [99.9,100.0]	0.02 [0.0,0.1]	0.02 [0.0,0.1]
Pres17	const.	early	real	96.79 [93.1,99.0]	2.78 [0.7,6.4]	0.27 [0.0,1.2]
Pres17	const.	early	marg.null	97.12 [94.5,99.3]	2.47 [0.4,4.9]	0.28 [0.0,1.1]
Pres17	const.	early	joint.null	97.36 [95.0,99.2]	2.24 [0.6,4.5]	0.30 [0.0,1.3]
Pres17	const.	early-out	real	98.92 [93.5,100.0]	0.62 [0.0,6.0]	0.28 [0.0,1.2]
Pres17	const.	early-out	marg.null	98.94 [92.5,100.0]	0.57 [0.0,7.0]	0.27 [0.0,1.2]
Pres17	const.	early-out	joint.null	97.37 [94.4,99.9]	2.23 [0.0,5.2]	0.29 [0.0,1.3]
Pres17	const.	early-total	real	99.08 [94.0,100.0]	0.45 [0.0,5.5]	0.28 [0.0,1.2]
Pres17	const.	early-total	marg.null	99.04 [97.3,100.0]	0.54 [0.0,1.9]	0.29 [0.0,1.4]
Pres17	const.	early-total	joint.null	99.07 [95.9,100.0]	0.46 [0.0,3.6]	0.28 [0.0,1.2]
Pres17	const.	sameday	real	99.32 [98.2,100.0]	0.28 [0.0,1.1]	0.28 [0.0,1.2]
Pres17	const.	sameday	marg.null	99.35 [98.2,100.0]	0.26 [0.0,1.2]	0.27 [0.0,1.2]
Pres17	const.	sameday	joint.null	99.34 [98.1,100.0]	0.28 [0.0,1.2]	0.27 [0.0,1.2]
Pres17	const.	total	real	77.67 [70.6,84.9]	21.89 [14.7,29.0]	0.29 [0.0,1.2]
Pres17	const.	total	marg.null	76.87 [70.7,83.1]	22.72 [16.6,28.8]	0.27 [0.0,1.3]

Table C.3 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres17	const.	total	joint.null	93.91 [73.0,98.5]	5.73 [1.3,26.7]	0.27 [0.0,1.2]
Pres17	dong	early	real	98.67 [97.0,99.5]	1.31 [0.5,3.0]	0.02 [0.0,0.1]
Pres17	dong	early	marg.null	99.18 [97.0,99.6]	0.79 [0.4,3.0]	0.02 [0.0,0.1]
Pres17	dong	early	joint.null	99.19 [97.0,99.7]	0.78 [0.3,3.0]	0.02 [0.0,0.1]
Pres17	dong	pooled	real	89.65 [88.0,94.8]	10.32 [5.1,11.9]	0.02 [0.0,0.1]
Pres17	dong	pooled	marg.null	89.46 [88.0,93.7]	10.51 [6.2,12.0]	0.02 [0.0,0.1]
Pres17	dong	pooled	joint.null	91.24 [88.0,94.2]	8.71 [5.8,11.9]	0.02 [0.0,0.1]
Pres17	dong	sameday	real	99.92 [99.8,100.0]	0.05 [0.0,0.1]	0.02 [0.0,0.1]
Pres17	dong	sameday	marg.null	99.87 [99.7,100.0]	0.10 [0.0,0.2]	0.02 [0.0,0.1]
Pres17	dong	sameday	joint.null	99.92 [99.8,100.0]	0.05 [0.0,0.1]	0.02 [0.0,0.1]
Gen20	const.	early	real	93.65 [90.2,96.7]	5.87 [3.1,9.3]	0.31 [0.0,1.4]
Gen20	const.	early	marg.null	94.76 [91.2,97.5]	4.84 [2.3,8.5]	0.26 [0.0,1.1]
Gen20	const.	early	joint.null	94.23 [88.6,97.5]	5.34 [2.1,10.9]	0.30 [0.0,1.2]
Gen20	const.	early-out	real	97.94 [95.0,99.7]	1.65 [0.1,4.5]	0.29 [0.0,1.2]
Gen20	const.	early-out	marg.null	97.74 [95.0,99.9]	1.81 [0.0,4.6]	0.28 [0.0,1.2]
Gen20	const.	early-out	joint.null	97.45 [94.1,99.9]	2.10 [0.0,5.5]	0.29 [0.0,1.3]
Gen20	const.	early-total	real	97.02 [92.6,99.3]	2.61 [0.4,6.8]	0.26 [0.0,1.1]
Gen20	const.	early-total	marg.null	96.78 [94.3,98.8]	2.83 [1.0,5.2]	0.29 [0.0,1.2]
Gen20	const.	early-total	joint.null	97.63 [95.1,99.4]	1.99 [0.4,4.4]	0.27 [0.0,1.1]
Gen20	const.	sameday	real	99.36 [98.2,100.0]	0.26 [0.0,1.2]	0.26 [0.0,1.1]
Gen20	const.	sameday	marg.null	99.20 [97.7,100.0]	0.42 [0.0,1.7]	0.26 [0.0,1.1]
Gen20	const.	sameday	joint.null	99.23 [97.9,100.0]	0.37 [0.0,1.5]	0.26 [0.0,1.2]
Gen20	const.	total	real	86.70 [81.8,91.1]	12.94 [8.3,17.7]	0.29 [0.0,1.1]
Gen20	const.	total	marg.null	88.60 [84.2,96.4]	11.02 [3.4,15.4]	0.26 [0.0,1.2]
Gen20	const.	total	joint.null	86.86 [81.3,91.5]	12.74 [8.0,18.0]	0.29 [0.0,1.2]
Gen20	dong	early	real	97.21 [96.4,98.0]	2.76 [2.0,3.6]	0.02 [0.0,0.1]
Gen20	dong	early	marg.null	97.43 [96.8,98.0]	2.54 [2.0,3.1]	0.02 [0.0,0.1]
Gen20	dong	early	joint.null	98.02 [97.5,98.5]	1.95 [1.5,2.5]	0.02 [0.0,0.1]

Table C.3 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen20	dong	pooled	real	95.06 [94.2,95.9]	4.91 [4.1,5.8]	0.02 [0.0,0.1]
Gen20	dong	pooled	marg.null	94.59 [93.7,95.4]	5.38 [4.6,6.3]	0.02 [0.0,0.1]
Gen20	dong	pooled	joint.null	94.74 [93.8,95.6]	5.24 [4.3,6.1]	0.02 [0.0,0.1]
Gen20	dong	sameday	real	99.87 [99.7,100.0]	0.10 [0.0,0.3]	0.02 [0.0,0.1]
Gen20	dong	sameday	marg.null	99.79 [99.6,100.0]	0.18 [0.0,0.4]	0.02 [0.0,0.1]
Gen20	dong	sameday	joint.null	99.84 [99.7,100.0]	0.13 [0.0,0.3]	0.02 [0.0,0.1]
Pres22	const.	early	real	98.21 [96.3,99.5]	1.37 [0.3,3.1]	0.28 [0.0,1.2]
Pres22	const.	early	marg.null	98.03 [95.9,99.7]	1.59 [0.1,3.6]	0.28 [0.0,1.1]
Pres22	const.	early	joint.null	97.65 [95.1,99.5]	1.97 [0.2,4.4]	0.27 [0.0,1.1]
Pres22	const.	early-out	real	95.92 [92.0,99.0]	3.70 [0.8,7.6]	0.26 [0.0,1.1]
Pres22	const.	early-out	marg.null	95.48 [92.4,99.7]	4.17 [0.0,7.0]	0.28 [0.0,1.2]
Pres22	const.	early-out	joint.null	94.72 [89.8,99.8]	4.85 [0.0,9.6]	0.30 [0.0,1.5]
Pres22	const.	early-total	real	94.19 [90.9,96.9]	5.41 [2.8,8.7]	0.30 [0.0,1.2]
Pres22	const.	early-total	marg.null	94.37 [90.2,98.0]	5.21 [1.7,9.3]	0.29 [0.0,1.3]
Pres22	const.	early-total	joint.null	94.46 [91.2,97.3]	5.15 [2.3,8.3]	0.27 [0.0,1.1]
Pres22	const.	sameday	real	97.76 [95.4,99.4]	1.82 [0.3,4.0]	0.27 [0.0,1.2]
Pres22	const.	sameday	marg.null	98.28 [96.1,99.9]	1.35 [0.0,3.4]	0.28 [0.0,1.1]
Pres22	const.	sameday	joint.null	98.80 [96.6,100.0]	0.76 [0.0,2.9]	0.27 [0.0,1.2]
Pres22	const.	total	real	76.30 [62.3,86.5]	20.91 [7.0,31.7]	5.03 [0.2,9.1]
Pres22	const.	total	marg.null	70.39 [63.5,76.1]	24.02 [18.3,30.4]	5.50 [2.7,8.8]
Pres22	const.	total	joint.null	72.78 [66.4,84.5]	20.33 [9.5,25.8]	6.63 [3.7,10.1]
Pres22	dong	early	real	96.76 [86.8,97.6]	3.21 [2.4,13.3]	0.02 [0.0,0.1]
Pres22	dong	early	marg.null	97.03 [90.6,97.8]	2.94 [2.2,9.3]	0.02 [0.0,0.1]
Pres22	dong	early	joint.null	97.68 [97.1,98.2]	2.29 [1.8,2.9]	0.02 [0.0,0.1]
Pres22	dong	pooled	real	81.32 [78.8,85.9]	18.58 [11.4,20.9]	0.16 [0.0,2.9]
Pres22	dong	pooled	marg.null	81.71 [79.7,90.6]	18.04 [9.4,20.0]	0.21 [0.0,0.4]
Pres22	dong	pooled	joint.null	80.96 [79.4,82.3]	18.93 [17.5,20.3]	0.08 [0.0,0.3]
Pres22	dong	sameday	real	99.38 [99.0,99.7]	0.59 [0.3,1.0]	0.02 [0.0,0.1]

Table C.3 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres22	dong	sameday	marg.null	99.31 [99.0,99.7]	0.67 [0.3,1.0]	0.02 [0.0,0.1]
Pres22	dong	sameday	joint.null	99.36 [99.0,99.8]	0.61 [0.2,1.0]	0.02 [0.0,0.1]
Gen24	const.	early	real	96.55 [91.6,99.4]	3.02 [0.4,7.9]	0.28 [0.0,1.2]
Gen24	const.	early	marg.null	95.14 [91.6,98.1]	4.51 [1.5,7.9]	0.27 [0.0,1.1]
Gen24	const.	early	joint.null	93.61 [86.1,99.1]	6.07 [0.5,13.3]	0.30 [0.0,1.3]
Gen24	const.	early-out	real	98.48 [96.1,99.9]	1.12 [0.0,3.4]	0.29 [0.0,1.2]
Gen24	const.	early-out	marg.null	97.09 [93.0,99.9]	2.46 [0.0,6.5]	0.29 [0.0,1.2]
Gen24	const.	early-out	joint.null	97.35 [94.8,100.0]	2.25 [0.0,4.7]	0.31 [0.0,1.4]
Gen24	const.	early-total	real	99.02 [97.4,100.0]	0.54 [0.0,2.0]	0.29 [0.0,1.3]
Gen24	const.	early-total	marg.null	98.81 [97.3,99.9]	0.78 [0.0,2.1]	0.27 [0.0,1.3]
Gen24	const.	early-total	joint.null	98.67 [96.5,99.9]	0.91 [0.0,2.9]	0.28 [0.0,1.2]
Gen24	const.	sameday	real	98.70 [97.0,99.8]	0.87 [0.1,2.4]	0.29 [0.0,1.2]
Gen24	const.	sameday	marg.null	98.59 [96.7,99.9]	0.98 [0.0,2.7]	0.30 [0.0,1.1]
Gen24	const.	sameday	joint.null	98.63 [96.9,99.8]	0.97 [0.0,2.5]	0.28 [0.0,1.2]
Gen24	const.	total	real	83.34 [78.2,88.2]	16.30 [11.2,21.1]	0.26 [0.0,1.1]
Gen24	const.	total	marg.null	85.96 [79.4,93.8]	13.69 [5.4,19.6]	0.27 [0.0,1.1]
Gen24	const.	total	joint.null	84.56 [76.2,90.6]	14.99 [8.8,23.1]	0.31 [0.0,1.3]
Gen24	dong	early	real	96.24 [93.4,98.3]	3.74 [1.7,6.6]	0.03 [0.0,0.1]
Gen24	dong	early	marg.null	97.54 [96.5,98.4]	2.43 [1.6,3.5]	0.02 [0.0,0.1]
Gen24	dong	early	joint.null	98.06 [94.9,98.8]	1.90 [1.2,5.1]	0.02 [0.0,0.1]
Gen24	dong	pooled	real	93.14 [91.7,94.8]	6.82 [5.2,8.2]	0.03 [0.0,0.1]
Gen24	dong	pooled	marg.null	92.30 [91.1,93.4]	7.66 [6.6,8.9]	0.02 [0.0,0.1]
Gen24	dong	pooled	joint.null	93.02 [91.9,94.0]	6.94 [6.0,8.1]	0.03 [0.0,0.1]
Gen24	dong	sameday	real	99.40 [99.0,99.8]	0.56 [0.2,0.9]	0.02 [0.0,0.1]
Gen24	dong	sameday	marg.null	99.25 [98.9,99.6]	0.72 [0.4,1.1]	0.02 [0.0,0.1]
Gen24	dong	sameday	joint.null	99.31 [98.9,99.6]	0.66 [0.4,1.0]	0.03 [0.0,0.1]
Pres25	const.	early	real	94.81 [90.6,97.8]	4.77 [2.0,9.0]	0.28 [0.0,1.2]
Pres25	const.	early	marg.null	93.14 [89.1,97.0]	6.47 [2.6,10.4]	0.26 [0.0,1.2]

Table C.3 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres25	const.	early	joint.null	95.29 [88.3,98.5]	4.33 [1.0,11.0]	0.29 [0.0,1.1]
Pres25	const.	early-out	real	94.61 [89.4,98.2]	4.97 [1.5,10.2]	0.28 [0.0,1.2]
Pres25	const.	early-out	marg.null	93.57 [90.0,99.9]	6.06 [0.0,9.5]	0.28 [0.0,1.2]
Pres25	const.	early-out	joint.null	91.75 [87.4,99.0]	7.90 [0.8,12.2]	0.25 [0.0,1.1]
Pres25	const.	early-total	real	92.18 [87.9,96.5]	7.32 [2.9,11.4]	0.45 [0.0,1.6]
Pres25	const.	early-total	marg.null	92.29 [88.5,95.5]	7.25 [4.1,10.9]	0.30 [0.0,1.3]
Pres25	const.	early-total	joint.null	90.66 [86.2,96.7]	8.92 [3.0,13.3]	0.28 [0.0,1.3]
Pres25	const.	sameday	real	98.46 [96.8,99.7]	1.14 [0.2,2.8]	0.28 [0.0,1.1]
Pres25	const.	sameday	marg.null	98.35 [96.1,99.8]	1.22 [0.0,3.2]	0.28 [0.0,1.3]
Pres25	const.	sameday	joint.null	98.66 [96.6,99.9]	0.92 [0.0,2.7]	0.27 [0.0,1.3]
Pres25	const.	total	real	72.21 [65.6,85.6]	22.38 [10.8,29.3]	4.77 [1.9,8.3]
Pres25	const.	total	marg.null	69.26 [61.6,81.4]	25.33 [16.3,33.4]	4.58 [0.9,8.2]
Pres25	const.	total	joint.null	68.72 [60.4,78.4]	25.10 [16.0,32.7]	5.95 [3.1,9.2]
Pres25	dong	early	real	91.55 [89.9,95.9]	8.42 [4.1,10.0]	0.02 [0.0,0.1]
Pres25	dong	early	marg.null	91.00 [84.4,94.8]	8.97 [5.2,15.4]	0.03 [0.0,0.2]
Pres25	dong	early	joint.null	93.14 [84.5,95.0]	6.84 [5.0,15.4]	0.03 [0.0,0.1]
Pres25	dong	pooled	real	78.05 [76.2,80.6]	21.53 [19.3,23.5]	0.37 [0.1,0.6]
Pres25	dong	pooled	marg.null	78.15 [76.3,92.1]	21.36 [7.9,23.2]	0.43 [0.0,0.7]
Pres25	dong	pooled	joint.null	77.92 [76.2,80.1]	21.69 [19.5,23.3]	0.36 [0.1,0.7]
Pres25	dong	sameday	real	99.40 [99.0,99.7]	0.57 [0.3,0.9]	0.02 [0.0,0.1]
Pres25	dong	sameday	marg.null	99.48 [99.2,99.7]	0.50 [0.2,0.8]	0.02 [0.0,0.1]
Pres25	dong	sameday	joint.null	99.30 [98.9,99.6]	0.67 [0.3,1.0]	0.02 [0.0,0.1]

Table C.4: Mixture-component probabilities, short-chain configuration, conservative-coded leader. Full coverage, all 11 elections.

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen08	const.	total-no-early	real	99.14 [97.7,100.0]	0.35 [0.0,1.5]	0.36 [0.0,1.6]
Gen08	const.	total-no-early	marg.null	99.17 [97.8,100.0]	0.35 [0.0,1.4]	0.33 [0.0,1.5]
Gen08	const.	total-no-early	joint.null	99.12 [97.4,100.0]	0.36 [0.0,1.7]	0.38 [0.0,1.6]
Gen08	dong	total-no-early	real	99.74 [99.5,99.9]	0.23 [0.1,0.5]	0.02 [0.0,0.1]
Gen08	dong	total-no-early	marg.null	99.65 [99.3,100.0]	0.31 [0.0,0.6]	0.03 [0.0,0.1]
Gen08	dong	total-no-early	joint.null	99.86 [99.7,100.0]	0.10 [0.0,0.3]	0.03 [0.0,0.1]
Gen12	const.	total-no-early	real	99.14 [97.6,100.0]	0.33 [0.0,1.4]	0.37 [0.0,1.7]
Gen12	const.	total-no-early	marg.null	98.81 [96.8,99.9]	0.68 [0.0,2.5]	0.34 [0.0,1.4]
Gen12	const.	total-no-early	joint.null	99.11 [97.6,100.0]	0.35 [0.0,1.6]	0.36 [0.0,1.6]
Gen12	dong	total-no-early	real	98.21 [97.6,98.9]	1.74 [1.1,2.4]	0.03 [0.0,0.1]
Gen12	dong	total-no-early	marg.null	98.23 [97.1,99.0]	1.74 [1.0,2.9]	0.02 [0.0,0.1]
Gen12	dong	total-no-early	joint.null	98.69 [98.2,99.1]	1.27 [0.9,1.7]	0.03 [0.0,0.1]
Pres02	const.	total-no-early	real	87.30 [82.5,92.3]	12.26 [7.6,17.4]	0.29 [0.0,1.2]
Pres02	const.	total-no-early	marg.null	90.17 [84.1,95.2]	9.43 [4.5,15.5]	0.30 [0.0,1.3]
Pres02	const.	total-no-early	joint.null	87.15 [82.3,91.6]	12.45 [8.0,17.2]	0.28 [0.0,1.2]
Pres02	dong	total-no-early	real	85.79 [84.7,87.0]	14.19 [13.0,15.3]	0.02 [0.0,0.1]
Pres02	dong	total-no-early	marg.null	85.24 [83.8,86.6]	14.74 [13.4,16.2]	0.03 [0.0,0.1]
Pres02	dong	total-no-early	joint.null	85.33 [83.8,86.5]	14.64 [13.3,16.1]	0.02 [0.0,0.1]
Pres07	const.	total-no-early	real	94.89 [91.5,98.0]	4.70 [1.7,8.1]	0.28 [0.0,1.2]
Pres07	const.	total-no-early	marg.null	93.78 [90.5,96.7]	5.80 [2.9,9.0]	0.31 [0.0,1.2]
Pres07	const.	total-no-early	joint.null	94.55 [90.0,98.3]	5.05 [1.2,9.3]	0.27 [0.0,1.2]
Pres07	dong	total-no-early	real	93.38 [92.4,94.4]	6.58 [5.6,7.5]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	marg.null	92.98 [92.0,93.8]	6.99 [6.1,7.9]	0.02 [0.0,0.1]
Pres07	dong	total-no-early	joint.null	93.64 [92.7,94.5]	6.34 [5.5,7.3]	0.02 [0.0,0.1]
Pres12	const.	total-no-early	real	77.70 [68.8,88.1]	21.73 [11.1,30.7]	0.45 [0.0,1.6]
Pres12	const.	total-no-early	marg.null	78.97 [71.1,87.5]	20.08 [11.9,28.6]	0.65 [0.0,1.9]

Table C.4 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres12	const.	total-no-early	joint.null	79.17 [67.9,87.9]	20.30 [10.8,31.7]	0.45 [0.0,2.3]
Pres12	dong	total-no-early	real	73.87 [72.1,89.8]	26.09 [10.2,27.8]	0.03 [0.0,0.1]
Pres12	dong	total-no-early	marg.null	84.44 [73.8,86.8]	15.43 [13.1,26.0]	0.09 [0.0,0.3]
Pres12	dong	total-no-early	joint.null	74.42 [72.4,88.0]	25.52 [12.0,27.6]	0.04 [0.0,0.1]
Gen16	const.	early	real	98.12 [95.6,99.8]	1.43 [0.1,3.9]	0.30 [0.0,1.3]
Gen16	const.	early	marg.null	97.56 [94.7,99.8]	2.05 [0.1,4.7]	0.29 [0.0,1.3]
Gen16	const.	early	joint.null	98.51 [96.6,99.8]	1.04 [0.0,2.8]	0.30 [0.0,1.3]
Gen16	const.	early-out	real	99.24 [97.8,100.0]	0.30 [0.0,1.3]	0.32 [0.0,1.5]
Gen16	const.	early-out	marg.null	99.27 [98.0,100.0]	0.30 [0.0,1.2]	0.30 [0.0,1.2]
Gen16	const.	early-out	joint.null	99.27 [98.0,100.0]	0.31 [0.0,1.2]	0.29 [0.0,1.3]
Gen16	const.	early-total	real	96.97 [94.2,99.2]	2.59 [0.5,5.2]	0.29 [0.0,1.3]
Gen16	const.	early-total	marg.null	97.05 [94.4,99.2]	2.48 [0.6,5.1]	0.34 [0.0,1.4]
Gen16	const.	early-total	joint.null	97.64 [93.4,99.7]	1.89 [0.1,6.2]	0.30 [0.0,1.3]
Gen16	const.	sameday	real	99.31 [97.9,100.0]	0.31 [0.0,1.4]	0.29 [0.0,1.2]
Gen16	const.	sameday	marg.null	99.29 [98.0,100.0]	0.30 [0.0,1.3]	0.31 [0.0,1.2]
Gen16	const.	sameday	joint.null	99.24 [97.9,100.0]	0.31 [0.0,1.4]	0.31 [0.0,1.4]
Gen16	const.	total	real	98.99 [97.3,100.0]	0.58 [0.0,2.1]	0.29 [0.0,1.2]
Gen16	const.	total	marg.null	98.71 [96.0,100.0]	0.83 [0.0,3.6]	0.27 [0.0,1.1]
Gen16	const.	total	joint.null	99.18 [97.6,100.0]	0.38 [0.0,1.5]	0.30 [0.0,1.5]
Gen16	dong	early	real	99.18 [98.0,99.9]	0.79 [0.1,2.0]	0.02 [0.0,0.1]
Gen16	dong	early	marg.null	99.74 [96.5,100.0]	0.23 [0.0,3.5]	0.02 [0.0,0.1]
Gen16	dong	early	joint.null	99.32 [98.8,99.8]	0.65 [0.1,1.2]	0.02 [0.0,0.1]
Gen16	dong	pooled	real	98.28 [97.7,98.8]	1.69 [1.1,2.2]	0.02 [0.0,0.1]
Gen16	dong	pooled	marg.null	98.17 [97.6,99.0]	1.80 [0.9,2.4]	0.02 [0.0,0.1]
Gen16	dong	pooled	joint.null	98.74 [98.2,99.2]	1.22 [0.8,1.7]	0.02 [0.0,0.1]
Gen16	dong	sameday	real	99.88 [99.7,100.0]	0.08 [0.0,0.2]	0.02 [0.0,0.1]
Gen16	dong	sameday	marg.null	99.83 [99.6,100.0]	0.14 [0.0,0.3]	0.02 [0.0,0.1]
Gen16	dong	sameday	joint.null	99.89 [99.7,100.0]	0.07 [0.0,0.3]	0.02 [0.0,0.1]

Table C.4 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres17	const.	early	real	98.06 [92.4,100.0]	1.41 [0.0,7.2]	0.27 [0.0,1.2]
Pres17	const.	early	marg.null	98.11 [95.1,100.0]	1.49 [0.0,4.4]	0.27 [0.0,1.1]
Pres17	const.	early	joint.null	98.71 [97.0,99.9]	0.89 [0.1,2.4]	0.28 [0.0,1.2]
Pres17	const.	early-out	real	99.15 [97.7,100.0]	0.44 [0.0,1.6]	0.28 [0.0,1.2]
Pres17	const.	early-out	marg.null	99.23 [97.9,100.0]	0.37 [0.0,1.6]	0.27 [0.0,1.1]
Pres17	const.	early-out	joint.null	99.13 [97.7,100.0]	0.46 [0.0,1.6]	0.27 [0.0,1.2]
Pres17	const.	early-total	real	99.09 [97.5,100.0]	0.50 [0.0,1.9]	0.29 [0.0,1.2]
Pres17	const.	early-total	marg.null	99.34 [98.1,100.0]	0.27 [0.0,1.1]	0.30 [0.0,1.3]
Pres17	const.	early-total	joint.null	99.16 [97.2,100.0]	0.42 [0.0,2.3]	0.27 [0.0,1.1]
Pres17	const.	sameday	real	99.18 [97.7,100.0]	0.40 [0.0,1.8]	0.27 [0.0,1.1]
Pres17	const.	sameday	marg.null	98.83 [96.7,100.0]	0.73 [0.0,2.8]	0.27 [0.0,1.2]
Pres17	const.	sameday	joint.null	99.35 [98.1,100.0]	0.26 [0.0,1.1]	0.28 [0.0,1.2]
Pres17	const.	total	real	87.06 [82.0,98.3]	12.57 [1.4,17.6]	0.28 [0.0,1.1]
Pres17	const.	total	marg.null	86.01 [81.1,90.4]	13.61 [9.2,18.5]	0.25 [0.0,1.2]
Pres17	const.	total	joint.null	91.36 [80.7,97.6]	8.15 [2.3,19.0]	0.27 [0.0,1.2]
Pres17	dong	early	real	99.65 [97.5,99.9]	0.32 [0.1,2.5]	0.02 [0.0,0.1]
Pres17	dong	early	marg.null	99.72 [98.9,100.0]	0.25 [0.0,1.1]	0.02 [0.0,0.1]
Pres17	dong	early	joint.null	99.18 [97.8,100.0]	0.75 [0.0,2.1]	0.02 [0.0,0.1]
Pres17	dong	pooled	real	91.10 [84.4,93.6]	8.90 [6.4,15.6]	0.02 [0.0,0.1]
Pres17	dong	pooled	marg.null	91.77 [88.3,93.2]	8.21 [6.8,11.7]	0.02 [0.0,0.1]
Pres17	dong	pooled	joint.null	92.78 [88.2,94.6]	7.19 [5.4,11.8]	0.02 [0.0,0.1]
Pres17	dong	sameday	real	99.88 [97.0,100.0]	0.09 [0.0,3.0]	0.02 [0.0,0.1]
Pres17	dong	sameday	marg.null	99.78 [99.6,99.9]	0.19 [0.1,0.4]	0.02 [0.0,0.1]
Pres17	dong	sameday	joint.null	99.90 [99.8,100.0]	0.07 [0.0,0.2]	0.02 [0.0,0.1]
Gen20	const.	early	real	98.90 [97.1,99.9]	0.70 [0.0,2.4]	0.29 [0.0,1.2]
Gen20	const.	early	marg.null	98.57 [96.5,99.9]	1.03 [0.0,3.0]	0.26 [0.0,1.2]
Gen20	const.	early	joint.null	98.75 [97.1,99.9]	0.84 [0.0,2.2]	0.28 [0.0,1.2]
Gen20	const.	early-out	real	98.97 [97.3,100.0]	0.62 [0.0,2.1]	0.27 [0.0,1.2]

Table C.4 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen20	const.	early-out	marg.null	99.22 [97.9,100.0]	0.39 [0.0,1.5]	0.28 [0.0,1.2]
Gen20	const.	early-out	joint.null	99.19 [97.6,100.0]	0.38 [0.0,1.8]	0.28 [0.0,1.3]
Gen20	const.	early-total	real	99.12 [96.9,100.0]	0.44 [0.0,2.7]	0.27 [0.0,1.1]
Gen20	const.	early-total	marg.null	98.79 [96.2,100.0]	0.77 [0.0,3.3]	0.28 [0.0,1.1]
Gen20	const.	early-total	joint.null	99.27 [98.0,100.0]	0.35 [0.0,1.4]	0.26 [0.0,1.1]
Gen20	const.	sameday	real	99.29 [98.1,100.0]	0.29 [0.0,1.2]	0.29 [0.0,1.2]
Gen20	const.	sameday	marg.null	99.31 [97.9,100.0]	0.29 [0.0,1.3]	0.28 [0.0,1.3]
Gen20	const.	sameday	joint.null	99.36 [98.2,100.0]	0.26 [0.0,1.1]	0.26 [0.0,1.2]
Gen20	const.	total	real	92.02 [88.0,99.6]	7.60 [0.3,11.5]	0.27 [0.0,1.3]
Gen20	const.	total	marg.null	90.95 [86.7,96.1]	8.67 [3.6,12.7]	0.28 [0.0,1.3]
Gen20	const.	total	joint.null	90.00 [85.5,93.8]	9.59 [5.6,13.9]	0.28 [0.0,1.2]
Gen20	dong	early	real	99.42 [99.0,99.7]	0.55 [0.2,0.9]	0.02 [0.0,0.1]
Gen20	dong	early	marg.null	99.48 [98.5,99.8]	0.49 [0.2,1.5]	0.02 [0.0,0.1]
Gen20	dong	early	joint.null	99.10 [98.7,99.5]	0.87 [0.5,1.2]	0.02 [0.0,0.1]
Gen20	dong	pooled	real	94.00 [93.1,94.8]	5.97 [5.2,6.8]	0.02 [0.0,0.1]
Gen20	dong	pooled	marg.null	93.40 [92.5,94.2]	6.57 [5.7,7.4]	0.02 [0.0,0.1]
Gen20	dong	pooled	joint.null	92.83 [91.8,93.8]	7.14 [6.1,8.1]	0.02 [0.0,0.1]
Gen20	dong	sameday	real	99.87 [99.7,100.0]	0.10 [0.0,0.2]	0.02 [0.0,0.1]
Gen20	dong	sameday	marg.null	99.83 [99.7,100.0]	0.14 [0.0,0.3]	0.02 [0.0,0.1]
Gen20	dong	sameday	joint.null	99.75 [99.5,99.9]	0.22 [0.1,0.4]	0.02 [0.0,0.1]
Pres22	const.	early	real	98.80 [97.2,99.9]	0.81 [0.0,2.2]	0.27 [0.0,1.2]
Pres22	const.	early	marg.null	98.82 [97.2,99.8]	0.77 [0.1,2.2]	0.28 [0.0,1.1]
Pres22	const.	early	joint.null	98.97 [97.4,99.9]	0.64 [0.0,2.0]	0.27 [0.0,1.2]
Pres22	const.	early-out	real	96.19 [90.0,99.5]	3.42 [0.3,9.5]	0.29 [0.0,1.2]
Pres22	const.	early-out	marg.null	97.87 [94.2,100.0]	1.68 [0.0,5.3]	0.28 [0.0,1.3]
Pres22	const.	early-out	joint.null	97.46 [94.8,99.6]	2.10 [0.2,4.7]	0.28 [0.0,1.2]
Pres22	const.	early-total	real	98.85 [97.2,99.9]	0.75 [0.0,2.1]	0.29 [0.0,1.2]
Pres22	const.	early-total	marg.null	98.85 [97.3,99.9]	0.77 [0.0,2.1]	0.27 [0.0,1.2]

Table C.4 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres22	const.	early-total	joint.null	99.17 [97.5,100.0]	0.44 [0.0,1.8]	0.29 [0.0,1.2]
Pres22	const.	sameday	real	99.23 [97.9,100.0]	0.36 [0.0,1.5]	0.29 [0.0,1.2]
Pres22	const.	sameday	marg.null	99.04 [97.7,99.9]	0.57 [0.0,1.8]	0.27 [0.0,1.1]
Pres22	const.	sameday	joint.null	99.20 [97.9,100.0]	0.36 [0.0,1.4]	0.30 [0.0,1.3]
Pres22	const.	total	real	67.75 [61.3,74.1]	31.02 [25.1,37.5]	1.14 [0.1,2.6]
Pres22	const.	total	marg.null	70.17 [61.6,88.0]	27.14 [8.9,37.8]	2.16 [0.0,6.3]
Pres22	const.	total	joint.null	75.83 [65.1,85.3]	23.34 [14.1,32.9]	0.61 [0.0,2.7]
Pres22	dong	early	real	99.50 [99.2,99.8]	0.47 [0.2,0.8]	0.02 [0.0,0.1]
Pres22	dong	early	marg.null	99.27 [92.7,99.6]	0.70 [0.4,7.3]	0.02 [0.0,0.1]
Pres22	dong	early	joint.null	97.97 [94.0,99.6]	1.95 [0.4,6.0]	0.02 [0.0,0.1]
Pres22	dong	pooled	real	78.62 [76.7,87.7]	21.35 [12.2,23.2]	0.02 [0.0,0.1]
Pres22	dong	pooled	marg.null	78.20 [76.1,87.3]	21.73 [12.7,23.8]	0.04 [0.0,0.2]
Pres22	dong	pooled	joint.null	78.03 [75.6,91.4]	21.94 [8.4,24.3]	0.04 [0.0,0.2]
Pres22	dong	sameday	real	99.81 [99.6,100.0]	0.16 [0.0,0.3]	0.02 [0.0,0.1]
Pres22	dong	sameday	marg.null	99.73 [99.5,99.9]	0.24 [0.1,0.4]	0.02 [0.0,0.1]
Pres22	dong	sameday	joint.null	99.64 [99.4,99.9]	0.33 [0.1,0.6]	0.02 [0.0,0.1]
Gen24	const.	early	real	98.92 [96.3,100.0]	0.61 [0.0,3.3]	0.28 [0.0,1.2]
Gen24	const.	early	marg.null	98.14 [96.0,100.0]	1.44 [0.0,3.5]	0.27 [0.0,1.2]
Gen24	const.	early	joint.null	98.41 [94.0,100.0]	1.14 [0.0,5.5]	0.29 [0.0,1.3]
Gen24	const.	early-out	real	98.63 [96.5,99.9]	0.97 [0.0,2.9]	0.27 [0.0,1.1]
Gen24	const.	early-out	marg.null	98.22 [95.8,99.8]	1.35 [0.0,3.7]	0.28 [0.0,1.2]
Gen24	const.	early-out	joint.null	98.74 [96.7,100.0]	0.84 [0.0,2.6]	0.30 [0.0,1.3]
Gen24	const.	early-total	real	99.08 [97.5,100.0]	0.51 [0.0,1.7]	0.29 [0.0,1.3]
Gen24	const.	early-total	marg.null	99.24 [97.8,100.0]	0.34 [0.0,1.6]	0.29 [0.0,1.2]
Gen24	const.	early-total	joint.null	99.22 [97.9,100.0]	0.37 [0.0,1.4]	0.28 [0.0,1.3]
Gen24	const.	sameday	real	99.29 [98.1,100.0]	0.29 [0.0,1.2]	0.29 [0.0,1.2]
Gen24	const.	sameday	marg.null	99.16 [97.8,100.0]	0.45 [0.0,1.6]	0.28 [0.0,1.2]
Gen24	const.	sameday	joint.null	99.21 [97.8,100.0]	0.36 [0.0,1.5]	0.29 [0.0,1.3]

Table C.4 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Gen24	const.	total	real	88.39 [82.5,92.7]	11.20 [6.7,16.7]	0.28 [0.0,1.2]
Gen24	const.	total	marg.null	88.38 [82.8,93.4]	11.21 [6.4,16.9]	0.28 [0.0,1.2]
Gen24	const.	total	joint.null	88.21 [83.8,92.2]	11.41 [7.6,15.9]	0.27 [0.0,1.1]
Gen24	dong	early	real	99.43 [93.5,99.8]	0.52 [0.2,6.4]	0.02 [0.0,0.1]
Gen24	dong	early	marg.null	99.33 [99.0,99.7]	0.64 [0.3,1.0]	0.02 [0.0,0.1]
Gen24	dong	early	joint.null	99.33 [99.0,99.6]	0.64 [0.3,1.0]	0.03 [0.0,0.1]
Gen24	dong	pooled	real	93.28 [91.9,94.6]	6.68 [5.3,8.0]	0.02 [0.0,0.1]
Gen24	dong	pooled	marg.null	93.36 [92.2,94.7]	6.60 [5.2,7.8]	0.03 [0.0,0.1]
Gen24	dong	pooled	joint.null	93.91 [92.7,95.0]	6.05 [4.9,7.2]	0.03 [0.0,0.1]
Gen24	dong	sameday	real	99.88 [99.7,100.0]	0.09 [0.0,0.2]	0.03 [0.0,0.1]
Gen24	dong	sameday	marg.null	99.76 [99.5,99.9]	0.20 [0.0,0.4]	0.03 [0.0,0.1]
Gen24	dong	sameday	joint.null	99.71 [98.3,99.9]	0.25 [0.1,1.6]	0.02 [0.0,0.1]
Pres25	const.	early	real	98.39 [96.0,99.8]	1.16 [0.0,3.4]	0.28 [0.0,1.3]
Pres25	const.	early	marg.null	99.36 [98.2,100.0]	0.27 [0.0,1.2]	0.27 [0.0,1.1]
Pres25	const.	early	joint.null	98.75 [96.8,99.9]	0.81 [0.0,2.7]	0.27 [0.0,1.3]
Pres25	const.	early-out	real	98.36 [96.3,99.9]	1.24 [0.0,3.1]	0.29 [0.0,1.2]
Pres25	const.	early-out	marg.null	98.44 [96.5,100.0]	1.16 [0.0,2.9]	0.29 [0.0,1.2]
Pres25	const.	early-out	joint.null	98.64 [95.9,100.0]	0.91 [0.0,3.7]	0.26 [0.0,1.1]
Pres25	const.	early-total	real	99.23 [97.6,100.0]	0.35 [0.0,1.6]	0.29 [0.0,1.3]
Pres25	const.	early-total	marg.null	99.32 [98.2,100.0]	0.27 [0.0,1.2]	0.29 [0.0,1.2]
Pres25	const.	early-total	joint.null	99.25 [97.9,100.0]	0.36 [0.0,1.4]	0.27 [0.0,1.2]
Pres25	const.	sameday	real	99.06 [97.5,100.0]	0.54 [0.0,1.9]	0.26 [0.0,1.1]
Pres25	const.	sameday	marg.null	98.91 [97.3,99.9]	0.69 [0.0,2.0]	0.26 [0.0,1.2]
Pres25	const.	sameday	joint.null	99.05 [97.6,99.9]	0.56 [0.0,1.8]	0.26 [0.0,1.2]
Pres25	const.	total	real	71.48 [63.8,89.6]	27.70 [10.1,35.3]	0.58 [0.0,1.8]
Pres25	const.	total	marg.null	70.63 [62.9,78.3]	28.18 [20.8,36.7]	0.89 [0.0,2.7]
Pres25	const.	total	joint.null	69.01 [61.6,82.3]	30.24 [17.3,37.4]	0.47 [0.0,1.8]
Pres25	dong	early	real	99.77 [99.6,99.9]	0.20 [0.1,0.4]	0.02 [0.0,0.1]

Table C.4 – continued

Elec.	Level	Channel	Source	$p_{\text{no-fraud}}$ % [CI]	$p_{\text{incr.}}$ % [CI]	$p_{\text{extr.}}$ % [CI]
Pres25	dong	early	marg.null	99.16 [97.5,100.0]	0.80 [0.0,2.5]	0.02 [0.0,0.1]
Pres25	dong	early	joint.null	99.78 [99.6,99.9]	0.19 [0.0,0.4]	0.02 [0.0,0.1]
Pres25	dong	pooled	real	80.40 [77.8,88.2]	19.56 [11.8,22.2]	0.02 [0.0,0.1]
Pres25	dong	pooled	marg.null	80.59 [78.9,90.8]	19.38 [9.1,21.0]	0.03 [0.0,0.1]
Pres25	dong	pooled	joint.null	80.14 [78.4,81.8]	19.83 [18.1,21.5]	0.02 [0.0,0.1]
Pres25	dong	sameday	real	99.15 [98.8,99.5]	0.82 [0.5,1.2]	0.02 [0.0,0.1]
Pres25	dong	sameday	marg.null	99.05 [98.7,99.4]	0.92 [0.6,1.3]	0.02 [0.0,0.1]
Pres25	dong	sameday	joint.null	99.14 [98.8,99.5]	0.83 [0.5,1.2]	0.02 [0.0,0.1]

C.2 Mebane (2023) multimodality diagnostics

Tables C.5–C.6 report $M(\pi_j)$ and $D(\pi_j)$, the multimodality diagnostics proposed by Mebane [2023], for each mixture component $j \in \{1, 2, 3\}$ (no-fraud, incremental fraud, extreme fraud) and every run listed in Tables C.1–C.2. $M(\pi_j)$ is the largest absolute difference between chain-specific posterior means for component j , and $D(\pi_j)$ is the Hartigan dip-test p -value [Hartigan and Hartigan, 1985] for unimodality on that component’s pooled posterior draws across chains. Cells marked † satisfy $M(\pi_j) > 0.01$ with $D(\pi_j) < 0.05$, the joint threshold Mebane [2023] treats as a substantive multimodality signal rather than an ordinary sampling artifact. This flag fires at comparable rates on real and fraud-free null data, so its presence here should not be read as evidence of fraud in any individual row.

Table C.5: Mebane (2023) multimodality diagnostics, short-chain configuration, Democratic-coded leader. Full coverage, all 11 elections.

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen08	const.	total-no-early	real	0.0012/0.996	0.0006/0.995	0.0009/0.993
Gen08	const.	total-no-early	marg.null	0.0010/0.911	0.0006/0.993	0.0009/0.926
Gen08	const.	total-no-early	joint.null	0.0016/0.996	0.0009/0.998	0.0014/0.992
Gen08	dong	total-no-early	real	0.0023/0.930	0.0022/0.916	0.0001/0.980
Gen08	dong	total-no-early	marg.null	0.0039/0.992	0.0038/0.994	0.0001/0.994
Gen08	dong	total-no-early	joint.null	0.0019/0.954	0.0019/1.000	0.0000/0.994
Gen12	const.	total-no-early	real	0.0018/0.995	0.0013/0.998	0.0013/0.996
Gen12	const.	total-no-early	marg.null	0.0006/0.991	0.0007/0.993	0.0003/0.992
Gen12	const.	total-no-early	joint.null	0.0025/0.991	0.0004/0.996	0.0026/0.992

Table C.5 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen12	dong	total-no-early	real	0.0037/0.933	0.0038/0.985	0.0001/1.000
Gen12	dong	total-no-early	marg.null	0.0013/0.998	0.0014/0.643	0.0001/0.934
Gen12	dong	total-no-early	joint.null	0.0017/0.990	0.0017/0.971	0.0000/0.995
Pres02	const.	total-no-early	real	0.0625/0.973	0.0629/0.954	0.0004/0.997
Pres02	const.	total-no-early	marg.null	0.0400/0.996	0.0401/0.993	0.0042/0.997
Pres02	const.	total-no-early	joint.null	0.0397/0.997	0.0443/0.994	0.0058/0.999
Pres02	dong	total-no-early	real	0.0283/0.006 [†]	0.0278/0.279	0.0023/0.982
Pres02	dong	total-no-early	marg.null	0.0404/0.000 [†]	0.0403/0.000 [†]	0.0006/0.995
Pres02	dong	total-no-early	joint.null	0.0480/0.000 [†]	0.0468/0.000 [†]	0.0012/0.995
Pres07	const.	total-no-early	real	0.0369/0.994	0.0367/0.884	0.0007/0.992
Pres07	const.	total-no-early	marg.null	0.0246/0.994	0.0247/0.994	0.0005/0.980
Pres07	const.	total-no-early	joint.null	0.0410/0.995	0.0410/1.000	0.0006/1.000
Pres07	dong	total-no-early	real	0.0114/0.996	0.0115/0.990	0.0001/0.990
Pres07	dong	total-no-early	marg.null	0.0287/0.000 [†]	0.0287/0.000 [†]	0.0000/1.000
Pres07	dong	total-no-early	joint.null	0.0084/0.990	0.0083/0.918	0.0001/0.997
Pres12	const.	total-no-early	real	0.0809/0.988	0.0845/0.856	0.0041/0.995
Pres12	const.	total-no-early	marg.null	0.0295/1.000	0.0292/0.994	0.0047/0.981
Pres12	const.	total-no-early	joint.null	0.0509/0.746	0.0508/0.981	0.0041/0.922
Pres12	dong	total-no-early	real	0.0959/0.000 [†]	0.0947/0.000 [†]	0.0018/0.000
Pres12	dong	total-no-early	marg.null	0.0823/0.000 [†]	0.0802/0.000 [†]	0.0021/0.000
Pres12	dong	total-no-early	joint.null	0.0348/0.000 [†]	0.0363/0.000 [†]	0.0015/0.998
Gen16	const.	early	real	0.0330/0.996	0.0332/0.115	0.0006/0.991
Gen16	const.	early	marg.null	0.0040/0.994	0.0043/0.997	0.0004/1.000
Gen16	const.	early	joint.null	0.0108/0.942	0.0097/0.993	0.0012/1.000
Gen16	const.	early-out	real	0.0004/1.000	0.0006/1.000	0.0003/0.999
Gen16	const.	early-out	marg.null	0.0133/0.990	0.0133/1.000	0.0004/0.998
Gen16	const.	early-out	joint.null	0.0011/1.000	0.0011/0.995	0.0005/0.997
Gen16	const.	early-total	real	0.0295/0.000 [†]	0.0298/0.000 [†]	0.0007/1.000
Gen16	const.	early-total	marg.null	0.0139/0.991	0.0138/0.998	0.0004/0.996
Gen16	const.	early-total	joint.null	0.0129/0.991	0.0132/0.894	0.0005/1.000
Gen16	const.	sameday	real	0.0009/0.993	0.0009/0.997	0.0008/0.999

Table C.5 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen16	const.	sameday	marg.null	0.0012/0.997	0.0008/0.993	0.0004/0.998
Gen16	const.	sameday	joint.null	0.0010/0.995	0.0009/0.993	0.0007/0.993
Gen16	const.	total	real	0.0127/0.993	0.0129/0.996	0.0003/0.924
Gen16	const.	total	marg.null	0.0138/0.991	0.0130/0.975	0.0009/0.999
Gen16	const.	total	joint.null	0.0179/0.798	0.0180/1.000	0.0005/0.993
Gen16	dong	early	real	0.0052/0.000	0.0052/0.000	0.0000/0.994
Gen16	dong	early	marg.null	0.0081/0.000	0.0081/0.000	0.0000/0.996
Gen16	dong	early	joint.null	0.0156/0.000 [†]	0.0155/0.000 [†]	0.0001/0.980
Gen16	dong	pooled	real	0.0016/0.990	0.0016/0.999	0.0000/0.995
Gen16	dong	pooled	marg.null	0.0028/0.734	0.0028/0.805	0.0001/0.995
Gen16	dong	pooled	joint.null	0.0013/0.993	0.0013/0.999	0.0003/0.980
Gen16	dong	sameday	real	0.0003/0.998	0.0003/1.000	0.0000/0.992
Gen16	dong	sameday	marg.null	0.0001/0.643	0.0001/0.999	0.0001/0.994
Gen16	dong	sameday	joint.null	0.0001/0.823	0.0001/0.999	0.0000/0.998
Pres17	const.	early	real	0.0284/0.997	0.0288/0.999	0.0006/1.000
Pres17	const.	early	marg.null	0.0211/0.996	0.0205/0.993	0.0007/0.993
Pres17	const.	early	joint.null	0.0112/0.984	0.0121/0.995	0.0009/0.996
Pres17	const.	early-out	real	0.0449/0.024 [†]	0.0445/0.000 [†]	0.0004/0.994
Pres17	const.	early-out	marg.null	0.0536/0.000 [†]	0.0536/0.000 [†]	0.0007/0.993
Pres17	const.	early-out	joint.null	0.0354/0.028 [†]	0.0358/0.000 [†]	0.0007/0.994
Pres17	const.	early-total	real	0.0404/0.004 [†]	0.0402/0.000 [†]	0.0006/0.996
Pres17	const.	early-total	marg.null	0.0090/0.997	0.0078/1.000	0.0012/0.996
Pres17	const.	early-total	joint.null	0.0247/0.999	0.0245/0.959	0.0002/0.994
Pres17	const.	sameday	real	0.0004/0.994	0.0007/0.995	0.0006/1.000
Pres17	const.	sameday	marg.null	0.0012/0.997	0.0010/0.999	0.0005/0.992
Pres17	const.	sameday	joint.null	0.0002/1.000	0.0004/0.970	0.0003/0.995
Pres17	const.	total	real	0.0706/0.675	0.0707/0.345	0.0012/0.995
Pres17	const.	total	marg.null	0.0424/0.999	0.0419/0.994	0.0005/0.993
Pres17	const.	total	joint.null	0.2111/0.000 [†]	0.2117/0.000 [†]	0.0009/0.994
Pres17	dong	early	real	0.0187/0.000 [†]	0.0186/0.000 [†]	0.0001/0.998
Pres17	dong	early	marg.null	0.0199/0.000 [†]	0.0198/0.000 [†]	0.0000/0.999
Pres17	dong	early	joint.null	0.0216/0.000 [†]	0.0216/0.000 [†]	0.0001/0.995
Pres17	dong	pooled	real	0.0559/0.000 [†]	0.0560/0.000 [†]	0.0000/0.992
Pres17	dong	pooled	marg.null	0.0438/0.000 [†]	0.0438/0.000 [†]	0.0000/0.981
Pres17	dong	pooled	joint.null	0.0492/0.000 [†]	0.0492/0.000 [†]	0.0003/0.994
Pres17	dong	sameday	real	0.0006/0.997	0.0006/0.981	0.0000/0.974
Pres17	dong	sameday	marg.null	0.0012/0.952	0.0011/0.901	0.0001/0.991
Pres17	dong	sameday	joint.null	0.0005/0.996	0.0006/0.929	0.0000/0.998
Gen20	const.	early	real	0.0187/0.990	0.0184/0.994	0.0015/0.990
Gen20	const.	early	marg.null	0.0244/0.996	0.0240/1.000	0.0006/0.995
Gen20	const.	early	joint.null	0.0490/1.000	0.0489/1.000	0.0004/0.998
Gen20	const.	early-out	real	0.0271/0.999	0.0270/0.960	0.0004/0.992
Gen20	const.	early-out	marg.null	0.0304/0.992	0.0311/0.164	0.0007/0.991
Gen20	const.	early-out	joint.null	0.0316/0.191	0.0315/0.001 [†]	0.0003/0.955
Gen20	const.	early-total	real	0.0400/1.000	0.0390/1.000	0.0010/0.994
Gen20	const.	early-total	marg.null	0.0092/0.995	0.0092/0.887	0.0004/0.999
Gen20	const.	early-total	joint.null	0.0167/0.994	0.0166/0.993	0.0006/0.997
Gen20	const.	sameday	real	0.0009/1.000	0.0008/0.997	0.0002/0.998

Table C.5 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen20	const.	sameday	marg.null	0.0083/0.991	0.0078/0.998	0.0008/0.991
Gen20	const.	sameday	joint.null	0.0048/0.951	0.0043/0.993	0.0006/0.995
Gen20	const.	total	real	0.0282/0.995	0.0284/1.000	0.0008/0.997
Gen20	const.	total	marg.null	0.0714/0.000 [†]	0.0716/0.000 [†]	0.0007/0.934
Gen20	const.	total	joint.null	0.0364/0.991	0.0353/0.998	0.0010/0.999
Gen20	dong	early	real	0.0081/0.948	0.0081/0.757	0.0001/1.000
Gen20	dong	early	marg.null	0.0034/0.996	0.0034/0.994	0.0000/0.993
Gen20	dong	early	joint.null	0.0034/0.991	0.0035/0.997	0.0001/0.968
Gen20	dong	pooled	real	0.0062/0.997	0.0061/0.999	0.0001/0.995
Gen20	dong	pooled	marg.null	0.0054/0.982	0.0053/0.990	0.0001/0.996
Gen20	dong	pooled	joint.null	0.0067/0.994	0.0066/0.996	0.0001/0.994
Gen20	dong	sameday	real	0.0015/0.408	0.0015/0.849	0.0000/1.000
Gen20	dong	sameday	marg.null	0.0017/0.992	0.0017/0.992	0.0000/0.993
Gen20	dong	sameday	joint.null	0.0013/0.992	0.0013/0.998	0.0000/0.999
Pres22	const.	early	real	0.0043/1.000	0.0043/0.996	0.0004/0.999
Pres22	const.	early	marg.null	0.0157/0.965	0.0161/0.997	0.0005/0.995
Pres22	const.	early	joint.null	0.0198/0.992	0.0193/0.994	0.0006/0.998
Pres22	const.	early-out	real	0.0403/0.997	0.0402/0.995	0.0004/0.993
Pres22	const.	early-out	marg.null	0.0445/0.000 [†]	0.0442/0.000 [†]	0.0003/0.995
Pres22	const.	early-out	joint.null	0.0672/0.000 [†]	0.0676/0.000 [†]	0.0009/0.998
Pres22	const.	early-total	real	0.0120/0.992	0.0122/0.946	0.0012/0.997
Pres22	const.	early-total	marg.null	0.0364/0.993	0.0363/0.939	0.0001/1.000
Pres22	const.	early-total	joint.null	0.0155/0.997	0.0163/0.991	0.0008/0.993
Pres22	const.	sameday	real	0.0127/0.943	0.0129/0.999	0.0005/1.000
Pres22	const.	sameday	marg.null	0.0197/0.954	0.0196/0.517	0.0003/1.000
Pres22	const.	sameday	joint.null	0.0163/0.999	0.0164/0.925	0.0008/0.999
Pres22	const.	total	real	0.1736/0.566	0.1876/0.000 [†]	0.0627/0.000 [†]
Pres22	const.	total	marg.null	0.0352/0.996	0.0384/0.993	0.0199/0.993
Pres22	const.	total	joint.null	0.1077/0.000 [†]	0.0922/0.000 [†]	0.0154/0.993
Pres22	dong	early	real	0.0984/0.000 [†]	0.0984/0.000 [†]	0.0000/1.000
Pres22	dong	early	marg.null	0.0626/0.000 [†]	0.0625/0.000 [†]	0.0000/0.996
Pres22	dong	early	joint.null	0.0035/0.892	0.0035/0.992	0.0000/0.991
Pres22	dong	pooled	real	0.0543/0.000 [†]	0.0791/0.000 [†]	0.0260/0.000 [†]
Pres22	dong	pooled	marg.null	0.0935/0.000 [†]	0.0907/0.000 [†]	0.0028/0.000
Pres22	dong	pooled	joint.null	0.0074/0.963	0.0059/0.994	0.0018/0.997
Pres22	dong	sameday	real	0.0030/0.979	0.0030/0.988	0.0000/0.996
Pres22	dong	sameday	marg.null	0.0030/0.999	0.0029/0.991	0.0001/0.998
Pres22	dong	sameday	joint.null	0.0043/1.000	0.0043/0.997	0.0000/0.999
Gen24	const.	early	real	0.0485/0.948	0.0489/0.994	0.0004/0.991
Gen24	const.	early	marg.null	0.0281/0.965	0.0281/0.941	0.0010/0.998
Gen24	const.	early	joint.null	0.0874/0.000 [†]	0.0870/0.000 [†]	0.0009/0.976
Gen24	const.	early-out	real	0.0213/0.883	0.0210/0.995	0.0008/0.996
Gen24	const.	early-out	marg.null	0.0488/0.000 [†]	0.0490/0.000 [†]	0.0004/0.996
Gen24	const.	early-out	joint.null	0.0278/0.029 [†]	0.0276/0.000 [†]	0.0007/0.952
Gen24	const.	early-total	real	0.0082/0.994	0.0085/1.000	0.0006/1.000
Gen24	const.	early-total	marg.null	0.0047/0.995	0.0044/0.992	0.0008/0.997
Gen24	const.	early-total	joint.null	0.0164/0.996	0.0168/1.000	0.0004/0.996
Gen24	const.	sameday	real	0.0049/0.993	0.0049/0.980	0.0003/1.000
Gen24	const.	sameday	marg.null	0.0126/0.993	0.0127/0.816	0.0006/0.992

Table C.5 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen24	const.	sameday	joint.null	0.0086/0.945	0.0088/0.988	0.0007/0.997
Gen24	const.	total	real	0.0231/0.992	0.0217/0.993	0.0016/0.991
Gen24	const.	total	marg.null	0.0840/0.993	0.0842/0.986	0.0002/0.996
Gen24	const.	total	joint.null	0.0786/0.990	0.0793/0.991	0.0010/0.995
Gen24	dong	early	real	0.0400/0.000 [†]	0.0400/0.000 [†]	0.0000/0.993
Gen24	dong	early	marg.null	0.0110/1.000	0.0110/0.998	0.0001/0.990
Gen24	dong	early	joint.null	0.0302/0.000 [†]	0.0302/0.000 [†]	0.0000/0.978
Gen24	dong	pooled	real	0.0185/0.990	0.0185/0.997	0.0002/0.990
Gen24	dong	pooled	marg.null	0.0069/0.998	0.0069/0.999	0.0000/1.000
Gen24	dong	pooled	joint.null	0.0072/0.993	0.0071/0.819	0.0001/0.998
Gen24	dong	sameday	real	0.0036/0.995	0.0036/0.994	0.0000/0.999
Gen24	dong	sameday	marg.null	0.0024/0.735	0.0025/0.996	0.0001/0.994
Gen24	dong	sameday	joint.null	0.0018/0.995	0.0018/0.991	0.0001/0.992
Pres25	const.	early	real	0.0322/0.998	0.0321/0.993	0.0007/1.000
Pres25	const.	early	marg.null	0.0392/0.998	0.0380/0.994	0.0012/0.991
Pres25	const.	early	joint.null	0.0694/0.990	0.0685/0.769	0.0014/0.994
Pres25	const.	early-out	real	0.0510/0.992	0.0519/0.992	0.0013/0.995
Pres25	const.	early-out	marg.null	0.0708/0.000 [†]	0.0705/0.000 [†]	0.0005/0.996
Pres25	const.	early-out	joint.null	0.0791/0.000 [†]	0.0790/0.000 [†]	0.0003/0.939
Pres25	const.	early-total	real	0.0367/0.997	0.0371/0.996	0.0041/0.996
Pres25	const.	early-total	marg.null	0.0191/0.815	0.0189/0.999	0.0006/0.995
Pres25	const.	early-total	joint.null	0.0546/0.007 [†]	0.0534/0.003 [†]	0.0012/0.998
Pres25	const.	sameday	real	0.0037/1.000	0.0036/0.941	0.0006/0.993
Pres25	const.	sameday	marg.null	0.0120/0.998	0.0126/0.886	0.0009/0.996
Pres25	const.	sameday	joint.null	0.0120/0.994	0.0120/0.994	0.0002/1.000
Pres25	const.	total	real	0.1164/0.000 [†]	0.1180/0.000 [†]	0.0236/0.991
Pres25	const.	total	marg.null	0.1186/0.173	0.0940/0.606	0.0414/0.967
Pres25	const.	total	joint.null	0.1039/0.955	0.0953/0.991	0.0086/0.980
Pres25	dong	early	real	0.0476/0.000 [†]	0.0476/0.000 [†]	0.0000/0.993
Pres25	dong	early	marg.null	0.0906/0.000 [†]	0.0898/0.000 [†]	0.0009/0.997
Pres25	dong	early	joint.null	0.0907/0.000 [†]	0.0901/0.000 [†]	0.0006/0.994
Pres25	dong	pooled	real	0.0246/0.996	0.0237/0.991	0.0023/0.996
Pres25	dong	pooled	marg.null	0.1426/0.000 [†]	0.1376/0.000 [†]	0.0050/0.000
Pres25	dong	pooled	joint.null	0.0198/1.000	0.0178/0.999	0.0029/0.996
Pres25	dong	sameday	real	0.0029/0.979	0.0029/0.985	0.0000/1.000
Pres25	dong	sameday	marg.null	0.0018/0.874	0.0018/0.999	0.0000/0.995
Pres25	dong	sameday	joint.null	0.0030/0.999	0.0029/0.996	0.0001/0.982

Table C.6: Mebane (2023) multimodality diagnostics, short-chain configuration, conservative-coded leader.
Full coverage, all 11 elections.

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Gen08	const.	total-no-early	real	0.0009/1.000	0.0004/0.990	0.0010/0.998
Gen08	const.	total-no-early	marg.null	0.0011/0.941	0.0008/0.994	0.0004/1.000
Gen08	const.	total-no-early	joint.null	0.0027/0.974	0.0014/0.996	0.0016/1.000
Gen08	dong	total-no-early	real	0.0013/0.972	0.0014/0.991	0.0000/0.983
Gen08	dong	total-no-early	marg.null	0.0036/0.915	0.0036/0.449	0.0000/0.998
Gen08	dong	total-no-early	joint.null	0.0015/0.994	0.0015/0.994	0.0000/0.998
Gen12	const.	total-no-early	real	0.0013/0.991	0.0005/0.997	0.0010/0.998
Gen12	const.	total-no-early	marg.null	0.0109/0.994	0.0106/0.859	0.0007/0.997
Gen12	const.	total-no-early	joint.null	0.0010/0.996	0.0005/0.998	0.0010/0.998
Gen12	dong	total-no-early	real	0.0061/0.924	0.0062/0.991	0.0001/0.994
Gen12	dong	total-no-early	marg.null	0.0120/0.996	0.0120/0.998	0.0000/0.985
Gen12	dong	total-no-early	joint.null	0.0015/0.995	0.0015/0.999	0.0001/1.000
Pres02	const.	total-no-early	real	0.0327/0.980	0.0332/0.991	0.0009/0.999
Pres02	const.	total-no-early	marg.null	0.0515/0.182	0.0513/0.020 [†]	0.0007/0.995
Pres02	const.	total-no-early	joint.null	0.0294/0.994	0.0296/0.996	0.0003/0.995
Pres02	dong	total-no-early	real	0.0024/0.993	0.0023/0.992	0.0000/0.996
Pres02	dong	total-no-early	marg.null	0.0103/0.987	0.0103/0.998	0.0003/0.995
Pres02	dong	total-no-early	joint.null	0.0104/0.996	0.0103/0.989	0.0001/0.991
Pres07	const.	total-no-early	real	0.0283/0.995	0.0285/0.997	0.0011/0.999
Pres07	const.	total-no-early	marg.null	0.0139/1.000	0.0131/0.997	0.0011/0.998
Pres07	const.	total-no-early	joint.null	0.0424/0.811	0.0421/0.924	0.0004/0.999
Pres07	dong	total-no-early	real	0.0070/1.000	0.0070/0.998	0.0001/0.991
Pres07	dong	total-no-early	marg.null	0.0041/0.994	0.0042/0.995	0.0001/0.998
Pres07	dong	total-no-early	joint.null	0.0072/0.970	0.0072/0.799	0.0000/0.988
Pres12	const.	total-no-early	real	0.1235/0.000 [†]	0.1277/0.000 [†]	0.0042/0.998
Pres12	const.	total-no-early	marg.null	0.0985/0.991	0.1027/0.969	0.0077/0.999

Table C.6 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres12	const.	total-no-early	joint.null	0.1324/0.000 [†]	0.1446/0.000 [†]	0.0122/0.997
Pres12	dong	total-no-early	real	0.1582/0.000 [†]	0.1582/0.000 [†]	0.0003/0.910
Pres12	dong	total-no-early	marg.null	0.1119/0.000 [†]	0.1116/0.000 [†]	0.0017/0.594
Pres12	dong	total-no-early	joint.null	0.1381/0.000 [†]	0.1378/0.000 [†]	0.0003/0.990
Gen16	const.	early	real	0.0176/0.999	0.0176/0.999	0.0001/1.000
Gen16	const.	early	marg.null	0.0256/0.994	0.0263/0.987	0.0007/0.975
Gen16	const.	early	joint.null	0.0090/0.994	0.0093/1.000	0.0008/0.999
Gen16	const.	early-out	real	0.0006/0.993	0.0007/1.000	0.0009/0.996
Gen16	const.	early-out	marg.null	0.0009/0.603	0.0002/0.996	0.0009/0.994
Gen16	const.	early-out	joint.null	0.0007/0.920	0.0004/0.995	0.0007/0.971
Gen16	const.	early-total	real	0.0177/0.991	0.0182/0.990	0.0006/0.994
Gen16	const.	early-total	marg.null	0.0167/0.994	0.0166/0.947	0.0008/0.992
Gen16	const.	early-total	joint.null	0.0391/1.000	0.0394/0.999	0.0008/0.997
Gen16	const.	sameday	real	0.0011/0.999	0.0006/0.996	0.0011/0.994
Gen16	const.	sameday	marg.null	0.0016/0.966	0.0012/0.999	0.0004/0.993
Gen16	const.	sameday	joint.null	0.0005/0.998	0.0003/0.995	0.0007/0.992
Gen16	const.	total	real	0.0090/0.918	0.0094/0.997	0.0003/0.990
Gen16	const.	total	marg.null	0.0222/0.968	0.0221/1.000	0.0005/1.000
Gen16	const.	total	joint.null	0.0055/0.981	0.0047/0.993	0.0013/0.998
Gen16	dong	early	real	0.0154/0.000 [†]	0.0154/0.000 [†]	0.0000/0.996
Gen16	dong	early	marg.null	0.0307/0.000 [†]	0.0307/0.000 [†]	0.0000/1.000
Gen16	dong	early	joint.null	0.0069/0.119	0.0069/0.049	0.0000/0.999
Gen16	dong	pooled	real	0.0034/0.986	0.0035/0.996	0.0001/0.998
Gen16	dong	pooled	marg.null	0.0081/0.993	0.0081/0.983	0.0000/1.000
Gen16	dong	pooled	joint.null	0.0040/0.986	0.0040/0.999	0.0000/0.714
Gen16	dong	sameday	real	0.0014/0.951	0.0014/1.000	0.0001/0.976
Gen16	dong	sameday	marg.null	0.0013/0.985	0.0013/0.812	0.0000/1.000
Gen16	dong	sameday	joint.null	0.0016/1.000	0.0016/0.995	0.0000/1.000
Pres17	const.	early	real	0.0550/0.991	0.0554/0.980	0.0005/0.999
Pres17	const.	early	marg.null	0.0277/0.169	0.0277/0.004 [†]	0.0007/0.869
Pres17	const.	early	joint.null	0.0049/0.995	0.0052/0.998	0.0006/0.995
Pres17	const.	early-out	real	0.0042/0.998	0.0046/0.944	0.0013/0.997
Pres17	const.	early-out	marg.null	0.0048/0.986	0.0046/0.998	0.0005/1.000
Pres17	const.	early-out	joint.null	0.0044/0.996	0.0042/0.993	0.0008/0.993
Pres17	const.	early-total	real	0.0082/0.996	0.0083/0.999	0.0006/0.994
Pres17	const.	early-total	marg.null	0.0004/0.980	0.0002/0.994	0.0005/0.982
Pres17	const.	early-total	joint.null	0.0122/0.997	0.0126/0.998	0.0006/1.000
Pres17	const.	sameday	real	0.0074/0.996	0.0076/0.992	0.0008/0.996
Pres17	const.	sameday	marg.null	0.0154/0.992	0.0160/0.975	0.0006/1.000
Pres17	const.	sameday	joint.null	0.0004/0.997	0.0003/0.995	0.0004/0.996
Pres17	const.	total	real	0.1192/0.000 [†]	0.1190/0.000 [†]	0.0003/0.992
Pres17	const.	total	marg.null	0.0243/0.992	0.0242/1.000	0.0009/1.000
Pres17	const.	total	joint.null	0.1262/0.000 [†]	0.1258/0.000 [†]	0.0006/1.000
Pres17	dong	early	real	0.0209/0.000 [†]	0.0209/0.000 [†]	0.0000/0.998
Pres17	dong	early	marg.null	0.0082/0.000	0.0082/0.000	0.0000/0.894

Table C.6 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres17	dong	early	joint.null	0.0182/0.000 [†]	0.0183/0.000 [†]	0.0001/1.000
Pres17	dong	pooled	real	0.0787/0.000 [†]	0.0787/0.000 [†]	0.0000/0.999
Pres17	dong	pooled	marg.null	0.0361/0.000 [†]	0.0361/0.000 [†]	0.0000/0.997
Pres17	dong	pooled	joint.null	0.0524/0.000 [†]	0.0524/0.000 [†]	0.0001/0.998
Pres17	dong	sameday	real	0.0271/0.000 [†]	0.0271/0.000 [†]	0.0000/0.956
Pres17	dong	sameday	marg.null	0.0008/0.997	0.0008/0.951	0.0000/0.991
Pres17	dong	sameday	joint.null	0.0009/0.991	0.0009/0.964	0.0001/0.999
Gen20	const.	early	real	0.0127/0.993	0.0121/0.867	0.0012/0.995
Gen20	const.	early	marg.null	0.0162/0.961	0.0166/0.999	0.0007/0.910
Gen20	const.	early	joint.null	0.0082/0.992	0.0079/0.951	0.0006/0.997
Gen20	const.	early-out	real	0.0088/1.000	0.0088/0.991	0.0007/0.996
Gen20	const.	early-out	marg.null	0.0040/0.956	0.0036/0.963	0.0004/0.999
Gen20	const.	early-out	joint.null	0.0082/0.998	0.0086/1.000	0.0007/0.993
Gen20	const.	early-total	real	0.0160/0.981	0.0159/0.997	0.0004/1.000
Gen20	const.	early-total	marg.null	0.0205/0.998	0.0204/0.992	0.0005/0.992
Gen20	const.	early-total	joint.null	0.0038/0.970	0.0039/0.993	0.0001/0.980
Gen20	const.	sameday	real	0.0008/0.955	0.0004/0.917	0.0007/0.994
Gen20	const.	sameday	marg.null	0.0026/0.991	0.0014/1.000	0.0012/0.990
Gen20	const.	sameday	joint.null	0.0005/0.999	0.0004/1.000	0.0006/0.992
Gen20	const.	total	real	0.0793/0.000 [†]	0.0796/0.000 [†]	0.0009/0.995
Gen20	const.	total	marg.null	0.0437/0.937	0.0433/0.991	0.0010/1.000
Gen20	const.	total	joint.null	0.0320/0.994	0.0315/0.995	0.0007/0.993
Gen20	dong	early	real	0.0040/0.999	0.0039/0.999	0.0001/1.000
Gen20	dong	early	marg.null	0.0094/0.000	0.0093/0.000	0.0001/0.996
Gen20	dong	early	joint.null	0.0023/0.998	0.0024/0.994	0.0001/1.000
Gen20	dong	pooled	real	0.0041/0.998	0.0041/0.999	0.0000/0.983
Gen20	dong	pooled	marg.null	0.0045/0.995	0.0045/0.997	0.0001/0.971
Gen20	dong	pooled	joint.null	0.0069/0.715	0.0069/0.984	0.0000/0.998
Gen20	dong	sameday	real	0.0014/0.904	0.0014/0.006	0.0000/0.977
Gen20	dong	sameday	marg.null	0.0011/0.941	0.0011/0.996	0.0000/0.994
Gen20	dong	sameday	joint.null	0.0016/0.993	0.0016/0.993	0.0001/0.997
Pres22	const.	early	real	0.0038/0.996	0.0042/0.991	0.0013/1.000
Pres22	const.	early	marg.null	0.0047/0.878	0.0049/0.995	0.0007/0.997
Pres22	const.	early	joint.null	0.0084/0.861	0.0082/0.918	0.0005/1.000
Pres22	const.	early-out	real	0.0629/0.570	0.0636/0.479	0.0008/0.973
Pres22	const.	early-out	marg.null	0.0353/0.011 [†]	0.0351/0.000 [†]	0.0006/0.993
Pres22	const.	early-out	joint.null	0.0204/0.823	0.0206/0.906	0.0015/0.993
Pres22	const.	early-total	real	0.0043/0.993	0.0042/0.994	0.0003/0.967
Pres22	const.	early-total	marg.null	0.0056/0.997	0.0050/0.998	0.0008/0.994
Pres22	const.	early-total	joint.null	0.0097/0.992	0.0084/0.994	0.0013/1.000
Pres22	const.	sameday	real	0.0045/0.991	0.0042/0.996	0.0003/0.992
Pres22	const.	sameday	marg.null	0.0030/0.991	0.0037/0.864	0.0008/0.999
Pres22	const.	sameday	joint.null	0.0043/0.992	0.0039/1.000	0.0005/0.991
Pres22	const.	total	real	0.0359/0.686	0.0324/0.979	0.0038/0.991
Pres22	const.	total	marg.null	0.1912/0.000 [†]	0.2234/0.000 [†]	0.0455/0.122
Pres22	const.	total	joint.null	0.1246/0.000 [†]	0.1083/0.000 [†]	0.0165/0.997
Pres22	dong	early	real	0.0024/0.992	0.0024/0.996	0.0000/0.999
Pres22	dong	early	marg.null	0.0625/0.000 [†]	0.0625/0.000 [†]	0.0000/0.993
Pres22	dong	early	joint.null	0.0505/0.000 [†]	0.0505/0.000 [†]	0.0001/0.990

Table C.6 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres22	dong	pooled	real	0.0921/0.000 [†]	0.0920/0.000 [†]	0.0001/0.998
Pres22	dong	pooled	marg.null	0.0946/0.000 [†]	0.0943/0.000 [†]	0.0009/0.998
Pres22	dong	pooled	joint.null	0.1424/0.000 [†]	0.1430/0.000 [†]	0.0009/1.000
Pres22	dong	sameday	real	0.0011/0.990	0.0011/0.997	0.0000/0.994
Pres22	dong	sameday	marg.null	0.0011/0.993	0.0011/0.991	0.0000/0.994
Pres22	dong	sameday	joint.null	0.0022/0.992	0.0022/0.997	0.0000/0.994
Gen24	const.	early	real	0.0201/0.991	0.0204/1.000	0.0003/0.992
Gen24	const.	early	marg.null	0.0171/0.824	0.0170/0.000 [†]	0.0003/0.993
Gen24	const.	early	joint.null	0.0405/0.973	0.0399/1.000	0.0009/0.991
Gen24	const.	early-out	real	0.0135/0.991	0.0132/0.995	0.0003/1.000
Gen24	const.	early-out	marg.null	0.0162/0.996	0.0166/0.994	0.0010/0.997
Gen24	const.	early-out	joint.null	0.0130/0.992	0.0128/0.983	0.0005/0.958
Gen24	const.	early-total	real	0.0039/1.000	0.0041/0.926	0.0003/0.998
Gen24	const.	early-total	marg.null	0.0058/1.000	0.0060/0.996	0.0004/0.570
Gen24	const.	early-total	joint.null	0.0033/0.995	0.0038/1.000	0.0005/0.990
Gen24	const.	sameday	real	0.0008/0.998	0.0009/0.992	0.0006/0.995
Gen24	const.	sameday	marg.null	0.0049/1.000	0.0048/0.992	0.0005/0.992
Gen24	const.	sameday	joint.null	0.0038/1.000	0.0038/1.000	0.0008/0.995
Gen24	const.	total	real	0.0448/0.985	0.0451/0.995	0.0008/0.997
Gen24	const.	total	marg.null	0.0530/0.978	0.0530/0.996	0.0003/0.999
Gen24	const.	total	joint.null	0.0236/0.995	0.0241/0.966	0.0007/0.997
Gen24	dong	early	real	0.0555/0.000 [†]	0.0555/0.000 [†]	0.0001/0.999
Gen24	dong	early	marg.null	0.0029/0.999	0.0029/0.990	0.0000/0.998
Gen24	dong	early	joint.null	0.0018/0.994	0.0018/0.991	0.0001/0.926
Gen24	dong	pooled	real	0.0127/0.966	0.0126/0.957	0.0000/0.961
Gen24	dong	pooled	marg.null	0.0117/0.990	0.0117/1.000	0.0000/0.992
Gen24	dong	pooled	joint.null	0.0106/0.990	0.0107/0.994	0.0001/1.000
Gen24	dong	sameday	real	0.0005/0.982	0.0004/0.553	0.0001/0.985
Gen24	dong	sameday	marg.null	0.0018/0.991	0.0018/0.996	0.0000/0.998
Gen24	dong	sameday	joint.null	0.0127/0.000 [†]	0.0128/0.000 [†]	0.0001/0.996
Pres25	const.	early	real	0.0169/0.987	0.0169/0.975	0.0015/0.995
Pres25	const.	early	marg.null	0.0015/0.991	0.0010/0.997	0.0007/0.993
Pres25	const.	early	joint.null	0.0125/0.959	0.0119/0.967	0.0009/1.000
Pres25	const.	early-out	real	0.0157/0.995	0.0159/0.013 [†]	0.0004/0.993
Pres25	const.	early-out	marg.null	0.0150/0.995	0.0151/0.190	0.0011/1.000
Pres25	const.	early-out	joint.null	0.0236/0.998	0.0240/0.954	0.0005/0.995
Pres25	const.	early-total	real	0.0051/0.998	0.0042/0.998	0.0009/1.000
Pres25	const.	early-total	marg.null	0.0002/0.991	0.0007/0.991	0.0007/0.993
Pres25	const.	early-total	joint.null	0.0051/0.997	0.0042/0.998	0.0011/0.997
Pres25	const.	sameday	real	0.0070/0.998	0.0071/0.997	0.0004/0.997
Pres25	const.	sameday	marg.null	0.0084/0.994	0.0079/0.996	0.0009/0.995
Pres25	const.	sameday	joint.null	0.0044/0.995	0.0043/0.997	0.0002/0.964
Pres25	const.	total	real	0.1836/0.000 [†]	0.1792/0.000 [†]	0.0043/0.947
Pres25	const.	total	marg.null	0.0706/0.993	0.0766/0.997	0.0132/0.993
Pres25	const.	total	joint.null	0.1340/0.000 [†]	0.1291/0.000 [†]	0.0050/0.984
Pres25	dong	early	real	0.0014/0.979	0.0014/0.998	0.0000/0.996
Pres25	dong	early	marg.null	0.0212/0.000 [†]	0.0212/0.000 [†]	0.0000/0.996
Pres25	dong	early	joint.null	0.0013/0.997	0.0014/0.991	0.0000/0.996
Pres25	dong	pooled	real	0.0880/0.000 [†]	0.0880/0.000 [†]	0.0000/0.995

Table C.6 – continued

Elec.	Level	Channel	Source	M_1/D_1 (no-fraud)	M_2/D_2 (incr.)	M_3/D_3 (extr.)
Pres25	dong	pooled	marg.null	0.1017/0.000 [†]	0.1020/0.000 [†]	0.0003/0.934
Pres25	dong	pooled	joint.null	0.0140/0.936	0.0141/0.941	0.0000/0.997
Pres25	dong	sameday	real	0.0017/0.998	0.0017/0.994	0.0000/1.000
Pres25	dong	sameday	marg.null	0.0025/0.980	0.0025/0.873	0.0001/0.998
Pres25	dong	sameday	joint.null	0.0029/0.995	0.0029/0.994	0.0001/0.995

C.3 Model-consistent self-null: complete results

Table C.7 lists all 116 cells of the model-consistent self-null analysis: every (election, level, channel) cell in the short-chain configuration under both leader codings, with population mixture weights fixed at $\pi = (1, 0, 0)$ by construction using the fitted qbl model’s class-1 hyperparameters. π_1 is the model’s posterior no-fraud probability on data drawn entirely from its class-1 process, so an unbiased procedure would recover values near 100% in every cell. Most cells do: the median π_1 is 98.1% with the Democratic-coded leader and 98.5% with the conservative-coded leader, and 41 and 44 of the 58 cells sit at or above 95%. The failure is concentrated in a substantial minority, with 11 and 10 cells below 90% and the worst at 75.6% and 75.2%. This is enough to produce the mean fraud shares of 4.6% and 4.2% reported in the main text and to generate fraud estimates on par with those obtained from the real dataset.

Table C.7: Model-consistent self-null results: π forced to $(1, 0, 0)$ using the qbl model’s own fitted class-1 (no-fraud) data-generating equations, short-chain configuration, Democratic- and conservative-coded leader. N is the unit count; “Flagged” is the number of units receiving a non-clean MAP classification.

Lead.	Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
dem	Gen08	const.	total-no-early	198	99.16	0.68	0/198	1.07*
dem	Gen08	dong	total-no-early	2733	99.65	0.34	2/2733	1.96*
dem	Gen12	const.	total-no-early	209	99.08	0.77	0/209	1.15*
dem	Gen12	dong	total-no-early	2923	99.22	0.77	5/2923	1.66*
dem	Gen16	const.	early	234	97.48	2.39	0/234	1.23*
dem	Gen16	const.	early-out	234	97.80	1.91	0/234	2.98*
dem	Gen16	const.	early-total	234	98.38	1.41	0/234	2.61*
dem	Gen16	const.	sameday	234	99.23	0.62	0/234	1.02
dem	Gen16	const.	total	234	98.67	1.15	0/234	2.05*
dem	Gen16	dong	early	3135	99.40	0.59	0/3135	2.61*
dem	Gen16	dong	pooled	3135	99.75	0.24	1/3135	1.29*
dem	Gen16	dong	sameday	3135	99.93	0.05	0/3135	1.51*
dem	Gen20	const.	early	253	94.37	5.50	0/253	1.47*
dem	Gen20	const.	early-out	253	97.41	2.43	0/253	1.51*
dem	Gen20	const.	early-total	253	96.82	3.05	0/253	1.14*
dem	Gen20	const.	sameday	253	99.35	0.53	0/253	1.02
dem	Gen20	const.	total	253	88.09	11.82	5/253	1.71*
dem	Gen20	dong	early	3480	98.45	1.53	1/3480	17.10*
dem	Gen20	dong	pooled	3480	95.89	4.10	29/3480	1.48*
dem	Gen20	dong	sameday	3480	99.91	0.08	0/3480	1.48*
dem	Gen24	const.	early	246	95.77	4.08	0/246	3.25*
dem	Gen24	const.	early-out	246	92.93	6.98	0/246	1.91*
dem	Gen24	const.	early-total	246	97.80	2.08	1/246	1.33*

Table C.7 – continued

Lead.	Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
dem	Gen24	const.	sameday	246	99.15	0.70	0/246	1.56*
dem	Gen24	const.	total	246	89.01	10.85	5/246	1.07*
dem	Gen24	dong	early	2739	98.04	1.95	1/2739	1.27*
dem	Gen24	dong	pooled	2739	94.27	5.72	45/2739	2.12*
dem	Gen24	dong	sameday	2739	99.77	0.21	0/2739	1.64*
dem	Pres02	const.	total-no-early	244	84.32	15.53	15/244	1.52*
dem	Pres02	dong	total-no-early	3515	85.28	14.71	213/3515	1.02
dem	Pres07	const.	total-no-early	248	98.11	1.74	1/248	2.65*
dem	Pres07	dong	total-no-early	3576	97.71	2.28	26/3576	1.21*
dem	Pres12	const.	total-no-early	251	87.51	12.41	4/251	2.40*
dem	Pres12	dong	total-no-early	3479	87.68	12.30	130/3479	12.12*
dem	Pres17	const.	early	250	98.05	1.81	0/250	1.90*
dem	Pres17	const.	early-out	250	97.99	1.83	0/250	2.78*
dem	Pres17	const.	early-total	250	99.10	0.72	0/250	3.49*
dem	Pres17	const.	sameday	250	99.35	0.54	0/250	1.00
dem	Pres17	const.	total	250	81.46	18.38	9/250	1.37*
dem	Pres17	dong	early	3491	99.39	0.60	1/3491	4.57*
dem	Pres17	dong	pooled	3491	91.31	8.69	47/3491	3.06*
dem	Pres17	dong	sameday	3491	99.93	0.06	0/3491	1.09*
dem	Pres22	const.	early	250	98.64	1.22	0/250	1.25*
dem	Pres22	const.	early-out	250	94.66	5.20	1/250	1.40*
dem	Pres22	const.	early-total	250	98.94	0.92	0/250	1.30*
dem	Pres22	const.	sameday	250	98.62	1.26	0/250	1.05*
dem	Pres22	const.	total	248	78.24	21.86	19/248	2.69*
dem	Pres22	dong	early	3509	99.28	0.72	1/3509	1.71*
dem	Pres22	dong	pooled	3509	84.01	15.98	218/3509	2.72*
dem	Pres22	dong	sameday	3509	99.77	0.22	0/3509	1.09*
dem	Pres25	const.	early	253	94.74	5.14	0/253	1.28*
dem	Pres25	const.	early-out	253	95.00	4.83	0/253	2.03*
dem	Pres25	const.	early-total	253	98.28	1.61	0/253	1.15*
dem	Pres25	const.	sameday	253	98.84	1.02	0/253	1.05*
dem	Pres25	const.	total	242	75.58	24.40	19/242	2.50*
dem	Pres25	dong	early	3553	98.33	1.66	2/3553	1.09*
dem	Pres25	dong	pooled	3553	86.00	13.90	164/3553	11.66*
dem	Pres25	dong	sameday	3553	99.65	0.35	1/3553	1.58*
con	Gen08	const.	total-no-early	198	98.91	0.96	0/198	1.15*
con	Gen08	dong	total-no-early	2733	99.69	0.29	2/2733	1.99*
con	Gen12	const.	total-no-early	194	99.12	0.70	0/194	1.01
con	Gen12	dong	total-no-early	2656	98.68	1.31	5/2656	1.35*
con	Gen16	const.	early	229	96.97	2.86	0/229	1.10*
con	Gen16	const.	early-out	229	99.04	0.83	0/229	1.12*
con	Gen16	const.	early-total	229	97.32	2.50	0/229	1.38*
con	Gen16	const.	sameday	229	99.17	0.71	0/229	1.21*
con	Gen16	const.	total	229	98.31	1.56	1/229	1.60*
con	Gen16	dong	early	3058	99.21	0.78	1/3058	2.03*
con	Gen16	dong	pooled	3058	97.96	2.04	21/3058	1.24*

Table C.7 – continued

Lead.	Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
con	Gen16	dong	sameday	3058	99.72	0.27	2/3058	1.10*
con	Gen20	const.	early	253	97.01	2.87	1/253	1.47*
con	Gen20	const.	early-out	253	98.54	1.32	0/253	1.15*
con	Gen20	const.	early-total	253	98.46	1.36	0/253	3.04*
con	Gen20	const.	sameday	253	99.09	0.77	0/253	1.39*
con	Gen20	const.	total	253	87.39	12.49	7/253	1.14*
con	Gen20	dong	early	3480	99.15	0.84	1/3480	1.44*
con	Gen20	dong	pooled	3480	94.60	5.40	67/3480	1.78*
con	Gen20	dong	sameday	3480	99.67	0.32	0/3480	1.72*
con	Gen24	const.	early	246	97.09	2.77	0/246	1.45*
con	Gen24	const.	early-out	246	98.34	1.54	0/246	1.33*
con	Gen24	const.	early-total	246	99.12	0.75	0/246	1.14*
con	Gen24	const.	sameday	246	99.16	0.68	0/246	1.61*
con	Gen24	const.	total	246	91.34	8.53	2/246	1.72*
con	Gen24	dong	early	2739	99.26	0.73	0/2739	9.06*
con	Gen24	dong	pooled	2739	95.04	4.95	40/2739	1.64*
con	Gen24	dong	sameday	2739	99.60	0.39	0/2739	1.24*
con	Pres02	const.	total-no-early	244	89.86	10.03	6/244	1.32*
con	Pres02	dong	total-no-early	3515	89.72	10.27	109/3515	4.70*
con	Pres07	const.	total-no-early	248	95.84	4.06	2/248	1.07*
con	Pres07	dong	total-no-early	3576	94.86	5.13	56/3576	1.47*
con	Pres12	const.	total-no-early	251	84.38	15.49	9/251	2.33*
con	Pres12	dong	total-no-early	3479	82.73	17.26	271/3479	2.15*
con	Pres17	const.	early	250	98.56	1.33	0/250	1.80*
con	Pres17	const.	early-out	250	98.55	1.31	0/250	1.74*
con	Pres17	const.	early-total	250	98.17	1.63	0/250	2.60*
con	Pres17	const.	sameday	250	97.23	2.57	0/250	3.36*
con	Pres17	const.	total	250	86.66	13.36	9/250	3.20*
con	Pres17	dong	early	3491	99.76	0.23	0/3491	4.79*
con	Pres17	dong	pooled	3491	91.48	8.50	113/3491	1.56*
con	Pres17	dong	sameday	3491	99.55	0.44	2/3491	2.17*
con	Pres22	const.	early	250	97.94	1.95	0/250	1.13*
con	Pres22	const.	early-out	250	97.70	2.16	0/250	1.45*
con	Pres22	const.	early-total	250	98.96	0.91	0/250	1.47*
con	Pres22	const.	sameday	250	99.13	0.74	0/250	1.18*
con	Pres22	const.	total	248	77.00	22.77	22/248	1.50*
con	Pres22	dong	early	3509	99.52	0.47	1/3509	1.19*
con	Pres22	dong	pooled	3509	83.97	16.03	243/3509	1.84*
con	Pres22	dong	sameday	3509	99.57	0.42	0/3509	1.73*
con	Pres25	const.	early	253	98.79	1.10	0/253	1.28*
con	Pres25	const.	early-out	253	98.11	1.74	0/253	2.16*
con	Pres25	const.	early-total	253	98.90	0.96	0/253	1.68*
con	Pres25	const.	sameday	253	98.83	1.04	0/253	1.10*
con	Pres25	const.	total	242	75.19	24.79	19/242	3.24*
con	Pres25	dong	early	3553	99.47	0.52	1/3553	6.03*
con	Pres25	dong	pooled	3553	85.13	14.88	150/3553	5.24*
con	Pres25	dong	sameday	3553	98.81	1.19	9/3553	1.67*

C.4 Latent-turnout form: complete results

Table C.8 lists the 348 latent-turnout runs of the main text’s robustness check (real data, marginal null and joint null for all 58 cells under both leader codings, short-chain configuration). Each run is paired with the observed-turnout run of the same cell, data source and leader coding. The latent-turnout form replaces the package’s leader-vote probability statement in all three classes with equation (A.4), and nothing else in the model changes.

Table C.8: Every eforensics run with the latent-turnout form, paired with the observed-turnout (package-default) run of the same cell, data source and leader coding (short-chain configuration). “Const.” is the constituency level.

Lead.	Elec.	Level	Channel	Source	n	Fraud share %		$\max \hat{R}$		Flagged	
						Obs.	Latent	Obs.	Latent	Obs.	Latent
con	Pres02	Const.	total-no-early	joint	244	12.7	6.2	1.39	2.00	9	0
con	Pres02	Const.	total-no-early	marginal	244	9.7	7.2	2.24	1.68	4	1
con	Pres02	Const.	total-no-early	real	244	12.6	6.6	1.40	2.44	9	2
con	Pres02	dong	total-no-early	joint	3515	14.7	8.0	1.42	1.55	222	66
con	Pres02	dong	total-no-early	marginal	3515	14.8	8.6	1.34	1.28	238	81
con	Pres02	dong	total-no-early	real	3515	14.2	8.4	1.02	1.75	217	52
con	Pres07	Const.	total-no-early	joint	248	5.3	3.3	2.19	1.17	2	2
con	Pres07	Const.	total-no-early	marginal	248	6.1	2.8	1.11	1.41	3	0
con	Pres07	Const.	total-no-early	real	248	5.0	3.1	1.50	1.43	6	1
con	Pres07	dong	total-no-early	joint	3576	6.3	3.6	1.36	1.17	86	19
con	Pres07	dong	total-no-early	marginal	3576	7.0	4.0	1.13	1.76	85	29
con	Pres07	dong	total-no-early	real	3576	6.6	3.8	1.42	1.17	77	27
con	Pres12	Const.	total-no-early	joint	251	20.7	17.8	3.67	3.22	22	8
con	Pres12	Const.	total-no-early	marginal	251	20.7	15.4	2.46	3.41	19	6
con	Pres12	Const.	total-no-early	real	251	22.2	13.4	3.40	3.04	19	3
con	Pres12	dong	total-no-early	joint	3479	25.6	18.3	12.64	1.14	325	228
con	Pres12	dong	total-no-early	marginal	3479	15.5	17.7	10.82	5.20	240	193
con	Pres12	dong	total-no-early	real	3479	26.1	18.4	15.44	11.71	350	144
con	Pres17	Const.	early	joint	250	1.2	1.2	1.11	1.81	0	0
con	Pres17	Const.	early	marginal	250	1.8	0.9	2.54	2.29	0	0
con	Pres17	Const.	early	real	250	1.7	0.6	3.75	1.01	0	0
con	Pres17	Const.	early-out	joint	250	0.7	0.8	1.17	1.29	0	0
con	Pres17	Const.	early-out	marginal	250	0.6	0.7	1.16	1.16	0	0
con	Pres17	Const.	early-out	real	250	0.7	0.8	1.20	1.47	0	0
con	Pres17	Const.	early-total	joint	250	0.7	0.7	1.94	1.88	0	0
con	Pres17	Const.	early-total	marginal	250	0.6	0.9	1.01	2.65	0	0
con	Pres17	Const.	early-total	real	250	0.8	1.3	1.44	1.70	0	0
con	Pres17	Const.	sameday	joint	250	0.5	0.6	1.00	1.17	0	0
con	Pres17	Const.	sameday	marginal	250	1.0	0.7	1.88	1.46	0	0
con	Pres17	Const.	sameday	real	250	0.7	0.6	1.44	1.01	0	0
con	Pres17	Const.	total	joint	250	8.4	5.7	4.55	3.09	2	2
con	Pres17	Const.	total	marginal	250	13.9	11.2	1.16	1.60	3	4
con	Pres17	Const.	total	real	250	12.8	10.4	4.03	2.55	2	3
con	Pres17	dong	early	joint	3491	0.8	1.2	7.30	4.44	0	0
con	Pres17	dong	early	marginal	3491	0.3	0.1	5.01	1.77	0	0
con	Pres17	dong	early	real	3491	0.3	0.2	9.87	8.04	1	0
con	Pres17	dong	pooled	joint	3491	7.2	3.7	7.10	3.06	51	17
con	Pres17	dong	pooled	marginal	3491	8.2	3.8	4.66	3.74	75	16
con	Pres17	dong	pooled	real	3491	8.9	3.2	10.21	2.88	132	9
con	Pres17	dong	sameday	joint	3491	0.1	0.0	1.46	1.02	0	0
con	Pres17	dong	sameday	marginal	3491	0.2	0.1	1.18	1.36	1	0
con	Pres17	dong	sameday	real	3491	0.1	0.1	12.71	1.09	0	0

Table C.8 – continued

Lead.	Elec.	Level	Channel	Source	n	Fraud share %		$\max \hat{R}$		Flagged	
						Obs.	Latent	Obs.	Latent	Obs.	Latent
con	Pres22	Const.	early	joint	250	0.9	0.7	1.26	1.11	0	0
con	Pres22	Const.	early	marginal	250	1.1	0.7	1.11	2.21	0	0
con	Pres22	Const.	early	real	250	1.1	1.3	1.06	3.41	0	0
con	Pres22	Const.	early-out	joint	250	2.4	1.2	1.78	1.53	0	0
con	Pres22	Const.	early-out	marginal	250	2.0	2.0	2.81	2.91	0	0
con	Pres22	Const.	early-out	real	250	3.7	2.1	3.04	2.17	0	0
con	Pres22	Const.	early-total	joint	250	0.7	0.6	1.51	1.19	0	0
con	Pres22	Const.	early-total	marginal	250	1.0	1.0	1.13	1.26	0	0
con	Pres22	Const.	early-total	real	250	1.0	0.8	1.09	1.11	0	0
con	Pres22	Const.	sameday	joint	250	0.7	0.5	1.14	1.02	0	0
con	Pres22	Const.	sameday	marginal	250	0.8	0.6	1.08	1.01	0	0
con	Pres22	Const.	sameday	real	250	0.7	0.6	1.17	1.00	0	0
con	Pres22	Const.	total	joint	248	23.9	28.0	3.45	6.84	19	10
con	Pres22	Const.	total	marginal	248	29.3	17.0	5.16	3.67	27	15
con	Pres22	Const.	total	real	248	32.1	23.9	1.24	7.36	30	11
con	Pres22	dong	early	joint	3509	2.0	0.1	12.72	1.17	0	0
con	Pres22	dong	early	marginal	3509	0.7	2.3	17.23	9.33	5	1
con	Pres22	dong	early	real	3509	0.5	0.2	1.54	16.20	0	0
con	Pres22	dong	pooled	joint	3509	22.0	15.5	13.52	8.87	248	178
con	Pres22	dong	pooled	marginal	3509	21.8	16.1	9.10	3.61	302	179
con	Pres22	dong	pooled	real	3509	21.4	9.8	8.97	9.14	272	68
con	Pres22	dong	sameday	joint	3509	0.4	0.1	1.56	1.57	1	0
con	Pres22	dong	sameday	marginal	3509	0.3	0.1	1.24	1.08	0	0
con	Pres22	dong	sameday	real	3509	0.2	0.1	1.38	1.24	0	0
con	Pres25	Const.	early	joint	253	1.1	0.8	1.63	1.37	0	0
con	Pres25	Const.	early	marginal	253	0.5	0.7	1.03	1.87	0	0
con	Pres25	Const.	early	real	253	1.4	2.0	1.71	3.42	0	0
con	Pres25	Const.	early-out	joint	253	1.2	0.9	2.24	1.69	0	0
con	Pres25	Const.	early-out	marginal	253	1.5	0.9	1.53	1.26	0	0
con	Pres25	Const.	early-out	real	253	1.5	0.7	1.72	2.81	0	0
con	Pres25	Const.	early-total	joint	253	0.6	0.6	1.16	1.16	0	0
con	Pres25	Const.	early-total	marginal	253	0.6	0.7	1.01	2.98	0	0
con	Pres25	Const.	early-total	real	253	0.6	0.5	1.14	1.03	0	0
con	Pres25	Const.	sameday	joint	253	0.8	0.8	1.08	1.19	0	0
con	Pres25	Const.	sameday	marginal	253	0.9	0.8	1.23	1.09	0	0
con	Pres25	Const.	sameday	real	253	0.8	0.6	1.27	1.09	0	0
con	Pres25	Const.	total	joint	242	30.7	40.0	3.02	8.76	25	27
con	Pres25	Const.	total	marginal	242	29.1	15.6	2.00	3.93	25	8
con	Pres25	Const.	total	real	242	28.3	12.2	4.42	8.20	19	10
con	Pres25	dong	early	joint	3553	0.2	0.1	1.43	7.11	0	0
con	Pres25	dong	early	marginal	3553	0.8	0.1	8.80	8.92	0	0
con	Pres25	dong	early	real	3553	0.2	0.1	1.46	1.18	0	0
con	Pres25	dong	pooled	joint	3553	19.9	11.2	1.57	19.14	264	37
con	Pres25	dong	pooled	marginal	3553	19.4	13.4	10.52	12.66	210	95
con	Pres25	dong	pooled	real	3553	19.6	14.1	8.29	9.20	266	149
con	Pres25	dong	sameday	joint	3553	0.9	0.4	1.47	1.18	8	0
con	Pres25	dong	sameday	marginal	3553	0.9	0.4	1.41	1.36	3	2
con	Pres25	dong	sameday	real	3553	0.8	0.3	1.17	1.43	2	0
con	Gen08	Const.	total-no-early	joint	198	0.7	0.7	1.03	1.00	0	0
con	Gen08	Const.	total-no-early	marginal	198	0.7	0.7	1.01	1.01	0	0
con	Gen08	Const.	total-no-early	real	198	0.7	0.7	1.01	1.01	0	0
con	Gen08	dong	total-no-early	joint	2733	0.1	0.1	1.79	1.29	0	1
con	Gen08	dong	total-no-early	marginal	2733	0.3	0.2	2.04	6.74	1	0
con	Gen08	dong	total-no-early	real	2733	0.2	0.2	1.24	1.15	3	2
con	Gen12	Const.	total-no-early	joint	194	0.7	0.7	1.01	1.01	0	0

Table C.8 – continued

Lead.	Elec.	Level	Channel	Source	n	Fraud share %		$\max \hat{R}$		Flagged	
						Obs.	Latent	Obs.	Latent	Obs.	Latent
con	Gen12	Const.	total-no-early	marginal	194	1.0	0.8	1.39	1.17	0	0
con	Gen12	Const.	total-no-early	real	194	0.7	0.7	1.02	1.01	0	0
con	Gen12	dong	total-no-early	joint	2656	1.3	0.6	1.06	1.13	10	2
con	Gen12	dong	total-no-early	marginal	2656	1.8	1.0	2.69	1.57	14	4
con	Gen12	dong	total-no-early	real	2656	1.8	0.9	1.64	2.32	18	4
con	Gen16	Const.	early	joint	229	1.3	0.8	1.27	3.38	0	0
con	Gen16	Const.	early	marginal	229	2.3	1.3	1.80	2.01	0	0
con	Gen16	Const.	early	real	229	1.7	0.8	1.73	1.16	0	0
con	Gen16	Const.	early-out	joint	229	0.6	0.6	1.00	1.02	0	0
con	Gen16	Const.	early-out	marginal	229	0.6	0.7	1.01	1.16	0	0
con	Gen16	Const.	early-out	real	229	0.6	0.6	1.01	1.01	0	0
con	Gen16	Const.	early-total	joint	229	2.2	1.0	2.53	2.63	0	0
con	Gen16	Const.	early-total	marginal	229	2.8	1.4	1.34	1.64	0	0
con	Gen16	Const.	early-total	real	229	2.9	1.8	1.35	2.40	0	0
con	Gen16	Const.	sameday	joint	229	0.6	0.6	1.01	1.03	0	0
con	Gen16	Const.	sameday	marginal	229	0.6	0.6	1.02	1.01	0	0
con	Gen16	Const.	sameday	real	229	0.6	0.6	1.02	1.01	0	0
con	Gen16	Const.	total	joint	229	0.7	0.6	1.22	1.00	0	0
con	Gen16	Const.	total	marginal	229	1.1	0.7	2.21	1.15	0	0
con	Gen16	Const.	total	real	229	0.9	0.7	1.42	1.14	0	0
con	Gen16	dong	early	joint	3058	0.7	0.4	3.02	2.61	0	0
con	Gen16	dong	early	marginal	3058	0.2	0.6	10.44	5.36	0	0
con	Gen16	dong	early	real	3058	0.8	0.1	5.13	4.03	0	0
con	Gen16	dong	pooled	joint	3058	1.2	0.6	1.55	1.02	9	1
con	Gen16	dong	pooled	marginal	3058	1.8	1.1	2.25	1.16	13	11
con	Gen16	dong	pooled	real	3058	1.7	0.8	1.34	1.12	19	2
con	Gen16	dong	sameday	joint	3058	0.1	0.1	2.03	1.42	0	0
con	Gen16	dong	sameday	marginal	3058	0.2	0.1	1.38	1.51	1	1
con	Gen16	dong	sameday	real	3058	0.1	0.1	1.74	1.17	0	0
con	Gen20	Const.	early	joint	253	1.1	0.7	1.27	1.88	0	0
con	Gen20	Const.	early	marginal	253	1.3	1.8	1.74	3.60	0	0
con	Gen20	Const.	early	real	253	1.0	0.7	1.52	1.15	0	0
con	Gen20	Const.	early-out	joint	253	0.7	0.6	1.57	1.19	0	0
con	Gen20	Const.	early-out	marginal	253	0.7	0.7	1.12	1.57	0	0
con	Gen20	Const.	early-out	real	253	0.9	0.6	1.51	1.01	0	0
con	Gen20	Const.	early-total	joint	253	0.6	0.6	1.17	1.20	0	0
con	Gen20	Const.	early-total	marginal	253	1.1	1.3	2.08	2.18	0	0
con	Gen20	Const.	early-total	real	253	0.7	0.6	2.18	1.18	0	0
con	Gen20	Const.	sameday	joint	253	0.5	0.6	1.01	1.02	0	0
con	Gen20	Const.	sameday	marginal	253	0.6	0.6	1.04	1.01	0	0
con	Gen20	Const.	sameday	real	253	0.6	0.6	1.01	1.02	0	0
con	Gen20	Const.	total	joint	253	9.9	4.2	1.38	1.04	8	2
con	Gen20	Const.	total	marginal	253	9.0	3.0	1.85	1.10	3	1
con	Gen20	Const.	total	real	253	7.9	3.5	3.49	1.32	2	0
con	Gen20	dong	early	joint	3480	0.9	0.2	1.37	4.34	0	0
con	Gen20	dong	early	marginal	3480	0.5	0.3	4.47	4.25	0	0
con	Gen20	dong	early	real	3480	0.6	0.2	2.06	3.29	1	1
con	Gen20	dong	pooled	joint	3480	7.2	3.6	1.35	2.16	92	20
con	Gen20	dong	pooled	marginal	3480	6.6	3.6	1.18	1.50	78	28
con	Gen20	dong	pooled	real	3480	6.0	3.3	1.12	1.23	71	20
con	Gen20	dong	sameday	joint	3480	0.2	0.0	1.53	1.00	0	0
con	Gen20	dong	sameday	marginal	3480	0.2	0.1	1.37	1.14	0	0
con	Gen20	dong	sameday	real	3480	0.1	0.0	1.69	1.00	0	0
con	Gen24	Const.	early	joint	246	1.4	0.8	2.96	2.45	0	0
con	Gen24	Const.	early	marginal	246	1.7	0.8	1.61	1.40	0	0

Table C.8 – continued

Lead.	Elec.	Level	Channel	Source	n	Fraud share %		$\max \hat{R}$		Flagged	
						Obs.	Latent	Obs.	Latent	Obs.	Latent
con	Gen24	Const.	early	real	246	0.9	0.9	2.27	2.39	0	0
con	Gen24	Const.	early-out	joint	246	1.1	1.1	1.46	1.12	0	0
con	Gen24	Const.	early-out	marginal	246	1.6	0.8	1.67	1.45	0	0
con	Gen24	Const.	early-out	real	246	1.2	1.2	1.54	1.92	0	0
con	Gen24	Const.	early-total	joint	246	0.7	0.6	1.14	1.02	0	0
con	Gen24	Const.	early-total	marginal	246	0.6	0.6	1.31	1.03	0	0
con	Gen24	Const.	early-total	real	246	0.8	0.7	1.13	1.54	0	0
con	Gen24	Const.	sameday	joint	246	0.7	0.5	1.14	1.01	0	0
con	Gen24	Const.	sameday	marginal	246	0.7	0.6	1.20	1.03	0	0
con	Gen24	Const.	sameday	real	246	0.6	0.5	1.01	1.05	0	0
con	Gen24	Const.	total	joint	246	11.7	4.9	1.18	2.36	6	1
con	Gen24	Const.	total	marginal	246	11.5	5.9	1.75	2.47	1	0
con	Gen24	Const.	total	real	246	11.5	5.7	1.59	3.12	6	0
con	Gen24	dong	early	joint	2739	0.7	0.1	1.18	8.39	0	0
con	Gen24	dong	early	marginal	2739	0.7	0.1	1.43	14.75	1	0
con	Gen24	dong	early	real	2739	0.5	1.0	14.76	6.18	0	0
con	Gen24	dong	pooled	joint	2739	6.1	2.8	1.72	2.55	45	11
con	Gen24	dong	pooled	marginal	2739	6.6	3.8	1.88	5.85	48	14
con	Gen24	dong	pooled	real	2739	6.7	3.2	2.02	1.50	61	16
con	Gen24	dong	sameday	joint	2739	0.3	0.1	6.35	1.05	0	0
con	Gen24	dong	sameday	marginal	2739	0.2	0.1	1.47	1.01	0	0
con	Gen24	dong	sameday	real	2739	0.1	0.1	1.06	1.04	0	0
dem	Pres02	Const.	total-no-early	joint	244	12.0	9.6	1.60	2.88	16	6
dem	Pres02	Const.	total-no-early	marginal	244	13.1	11.5	1.46	1.20	20	16
dem	Pres02	Const.	total-no-early	real	244	14.3	10.3	1.89	1.29	17	10
dem	Pres02	dong	total-no-early	joint	3515	12.5	10.0	6.27	1.66	218	153
dem	Pres02	dong	total-no-early	marginal	3515	15.1	11.3	4.38	1.96	316	192
dem	Pres02	dong	total-no-early	real	3515	13.9	10.7	2.95	1.22	304	163
dem	Pres07	Const.	total-no-early	joint	248	9.0	4.5	1.69	1.80	8	2
dem	Pres07	Const.	total-no-early	marginal	248	6.3	5.0	1.30	1.59	8	6
dem	Pres07	Const.	total-no-early	real	248	6.7	4.0	1.61	1.51	8	3
dem	Pres07	dong	total-no-early	joint	3576	7.3	4.2	1.52	4.36	136	45
dem	Pres07	dong	total-no-early	marginal	3576	6.0	4.5	4.78	2.03	114	50
dem	Pres07	dong	total-no-early	real	3576	7.1	4.2	1.97	1.08	154	49
dem	Pres12	Const.	total-no-early	joint	251	19.1	12.4	1.66	1.68	14	14
dem	Pres12	Const.	total-no-early	marginal	251	18.2	13.3	1.22	4.65	18	6
dem	Pres12	Const.	total-no-early	real	251	16.8	17.1	2.43	3.14	17	11
dem	Pres12	dong	total-no-early	joint	3479	17.6	13.0	3.47	6.99	325	160
dem	Pres12	dong	total-no-early	marginal	3479	17.6	13.4	8.33	7.47	264	192
dem	Pres12	dong	total-no-early	real	3479	17.3	11.0	9.90	8.64	275	118
dem	Pres17	Const.	early	joint	250	2.5	0.7	1.24	1.18	0	0
dem	Pres17	Const.	early	marginal	250	2.8	0.9	1.47	1.21	0	0
dem	Pres17	Const.	early	real	250	3.0	0.8	2.03	1.43	0	0
dem	Pres17	Const.	early-out	joint	250	2.5	0.6	2.43	1.01	0	0
dem	Pres17	Const.	early-out	marginal	250	0.9	1.4	4.30	1.91	0	0
dem	Pres17	Const.	early-out	real	250	0.9	1.4	3.92	1.44	0	0
dem	Pres17	Const.	early-total	joint	250	0.8	0.6	2.82	1.23	0	0
dem	Pres17	Const.	early-total	marginal	250	0.8	0.8	1.38	1.81	0	0
dem	Pres17	Const.	early-total	real	250	0.7	0.7	3.98	4.81	0	0
dem	Pres17	Const.	sameday	joint	250	0.6	0.6	1.00	1.02	0	0
dem	Pres17	Const.	sameday	marginal	250	0.5	0.6	1.01	1.01	0	0
dem	Pres17	Const.	sameday	real	250	0.6	0.5	1.01	1.03	0	0
dem	Pres17	Const.	total	joint	250	6.0	17.6	7.52	6.49	2	12
dem	Pres17	Const.	total	marginal	250	23.0	18.1	1.36	4.84	15	4
dem	Pres17	Const.	total	real	250	22.2	16.8	2.03	3.60	5	5

Table C.8 – continued

Lead.	Elec.	Level	Channel	Source	n	Fraud share %		$\max \hat{R}$		Flagged	
						Obs.	Latent	Obs.	Latent	Obs.	Latent
dem	Pres17	dong	early	joint	3491	0.8	0.8	6.97	5.70	0	0
dem	Pres17	dong	early	marginal	3491	0.8	1.1	6.00	6.36	0	0
dem	Pres17	dong	early	real	3491	1.3	0.6	5.42	7.97	0	0
dem	Pres17	dong	pooled	joint	3491	8.7	6.3	7.17	1.13	53	39
dem	Pres17	dong	pooled	marginal	3491	10.5	6.7	5.69	8.96	67	26
dem	Pres17	dong	pooled	real	3491	10.3	6.7	7.09	1.68	56	39
dem	Pres17	dong	sameday	joint	3491	0.1	0.1	1.25	1.20	0	0
dem	Pres17	dong	sameday	marginal	3491	0.1	0.1	1.54	1.14	0	0
dem	Pres17	dong	sameday	real	3491	0.1	0.0	1.27	1.20	0	0
dem	Pres22	Const.	early	joint	250	2.2	0.9	1.49	2.62	0	0
dem	Pres22	Const.	early	marginal	250	1.9	0.8	1.47	4.59	0	0
dem	Pres22	Const.	early	real	250	1.7	1.2	1.05	1.75	1	0
dem	Pres22	Const.	early-out	joint	250	5.1	1.7	3.36	1.40	1	0
dem	Pres22	Const.	early-out	marginal	250	4.5	0.8	2.47	1.72	0	0
dem	Pres22	Const.	early-out	real	250	4.0	1.1	2.11	1.51	1	0
dem	Pres22	Const.	early-total	joint	250	5.4	3.2	1.22	1.98	4	2
dem	Pres22	Const.	early-total	marginal	250	5.5	3.2	1.96	1.75	7	0
dem	Pres22	Const.	early-total	real	250	5.7	2.7	1.11	1.49	9	0
dem	Pres22	Const.	sameday	joint	250	1.0	0.5	1.80	1.01	0	0
dem	Pres22	Const.	sameday	marginal	250	1.6	0.6	1.80	1.01	0	0
dem	Pres22	Const.	sameday	real	250	2.1	0.6	1.33	1.02	0	0
dem	Pres22	Const.	total	joint	248	27.0	23.9	2.71	4.09	28	21
dem	Pres22	Const.	total	marginal	248	29.5	28.5	1.25	3.66	33	30
dem	Pres22	Const.	total	real	248	25.9	19.2	4.30	6.52	23	21
dem	Pres22	dong	early	joint	3509	2.3	0.9	1.29	2.53	16	6
dem	Pres22	dong	early	marginal	3509	3.0	1.1	12.10	10.98	39	11
dem	Pres22	dong	early	real	3509	3.2	0.9	17.34	20.37	63	8
dem	Pres22	dong	pooled	joint	3509	19.0	12.8	2.24	1.44	405	207
dem	Pres22	dong	pooled	marginal	3509	18.2	13.6	9.31	4.75	329	187
dem	Pres22	dong	pooled	real	3509	18.7	15.3	11.93	11.46	414	248
dem	Pres22	dong	sameday	joint	3509	0.6	0.1	1.90	5.32	0	0
dem	Pres22	dong	sameday	marginal	3509	0.7	0.1	1.65	1.08	0	0
dem	Pres22	dong	sameday	real	3509	0.6	0.1	1.73	4.18	0	0
dem	Pres25	Const.	early	joint	253	4.6	2.5	3.14	1.70	0	0
dem	Pres25	Const.	early	marginal	253	6.7	1.9	1.70	1.16	1	1
dem	Pres25	Const.	early	real	253	5.1	2.4	1.71	4.51	1	1
dem	Pres25	Const.	early-out	joint	253	8.2	2.1	3.07	1.30	0	0
dem	Pres25	Const.	early-out	marginal	253	6.3	2.5	3.35	1.75	0	0
dem	Pres25	Const.	early-out	real	253	5.2	1.8	2.53	2.39	0	0
dem	Pres25	Const.	early-total	joint	253	9.2	7.0	2.22	1.16	14	7
dem	Pres25	Const.	early-total	marginal	253	7.5	5.3	1.15	2.29	10	3
dem	Pres25	Const.	early-total	real	253	7.8	5.5	1.70	1.20	7	7
dem	Pres25	Const.	sameday	joint	253	1.2	0.6	1.48	1.03	0	0
dem	Pres25	Const.	sameday	marginal	253	1.5	0.6	1.60	1.04	0	0
dem	Pres25	Const.	sameday	real	253	1.4	0.6	1.04	1.12	0	0
dem	Pres25	Const.	total	joint	242	31.1	27.2	2.26	5.15	37	24
dem	Pres25	Const.	total	marginal	242	29.9	25.0	2.73	2.21	31	25
dem	Pres25	Const.	total	real	242	27.2	24.3	2.79	3.18	29	26
dem	Pres25	dong	early	joint	3553	6.9	1.9	11.87	1.28	145	10
dem	Pres25	dong	early	marginal	3553	9.0	2.3	9.92	1.21	210	19
dem	Pres25	dong	early	real	3553	8.4	2.3	5.98	7.12	179	20
dem	Pres25	dong	pooled	joint	3553	22.1	15.1	1.93	4.58	445	230
dem	Pres25	dong	pooled	marginal	3553	21.8	16.1	14.12	1.64	350	269
dem	Pres25	dong	pooled	real	3553	21.9	14.8	2.21	11.67	463	159
dem	Pres25	dong	sameday	joint	3553	0.7	0.1	1.56	2.75	1	0

Table C.8 – continued

Lead.	Elec.	Level	Channel	Source	n	Fraud share %		$\max \hat{R}$		Flagged	
						Obs.	Latent	Obs.	Latent	Obs.	Latent
dem	Pres25	dong	sameday	marginal	3553	0.5	0.1	1.45	1.00	1	1
dem	Pres25	dong	sameday	real	3553	0.6	0.0	1.73	1.00	0	0
dem	Gen08	Const.	total-no-early	joint	198	0.7	0.7	1.02	1.02	0	0
dem	Gen08	Const.	total-no-early	marginal	198	0.7	0.7	1.01	1.01	0	0
dem	Gen08	Const.	total-no-early	real	198	0.7	0.7	1.00	1.01	0	0
dem	Gen08	dong	total-no-early	joint	2733	0.3	0.1	1.50	1.04	0	0
dem	Gen08	dong	total-no-early	marginal	2733	0.8	0.4	1.68	1.37	5	0
dem	Gen08	dong	total-no-early	real	2733	0.6	0.4	1.32	1.22	4	2
dem	Gen12	Const.	total-no-early	joint	209	0.7	0.7	1.06	1.01	0	0
dem	Gen12	Const.	total-no-early	marginal	209	0.7	0.7	1.00	1.01	0	0
dem	Gen12	Const.	total-no-early	real	209	0.7	0.7	1.02	1.01	0	0
dem	Gen12	dong	total-no-early	joint	2923	0.3	0.2	1.41	1.60	0	0
dem	Gen12	dong	total-no-early	marginal	2923	0.6	0.3	1.12	1.50	4	2
dem	Gen12	dong	total-no-early	real	2923	0.5	0.2	1.84	1.38	4	0
dem	Gen16	Const.	early	joint	234	1.0	0.8	1.29	2.73	0	0
dem	Gen16	Const.	early	marginal	234	1.6	0.9	1.04	1.39	0	0
dem	Gen16	Const.	early	real	234	1.9	0.9	2.32	1.41	0	0
dem	Gen16	Const.	early-out	joint	234	0.6	0.6	1.02	1.01	0	0
dem	Gen16	Const.	early-out	marginal	234	0.8	0.6	1.85	1.02	0	0
dem	Gen16	Const.	early-out	real	234	0.6	0.6	1.01	1.02	0	0
dem	Gen16	Const.	early-total	joint	234	2.9	1.5	1.26	1.84	0	0
dem	Gen16	Const.	early-total	marginal	234	3.7	1.1	1.21	1.09	0	0
dem	Gen16	Const.	early-total	real	234	2.8	0.9	2.04	1.73	0	0
dem	Gen16	Const.	sameday	joint	234	0.6	0.6	1.02	1.01	0	0
dem	Gen16	Const.	sameday	marginal	234	0.6	0.6	1.01	1.02	0	0
dem	Gen16	Const.	sameday	real	234	0.6	0.6	1.01	1.01	0	0
dem	Gen16	Const.	total	joint	234	1.4	0.7	1.82	1.21	1	0
dem	Gen16	Const.	total	marginal	234	2.1	0.7	1.26	1.16	1	0
dem	Gen16	Const.	total	real	234	1.0	0.6	1.54	1.01	0	0
dem	Gen16	dong	early	joint	3135	0.2	0.1	7.68	3.91	0	0
dem	Gen16	dong	early	marginal	3135	0.9	0.1	3.35	1.55	0	0
dem	Gen16	dong	early	real	3135	0.5	0.4	3.13	4.80	0	0
dem	Gen16	dong	pooled	joint	3135	0.3	0.1	1.26	1.91	3	0
dem	Gen16	dong	pooled	marginal	3135	0.6	0.3	1.64	1.60	2	0
dem	Gen16	dong	pooled	real	3135	0.3	0.2	1.34	1.48	2	0
dem	Gen16	dong	sameday	joint	3135	0.0	0.0	1.01	1.01	0	0
dem	Gen16	dong	sameday	marginal	3135	0.0	0.0	1.01	1.01	0	0
dem	Gen16	dong	sameday	real	3135	0.1	0.0	1.12	1.01	0	0
dem	Gen20	Const.	early	joint	253	5.6	1.7	2.23	4.95	0	0
dem	Gen20	Const.	early	marginal	253	5.1	0.8	1.49	1.14	0	0
dem	Gen20	Const.	early	real	253	6.2	1.2	1.25	2.23	0	0
dem	Gen20	Const.	early-out	joint	253	2.4	1.3	2.46	1.51	0	0
dem	Gen20	Const.	early-out	marginal	253	2.1	1.0	2.22	1.49	0	0
dem	Gen20	Const.	early-out	real	253	1.9	1.6	2.14	1.72	0	0
dem	Gen20	Const.	early-total	joint	253	2.3	1.6	1.40	2.92	0	0
dem	Gen20	Const.	early-total	marginal	253	3.1	0.6	1.12	1.19	0	0
dem	Gen20	Const.	early-total	real	253	2.9	0.7	2.33	1.44	0	0
dem	Gen20	Const.	sameday	joint	253	0.6	0.6	1.15	1.08	0	0
dem	Gen20	Const.	sameday	marginal	253	0.7	0.5	1.50	1.02	0	0
dem	Gen20	Const.	sameday	real	253	0.5	0.5	1.01	1.01	0	0
dem	Gen20	Const.	total	joint	253	13.0	6.8	1.49	1.19	9	3
dem	Gen20	Const.	total	marginal	253	11.3	7.0	2.53	3.41	4	2
dem	Gen20	Const.	total	real	253	13.2	6.8	1.26	1.38	7	1
dem	Gen20	dong	early	joint	3480	2.0	0.8	1.29	12.72	2	0
dem	Gen20	dong	early	marginal	3480	2.6	0.9	1.18	12.49	3	0

Table C.8 – continued

Lead.	Elec.	Level	Channel	Source	n	Fraud share %		$\max \hat{R}$		Flagged	
						Obs.	Latent	Obs.	Latent	Obs.	Latent
dem	Gen20	dong	early	real	3480	2.8	1.0	1.83	13.05	8	0
dem	Gen20	dong	pooled	joint	3480	5.3	2.5	1.34	1.68	47	2
dem	Gen20	dong	pooled	marginal	3480	5.4	2.8	1.21	1.79	53	9
dem	Gen20	dong	pooled	real	3480	4.9	2.7	1.45	1.02	40	12
dem	Gen20	dong	sameday	joint	3480	0.1	0.0	1.56	1.02	1	0
dem	Gen20	dong	sameday	marginal	3480	0.2	0.0	1.62	1.01	0	0
dem	Gen20	dong	sameday	real	3480	0.1	0.0	1.92	1.01	0	0
dem	Gen24	Const.	early	joint	246	6.4	0.7	4.38	3.68	0	0
dem	Gen24	Const.	early	marginal	246	4.8	0.8	1.56	2.92	1	0
dem	Gen24	Const.	early	real	246	3.3	0.6	2.59	1.01	0	0
dem	Gen24	Const.	early-out	joint	246	2.5	0.9	2.12	1.41	0	0
dem	Gen24	Const.	early-out	marginal	246	2.7	1.4	2.83	2.56	0	0
dem	Gen24	Const.	early-out	real	246	1.4	1.2	1.90	2.08	0	0
dem	Gen24	Const.	early-total	joint	246	1.2	0.6	1.72	1.01	0	0
dem	Gen24	Const.	early-total	marginal	246	1.1	0.6	1.10	1.01	0	0
dem	Gen24	Const.	early-total	real	246	0.8	0.6	1.40	1.02	0	0
dem	Gen24	Const.	sameday	joint	246	1.2	0.6	1.27	1.01	0	0
dem	Gen24	Const.	sameday	marginal	246	1.3	0.6	1.42	1.02	0	0
dem	Gen24	Const.	sameday	real	246	1.2	0.5	1.10	1.01	0	0
dem	Gen24	Const.	total	joint	246	15.3	6.0	2.37	3.41	8	3
dem	Gen24	Const.	total	marginal	246	14.0	9.2	2.72	3.04	9	5
dem	Gen24	Const.	total	real	246	16.6	12.6	1.13	4.22	10	9
dem	Gen24	dong	early	joint	2739	1.9	0.1	6.55	1.20	0	0
dem	Gen24	dong	early	marginal	2739	2.5	0.5	2.30	12.06	3	0
dem	Gen24	dong	early	real	2739	3.8	0.1	7.32	16.86	5	0
dem	Gen24	dong	pooled	joint	2739	7.0	3.7	1.26	2.59	61	16
dem	Gen24	dong	pooled	marginal	2739	7.7	3.8	1.24	2.56	78	23
dem	Gen24	dong	pooled	real	2739	6.9	3.5	2.38	1.74	60	16
dem	Gen24	dong	sameday	joint	2739	0.7	0.1	1.19	1.01	0	0
dem	Gen24	dong	sameday	marginal	2739	0.7	0.1	1.28	1.00	0	0
dem	Gen24	dong	sameday	real	2739	0.6	0.1	1.75	1.00	0	0

Table C.9 lists the latent-turnout model-consistent self-null under both leader codings. It is built in the same way as the observed-turnout self-null of Table C.7, except that both the simulation of the synthetic data and the two model fits use the latent-turnout form.

In both self-nulls, a simulated unit whose leader votes exceed its votes cast, which the package rejects as invalid data (see Appendix A), has its abstentions and leader votes redrawn together until they are feasible.

Table C.9: Latent-turnout form: model-consistent self-null results: π forced to $(1, 0, 0)$ using the qbl model’s own fitted class-1 (no-fraud) data-generating equations, short-chain configuration, Democratic- and conservative-coded leader. N is the unit count; “Flagged” is the number of units receiving a non-clean MAP classification. The synthetic data are simulated from, and refit with, the latent-turnout form.

Lead.	Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
dem	Gen08	const.	total-no-early	198	99.14	0.71	0/198	1.01
dem	Gen08	dong	total-no-early	2733	99.80	0.19	0/2733	2.02*
dem	Gen12	const.	total-no-early	209	99.20	0.65	0/209	1.01

Table C.9 – continued

Lead.	Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
dem	Gen12	dong	total-no-early	2923	99.63	0.36	1/2923	1.05*
dem	Gen16	const.	early	234	98.50	1.32	0/234	2.14*
dem	Gen16	const.	early-out	234	98.90	0.99	0/234	1.22*
dem	Gen16	const.	early-total	234	98.12	1.75	0/234	1.15*
dem	Gen16	const.	sameday	234	99.31	0.57	0/234	1.01
dem	Gen16	const.	total	234	98.78	1.05	0/234	1.42*
dem	Gen16	dong	early	3135	99.66	0.32	0/3135	5.24*
dem	Gen16	dong	pooled	3135	99.86	0.13	0/3135	1.57*
dem	Gen16	dong	sameday	3135	99.95	0.05	0/3135	1.00
dem	Gen20	const.	early	253	97.66	2.17	0/253	2.21*
dem	Gen20	const.	early-out	253	98.77	1.10	0/253	1.36*
dem	Gen20	const.	early-total	253	98.01	1.82	0/253	4.49*
dem	Gen20	const.	sameday	253	99.32	0.56	0/253	1.01
dem	Gen20	const.	total	253	92.98	6.85	2/253	2.92*
dem	Gen20	dong	early	3480	99.88	0.11	0/3480	2.95*
dem	Gen20	dong	pooled	3480	97.74	2.25	8/3480	2.04*
dem	Gen20	dong	sameday	3480	99.95	0.04	0/3480	1.01
dem	Gen24	const.	early	246	98.94	0.89	0/246	1.59*
dem	Gen24	const.	early-out	246	98.80	1.03	0/246	1.97*
dem	Gen24	const.	early-total	246	99.35	0.55	0/246	1.01
dem	Gen24	const.	sameday	246	99.31	0.59	0/246	1.07*
dem	Gen24	const.	total	246	89.42	10.48	5/246	2.06*
dem	Gen24	dong	early	2739	99.92	0.06	0/2739	3.55*
dem	Gen24	dong	pooled	2739	96.90	3.08	6/2739	1.99*
dem	Gen24	dong	sameday	2739	99.94	0.05	0/2739	1.01
dem	Pres02	const.	total-no-early	244	89.87	10.04	3/244	1.36*
dem	Pres02	dong	total-no-early	3515	84.72	15.25	202/3515	2.92*
dem	Pres07	const.	total-no-early	248	98.75	1.13	0/248	1.47*
dem	Pres07	dong	total-no-early	3576	98.69	1.30	12/3576	1.24*
dem	Pres12	const.	total-no-early	251	89.95	9.94	2/251	4.80*
dem	Pres12	dong	total-no-early	3479	88.55	11.44	116/3479	1.82*
dem	Pres17	const.	early	250	98.48	1.35	0/250	3.07*
dem	Pres17	const.	early-out	250	98.26	1.56	0/250	2.74*
dem	Pres17	const.	early-total	250	99.15	0.67	0/250	1.58*
dem	Pres17	const.	sameday	250	99.32	0.57	0/250	1.00
dem	Pres17	const.	total	250	88.06	11.83	2/250	2.91*
dem	Pres17	dong	early	3491	98.93	1.06	2/3491	18.27*
dem	Pres17	dong	pooled	3491	96.17	3.81	10/3491	1.68*
dem	Pres17	dong	sameday	3491	99.94	0.05	0/3491	1.19*
dem	Pres22	const.	early	250	99.35	0.53	0/250	1.01
dem	Pres22	const.	early-out	250	96.87	3.06	1/250	3.22*
dem	Pres22	const.	early-total	250	99.25	0.62	0/250	1.18*
dem	Pres22	const.	sameday	250	99.30	0.57	0/250	1.04
dem	Pres22	const.	total	248	78.90	20.85	25/248	1.66*
dem	Pres22	dong	early	3509	99.92	0.07	0/3509	1.09*
dem	Pres22	dong	pooled	3509	88.84	11.16	74/3509	4.64*
dem	Pres22	dong	sameday	3509	99.94	0.05	0/3509	1.22*
dem	Pres25	const.	early	253	98.82	1.05	0/253	1.10*
dem	Pres25	const.	early-out	253	98.80	1.05	0/253	1.71*

Table C.9 – continued

Lead.	Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
dem	Pres25	const.	early-total	253	98.68	1.23	1/253	1.39*
dem	Pres25	const.	sameday	253	99.24	0.65	0/253	1.23*
dem	Pres25	const.	total	242	80.86	18.87	12/242	3.34*
dem	Pres25	dong	early	3553	99.82	0.17	0/3553	13.51*
dem	Pres25	dong	pooled	3553	85.11	14.88	186/3553	1.59*
dem	Pres25	dong	sameday	3553	99.86	0.13	0/3553	1.50*
con	Gen08	const.	total-no-early	198	98.90	0.96	0/198	1.11*
con	Gen08	dong	total-no-early	2733	99.82	0.17	0/2733	3.19*
con	Gen12	const.	total-no-early	194	99.13	0.71	0/194	1.00
con	Gen12	dong	total-no-early	2656	99.19	0.80	0/2656	1.45*
con	Gen16	const.	early	229	97.86	1.93	0/229	3.03*
con	Gen16	const.	early-out	229	99.25	0.62	0/229	1.01
con	Gen16	const.	early-total	229	98.25	1.64	0/229	1.34*
con	Gen16	const.	sameday	229	99.26	0.61	0/229	1.01
con	Gen16	const.	total	229	98.43	1.45	0/229	1.18*
con	Gen16	dong	early	3058	99.67	0.32	0/3058	5.26*
con	Gen16	dong	pooled	3058	98.86	1.13	3/3058	1.23*
con	Gen16	dong	sameday	3058	99.88	0.11	0/3058	1.17*
con	Gen20	const.	early	253	99.33	0.54	0/253	1.02
con	Gen20	const.	early-out	253	98.78	1.02	0/253	2.52*
con	Gen20	const.	early-total	253	99.35	0.55	0/253	1.01
con	Gen20	const.	sameday	253	99.33	0.54	0/253	1.01
con	Gen20	const.	total	253	96.85	2.98	0/253	1.96*
con	Gen20	dong	early	3480	99.56	0.42	0/3480	7.43*
con	Gen20	dong	pooled	3480	96.72	3.27	23/3480	1.94*
con	Gen20	dong	sameday	3480	99.93	0.06	0/3480	1.37*
con	Gen24	const.	early	246	97.12	2.67	0/246	3.54*
con	Gen24	const.	early-out	246	98.96	0.88	0/246	2.36*
con	Gen24	const.	early-total	246	99.21	0.66	0/246	1.14*
con	Gen24	const.	sameday	246	99.27	0.61	0/246	1.01
con	Gen24	const.	total	246	93.72	6.16	3/246	1.52*
con	Gen24	dong	early	2739	99.90	0.08	0/2739	5.75*
con	Gen24	dong	pooled	2739	97.24	2.75	10/2739	1.04
con	Gen24	dong	sameday	2739	99.94	0.05	0/2739	1.02
con	Pres02	const.	total-no-early	244	92.79	7.04	4/244	1.43*
con	Pres02	dong	total-no-early	3515	93.28	6.72	59/3515	1.59*
con	Pres07	const.	total-no-early	248	96.91	2.96	0/248	1.61*
con	Pres07	dong	total-no-early	3576	97.05	2.94	15/3576	1.35*
con	Pres12	const.	total-no-early	251	89.39	10.47	10/251	1.37*
con	Pres12	dong	total-no-early	3479	84.85	15.11	239/3479	7.31*
con	Pres17	const.	early	250	98.41	1.40	0/250	2.62*
con	Pres17	const.	early-out	250	98.94	0.93	0/250	1.42*
con	Pres17	const.	early-total	250	97.73	2.00	0/250	3.25*
con	Pres17	const.	sameday	250	99.25	0.62	0/250	1.20*
con	Pres17	const.	total	250	88.53	11.37	5/250	1.65*
con	Pres17	dong	early	3491	99.71	0.28	0/3491	3.49*

Table C.9 – continued

Lead.	Elec.	Level	Channel	N	π_1 (no fraud) %	Fraud share %	Flagged	$\hat{R}$
con	Pres17	dong	pooled	3491	95.73	4.26	38/3491	1.17*
con	Pres17	dong	sameday	3491	99.75	0.24	0/3491	1.21*
con	Pres22	const.	early	250	98.90	0.90	0/250	2.30*
con	Pres22	const.	early-out	250	99.09	0.77	0/250	1.34*
con	Pres22	const.	early-total	250	99.31	0.57	0/250	1.01
con	Pres22	const.	sameday	250	99.36	0.52	0/250	1.01
con	Pres22	const.	total	248	80.07	19.65	16/248	2.26*
con	Pres22	dong	early	3509	99.90	0.09	0/3509	1.22*
con	Pres22	dong	pooled	3509	88.99	11.00	93/3509	1.12*
con	Pres22	dong	sameday	3509	99.92	0.08	0/3509	1.22*
con	Pres25	const.	early	253	99.11	0.72	0/253	2.61*
con	Pres25	const.	early-out	253	98.24	1.59	0/253	2.02*
con	Pres25	const.	early-total	253	99.34	0.55	0/253	1.01
con	Pres25	const.	sameday	253	99.26	0.61	0/253	1.10*
con	Pres25	const.	total	242	85.91	13.91	12/242	1.42*
con	Pres25	dong	early	3553	99.38	0.61	0/3553	8.55*
con	Pres25	dong	pooled	3553	92.14	7.85	31/3553	8.36*
con	Pres25	dong	sameday	3553	99.54	0.45	1/3553	1.21*

C.5 PSRF stopping-rule check

Table C.10 lists the 210 constituency-level runs refitted with the package’s PSRF stopping rule imposed as well (all 35 constituency cells, for the real data and both nulls under both leader codings), starting from the short-chain configuration. After the initial run, the package checks whether the PSRF of every mixture weight is below 1.05 and, if not, extends every chain by another burn-in and sampling block, up to ten times, before drawing the samples that are kept. The rule was met in 37 runs (17.6%), 36 of them at the first check, so the extensions produced convergence in only one run. The other 173 used all ten extensions without meeting it. Of the runs that met the rule, 35 have an estimated fraud share of 1% or less, and the largest is 2.0% (the real data of the 2022 presidential election, early channel, Democratic-coded). Across all 210 runs, fraud shares correlate with those of the corresponding short-chain runs at $r = 0.96$, with a median absolute difference of 0.4 percentage points. The main text uses these runs to check whether longer chains would change the results (its section “MCMC configuration and chain length”).

Table C.10: Constituency-level runs under the package’s PSRF stopping rule (Democratic-coded runs first, then conservative-coded). “Ext.” is the number of chain extensions (at most 10); “Met” is whether the rule, $\text{PSRF} < 1.05$ for every mixture weight, was eventually met; “Final PSRF” is the largest mixture-weight PSRF at the package’s last check; $\hat{R}$ (kept) is our maximum $\hat{R}$ over the mixture weights on the retained draws. The two differ slightly because the package checks the draws before the final sampling block.

Elec.	Channel	Source	Leader	Ext.	Met	Final PSRF	$\hat{R}$ (kept)	Fraud %	Flagged
Pres02	total-no-early	real	Dem.	10	no	1.18	1.21	13.49	16/244
Pres02	total-no-early	marg.null	Dem.	10	no	1.83	1.85	14.45	19/244
Pres02	total-no-early	joint.null	Dem.	10	no	2.37	2.32	12.52	15/244
Pres07	total-no-early	real	Dem.	10	no	1.15	1.16	8.24	10/248
Pres07	total-no-early	marg.null	Dem.	10	no	1.20	1.16	6.89	7/248
Pres07	total-no-early	joint.null	Dem.	10	no	1.34	1.28	7.96	10/248
Gen08	total-no-early	real	Dem.	0	yes	1.00	1.01	0.67	0/198
Gen08	total-no-early	marg.null	Dem.	0	yes	1.01	1.01	0.68	0/198
Gen08	total-no-early	joint.null	Dem.	0	yes	1.00	1.02	0.71	0/198
Gen12	total-no-early	real	Dem.	0	yes	1.01	1.01	0.65	0/209

Table C.10 – continued

Elec.	Channel	Source	Leader	Ext.	Met	Final PSRF	$\hat{R}$ (kept)	Fraud %	Flagged
Gen12	total-no-early	marg.null	Dem.	0	yes	1.01	1.02	0.66	0/209
Gen12	total-no-early	joint.null	Dem.	0	yes	1.00	1.00	0.66	0/209
Pres12	total-no-early	real	Dem.	10	no	1.21	1.21	19.65	21/251
Pres12	total-no-early	marg.null	Dem.	10	no	2.52	2.52	14.60	10/251
Pres12	total-no-early	joint.null	Dem.	10	no	2.60	2.42	17.61	15/251
Gen16	early	real	Dem.	10	no	1.71	1.73	1.23	0/234
Gen16	early	marg.null	Dem.	10	no	1.65	1.49	1.93	0/234
Gen16	early	joint.null	Dem.	10	no	1.59	1.83	1.59	0/234
Gen16	early-out	real	Dem.	0	yes	1.00	1.02	0.61	0/234
Gen16	early-out	marg.null	Dem.	0	yes	1.00	1.02	0.62	0/234
Gen16	early-out	joint.null	Dem.	0	yes	1.01	1.00	0.58	0/234
Gen16	early-total	real	Dem.	10	no	1.22	1.28	3.24	0/234
Gen16	early-total	marg.null	Dem.	10	no	1.31	1.28	3.61	0/234
Gen16	early-total	joint.null	Dem.	10	no	1.32	1.28	3.56	0/234
Gen16	sameday	real	Dem.	0	yes	1.01	1.03	0.59	0/234
Gen16	sameday	marg.null	Dem.	0	yes	1.00	1.00	0.59	0/234
Gen16	sameday	joint.null	Dem.	0	yes	1.00	1.06	0.60	0/234
Gen16	total	real	Dem.	10	no	1.51	1.51	1.03	0/234
Gen16	total	marg.null	Dem.	10	no	1.82	1.40	1.46	0/234
Gen16	total	joint.null	Dem.	10	no	1.11	1.23	0.70	0/234
Pres17	early	real	Dem.	10	no	1.29	1.43	1.59	0/250
Pres17	early	marg.null	Dem.	10	no	2.01	2.33	2.98	0/250
Pres17	early	joint.null	Dem.	10	no	2.10	2.17	2.01	0/250
Pres17	early-out	real	Dem.	10	no	3.48	3.78	2.26	0/250
Pres17	early-out	marg.null	Dem.	10	no	2.15	2.41	2.30	0/250
Pres17	early-out	joint.null	Dem.	10	no	1.50	1.32	3.13	0/250
Pres17	early-total	real	Dem.	0	yes	1.00	1.01	0.60	0/250
Pres17	early-total	marg.null	Dem.	0	yes	1.00	1.02	0.57	0/250
Pres17	early-total	joint.null	Dem.	0	yes	1.01	1.01	0.55	0/250
Pres17	sameday	real	Dem.	0	yes	1.01	1.03	0.54	0/250
Pres17	sameday	marg.null	Dem.	0	yes	1.00	1.01	0.54	0/250
Pres17	sameday	joint.null	Dem.	0	yes	1.00	1.02	0.58	0/250
Pres17	total	real	Dem.	10	no	6.40	7.27	14.61	5/250
Pres17	total	marg.null	Dem.	10	no	2.76	2.74	21.94	10/250
Pres17	total	joint.null	Dem.	10	no	6.37	5.37	20.82	11/250
Gen20	early	real	Dem.	10	no	2.30	2.25	3.47	0/253
Gen20	early	marg.null	Dem.	10	no	2.55	2.80	2.94	0/253
Gen20	early	joint.null	Dem.	10	no	2.76	2.70	4.89	0/253
Gen20	early-out	real	Dem.	10	no	2.39	3.14	1.49	0/253
Gen20	early-out	marg.null	Dem.	10	no	2.37	2.98	1.59	0/253
Gen20	early-out	joint.null	Dem.	10	no	1.74	2.19	1.02	0/253
Gen20	early-total	real	Dem.	10	no	1.72	1.67	2.31	0/253
Gen20	early-total	marg.null	Dem.	10	no	2.64	2.09	2.60	0/253
Gen20	early-total	joint.null	Dem.	10	no	2.24	2.34	2.32	0/253
Gen20	sameday	real	Dem.	10	no	1.11	1.14	0.64	0/253
Gen20	sameday	marg.null	Dem.	10	no	1.21	1.16	0.74	0/253
Gen20	sameday	joint.null	Dem.	10	no	1.12	1.18	0.63	0/253
Gen20	total	real	Dem.	10	no	1.79	1.81	11.24	3/253
Gen20	total	marg.null	Dem.	10	no	2.92	2.54	10.96	1/253
Gen20	total	joint.null	Dem.	10	no	1.64	1.59	14.40	8/253
Pres22	early	real	Dem.	2	yes	1.05	1.05	1.96	0/250
Pres22	early	marg.null	Dem.	10	no	3.90	4.84	4.42	1/250
Pres22	early	joint.null	Dem.	10	no	1.52	1.59	1.58	0/250
Pres22	early-out	real	Dem.	10	no	3.47	3.18	3.84	0/250
Pres22	early-out	marg.null	Dem.	10	no	1.34	1.32	4.20	1/250
Pres22	early-out	joint.null	Dem.	10	no	3.01	2.39	3.85	0/250

Table C.10 – continued

Elec.	Channel	Source	Leader	Ext.	Met	Final PSRF	$\hat{R}$ (kept)	Fraud %	Flagged
Pres22	early-total	real	Dem.	10	no	1.52	1.39	5.39	5/250
Pres22	early-total	marg.null	Dem.	10	no	1.12	1.13	4.34	4/250
Pres22	early-total	joint.null	Dem.	10	no	1.38	1.30	5.50	4/250
Pres22	sameday	real	Dem.	10	no	1.13	1.16	1.90	1/250
Pres22	sameday	marg.null	Dem.	10	no	1.24	1.23	2.33	0/250
Pres22	sameday	joint.null	Dem.	10	no	1.86	1.68	2.04	0/250
Pres22	total	real	Dem.	10	no	2.12	2.12	30.23	36/248
Pres22	total	marg.null	Dem.	10	no	3.99	3.92	27.88	32/248
Pres22	total	joint.null	Dem.	10	no	5.73	5.82	20.25	23/248
Gen24	early	real	Dem.	10	no	2.08	2.20	3.79	0/246
Gen24	early	marg.null	Dem.	10	no	1.59	1.63	3.74	0/246
Gen24	early	joint.null	Dem.	10	no	1.54	1.46	3.93	0/246
Gen24	early-out	real	Dem.	10	no	3.61	3.19	5.25	0/246
Gen24	early-out	marg.null	Dem.	10	no	2.69	2.70	3.63	0/246
Gen24	early-out	joint.null	Dem.	10	no	3.35	3.97	3.38	0/246
Gen24	early-total	real	Dem.	10	no	1.70	1.36	1.27	0/246
Gen24	early-total	marg.null	Dem.	10	no	2.16	2.63	1.86	0/246
Gen24	early-total	joint.null	Dem.	10	no	2.03	1.50	1.62	0/246
Gen24	sameday	real	Dem.	10	no	1.59	1.26	1.18	0/246
Gen24	sameday	marg.null	Dem.	10	no	1.11	1.09	1.17	0/246
Gen24	sameday	joint.null	Dem.	10	no	1.19	1.15	1.38	0/246
Gen24	total	real	Dem.	10	no	1.11	1.10	14.02	9/246
Gen24	total	marg.null	Dem.	10	no	1.64	1.68	15.75	10/246
Gen24	total	joint.null	Dem.	10	no	1.07	1.09	18.45	13/246
Pres25	early	real	Dem.	10	no	3.27	4.30	5.59	3/253
Pres25	early	marg.null	Dem.	10	no	3.17	2.78	5.49	1/253
Pres25	early	joint.null	Dem.	10	no	1.06	1.08	6.46	4/253
Pres25	early-out	real	Dem.	10	no	2.03	2.47	8.03	1/253
Pres25	early-out	marg.null	Dem.	10	no	2.84	2.62	2.85	1/253
Pres25	early-out	joint.null	Dem.	10	no	2.33	1.87	6.63	0/253
Pres25	early-total	real	Dem.	10	no	1.57	1.68	8.16	12/253
Pres25	early-total	marg.null	Dem.	10	no	1.56	1.52	8.23	13/253
Pres25	early-total	joint.null	Dem.	10	no	1.52	1.58	7.86	11/253
Pres25	sameday	real	Dem.	10	no	1.74	1.68	1.57	0/253
Pres25	sameday	marg.null	Dem.	10	no	1.31	1.18	1.43	0/253
Pres25	sameday	joint.null	Dem.	10	no	1.14	1.10	1.73	1/253
Pres25	total	real	Dem.	10	no	1.21	1.18	29.66	36/242
Pres25	total	marg.null	Dem.	10	no	6.29	6.23	29.96	23/242
Pres25	total	joint.null	Dem.	10	no	5.93	6.43	22.65	18/242
Pres02	total-no-early	real	Con.	10	no	1.60	1.62	12.71	14/244
Pres02	total-no-early	marg.null	Con.	10	no	1.45	1.48	14.77	13/244
Pres02	total-no-early	joint.null	Con.	10	no	2.98	2.61	11.38	10/244
Pres07	total-no-early	real	Con.	10	no	1.61	1.52	5.54	4/248
Pres07	total-no-early	marg.null	Con.	10	no	1.98	2.02	4.40	2/248
Pres07	total-no-early	joint.null	Con.	10	no	1.91	2.11	4.77	3/248
Gen08	total-no-early	real	Con.	0	yes	1.00	1.03	0.71	0/198
Gen08	total-no-early	marg.null	Con.	0	yes	1.00	1.04	0.70	0/198
Gen08	total-no-early	joint.null	Con.	0	yes	1.01	1.03	0.71	0/198
Gen12	total-no-early	real	Con.	10	no	1.12	1.23	0.83	0/194
Gen12	total-no-early	marg.null	Con.	0	yes	1.01	1.01	1.17	1/194
Gen12	total-no-early	joint.null	Con.	0	yes	1.01	1.03	0.77	0/194
Pres12	total-no-early	real	Con.	10	no	2.10	2.10	24.97	24/251
Pres12	total-no-early	marg.null	Con.	10	no	2.74	2.75	23.09	17/251
Pres12	total-no-early	joint.null	Con.	10	no	1.81	1.80	26.63	25/251
Gen16	early	real	Con.	10	no	2.08	1.79	1.79	0/229
Gen16	early	marg.null	Con.	10	no	1.60	1.44	1.23	0/229

Table C.10 – continued

Elec.	Channel	Source	Leader	Ext.	Met	Final PSRF	$\hat{R}$ (kept)	Fraud %	Flagged
Gen16	early	joint.null	Con.	10	no	1.23	1.26	1.30	0/229
Gen16	early-out	real	Con.	0	yes	1.01	1.01	0.60	0/229
Gen16	early-out	marg.null	Con.	10	no	1.11	1.14	0.72	0/229
Gen16	early-out	joint.null	Con.	0	yes	1.00	1.01	0.62	0/229
Gen16	early-total	real	Con.	10	no	1.67	1.35	2.04	0/229
Gen16	early-total	marg.null	Con.	10	no	2.17	2.24	1.67	0/229
Gen16	early-total	joint.null	Con.	10	no	1.59	1.39	1.87	0/229
Gen16	sameday	real	Con.	0	yes	1.00	1.00	0.61	0/229
Gen16	sameday	marg.null	Con.	0	yes	1.01	1.03	0.61	0/229
Gen16	sameday	joint.null	Con.	0	yes	1.00	1.02	0.59	0/229
Gen16	total	real	Con.	10	no	1.22	1.21	0.77	0/229
Gen16	total	marg.null	Con.	10	no	1.20	1.35	1.18	0/229
Gen16	total	joint.null	Con.	10	no	1.26	1.57	0.75	0/229
Pres17	early	real	Con.	10	no	1.40	1.92	0.69	0/250
Pres17	early	marg.null	Con.	10	no	1.86	2.54	1.33	0/250
Pres17	early	joint.null	Con.	10	no	1.57	1.38	1.76	0/250
Pres17	early-out	real	Con.	10	no	1.11	1.20	0.63	0/250
Pres17	early-out	marg.null	Con.	0	yes	1.00	1.01	0.57	0/250
Pres17	early-out	joint.null	Con.	10	no	1.21	1.15	0.73	0/250
Pres17	early-total	real	Con.	10	no	1.40	1.87	0.68	0/250
Pres17	early-total	marg.null	Con.	0	yes	1.01	1.02	0.57	0/250
Pres17	early-total	joint.null	Con.	10	no	2.40	2.84	1.76	0/250
Pres17	sameday	real	Con.	10	no	2.82	3.79	1.80	0/250
Pres17	sameday	marg.null	Con.	10	no	2.35	4.59	0.73	0/250
Pres17	sameday	joint.null	Con.	10	no	1.84	3.18	0.76	0/250
Pres17	total	real	Con.	10	no	1.54	1.42	16.72	3/250
Pres17	total	marg.null	Con.	10	no	1.35	1.36	14.83	4/250
Pres17	total	joint.null	Con.	10	no	1.94	1.97	13.19	7/250
Gen20	early	real	Con.	10	no	1.58	1.43	1.69	0/253
Gen20	early	marg.null	Con.	10	no	1.20	1.21	2.47	0/253
Gen20	early	joint.null	Con.	10	no	1.25	1.30	1.77	0/253
Gen20	early-out	real	Con.	10	no	1.49	1.28	0.98	0/253
Gen20	early-out	marg.null	Con.	10	no	1.27	1.51	0.72	0/253
Gen20	early-out	joint.null	Con.	10	no	1.50	1.66	0.87	0/253
Gen20	early-total	real	Con.	10	no	1.36	1.42	0.82	0/253
Gen20	early-total	marg.null	Con.	10	no	2.01	2.39	1.22	0/253
Gen20	early-total	joint.null	Con.	10	no	2.06	1.96	1.46	0/253
Gen20	sameday	real	Con.	0	yes	1.00	1.01	0.59	0/253
Gen20	sameday	marg.null	Con.	0	yes	1.00	1.06	0.55	0/253
Gen20	sameday	joint.null	Con.	0	yes	1.00	1.01	0.55	0/253
Gen20	total	real	Con.	10	no	1.63	1.69	10.11	5/253
Gen20	total	marg.null	Con.	10	no	1.57	1.61	10.06	10/253
Gen20	total	joint.null	Con.	10	no	1.42	1.38	8.15	5/253
Pres22	early	real	Con.	10	no	1.35	1.39	0.82	0/250
Pres22	early	marg.null	Con.	10	no	1.60	1.90	0.99	0/250
Pres22	early	joint.null	Con.	10	no	1.36	1.21	0.97	0/250
Pres22	early-out	real	Con.	10	no	2.14	2.15	3.94	0/250
Pres22	early-out	marg.null	Con.	10	no	2.08	1.63	2.48	0/250
Pres22	early-out	joint.null	Con.	10	no	2.76	2.35	3.77	0/250
Pres22	early-total	real	Con.	10	no	1.37	1.40	0.82	0/250
Pres22	early-total	marg.null	Con.	10	no	1.50	1.44	0.91	0/250
Pres22	early-total	joint.null	Con.	10	no	1.12	1.15	0.63	0/250
Pres22	sameday	real	Con.	10	no	1.23	1.19	0.75	0/250
Pres22	sameday	marg.null	Con.	10	no	1.38	1.39	0.81	0/250
Pres22	sameday	joint.null	Con.	10	no	1.11	1.26	0.62	0/250
Pres22	total	real	Con.	10	no	2.00	1.99	30.70	31/248

Table C.10 – continued

Elec.	Channel	Source	Leader	Ext.	Met	Final PSRF	$\hat{R}$ (kept)	Fraud %	Flagged
Pres22	total	marg.null	Con.	10	no	1.40	1.42	32.19	26/248
Pres22	total	joint.null	Con.	10	no	4.83	4.77	25.46	21/248
Gen24	early	real	Con.	10	no	2.07	2.10	2.26	0/246
Gen24	early	marg.null	Con.	10	no	1.94	1.58	1.47	0/246
Gen24	early	joint.null	Con.	10	no	1.82	2.18	1.02	0/246
Gen24	early-out	real	Con.	10	no	1.69	2.35	0.89	0/246
Gen24	early-out	marg.null	Con.	10	no	2.12	3.97	0.72	0/246
Gen24	early-out	joint.null	Con.	10	no	1.22	1.10	0.81	0/246
Gen24	early-total	real	Con.	10	no	1.12	1.20	0.65	0/246
Gen24	early-total	marg.null	Con.	10	no	1.12	1.19	0.67	0/246
Gen24	early-total	joint.null	Con.	0	yes	1.01	1.01	0.58	0/246
Gen24	sameday	real	Con.	10	no	1.12	1.17	0.66	0/246
Gen24	sameday	marg.null	Con.	0	yes	1.01	1.01	0.58	0/246
Gen24	sameday	joint.null	Con.	0	yes	1.01	1.01	0.57	0/246
Gen24	total	real	Con.	10	no	1.29	1.27	10.56	2/246
Gen24	total	marg.null	Con.	10	no	2.90	2.77	11.63	6/246
Gen24	total	joint.null	Con.	10	no	2.20	2.26	9.79	3/246
Pres25	early	real	Con.	10	no	1.24	1.29	1.20	0/253
Pres25	early	marg.null	Con.	10	no	1.50	1.24	1.07	0/253
Pres25	early	joint.null	Con.	10	no	1.29	1.21	1.40	0/253
Pres25	early-out	real	Con.	10	no	1.22	1.19	0.74	0/253
Pres25	early-out	marg.null	Con.	10	no	2.16	3.02	1.22	0/253
Pres25	early-out	joint.null	Con.	10	no	1.92	1.60	1.38	0/253
Pres25	early-total	real	Con.	10	no	1.22	1.13	0.73	0/253
Pres25	early-total	marg.null	Con.	10	no	1.12	1.18	0.64	0/253
Pres25	early-total	joint.null	Con.	10	no	1.11	1.18	0.63	0/253
Pres25	sameday	real	Con.	10	no	1.22	1.17	0.74	0/253
Pres25	sameday	marg.null	Con.	10	no	1.85	1.67	1.29	0/253
Pres25	sameday	joint.null	Con.	10	no	1.12	1.18	0.62	0/253
Pres25	total	real	Con.	10	no	6.23	6.43	19.61	5/242
Pres25	total	marg.null	Con.	10	no	5.69	5.66	20.29	7/242
Pres25	total	joint.null	Con.	10	no	4.44	4.23	25.38	11/242

D Klimek-type fingerprint analysis

D.1 Method and results in detail

This subsection expands the Klimek-type analysis summarized in the main text (section “Does this extend to other election forensics methods?”). The concern tested there is that ordinary demographic and regional differences between units may look like a fraud signature to any method that uses only the joint distribution of turnout and vote shares. We check whether this applies to the model of Klimek et al. [2012], fitted as in Klimek et al. [2018], using the real data and the joint-null data.

How the model works. The model assumes that without fraud, every unit’s turnout and leader share come from one nationwide pair of Normal distributions. Their centers are the modes of the empirical distributions: the most common turnout and the most common leader share among all units of the election. No province or regional information is used. The model also assumes that fraud only pushes units upward toward higher turnout and a higher leader share. The empirical spread of units below each mode, where fraud has less effect, is therefore treated as true variation and sets the width of the fraud-free distributions, which are assumed to be symmetric. The additional spread above the mode is attributed to fraud. These centers and widths are computed from the observed data before any estimation.

Fraud is then added to simulated fraud-free units. With probability f_i , a unit suffers incremental fraud: a

proportion x_i of its abstainers and a proportion x_i^α of its opposition voters are counted for the leader, where $x_i \sim |N(0, \theta)|$ and θ is the spread of leader shares above the mode. With probability f_e , a unit suffers extreme fraud, in which nearly all abstainers and opposition voters are counted for the leader. The fitted f_i , f_e and α are the values for which the simulated two-dimensional histogram of turnout against leader share (the election’s “fingerprint”, at 1% resolution) comes closest to the observed one.

Sweeps. Because the simulated units are random, the fit depends on the simulation. Following Klimek et al. [2018], we therefore repeat it 200 times, each time with a new simulated set of units. Each repetition is a *sweep*, and we report the mean and standard deviation of the fitted parameters across the 200 sweeps. Our only departure from Klimek et al. [2018] is computational: within each sweep we compute the best-fitting values directly instead of scoring candidate values one at a time, which gives the same answer (Appendix D.2).

Accuracy on simulated data. Appendix D.3 tests whether the procedure recovers known fraud shares on data simulated from the model itself. It often does not. At dong-sized samples (2,700–3,500 units), it reports 16–30% incremental fraud on data with none and 8% where the true share is 30%. At polling-district sizes (13,500 units), it recovers a true 30% more closely (35%) but still reports 28% on data with none. The estimates are therefore imprecise even when the data are generated by the model itself.

Data. The Klimek analysis serves only to check whether the critique extends beyond eforensics, so we restrict it to the five elections held before early voting: the 2002, 2007 and 2012 presidential elections and the 2008 and 2012 general elections. In these elections every regular ballot is cast and counted at its polling district, so total turnout is observed at the finest published level. We fit the model at two levels: the dong (2,656–3,576 units per election) and the polling district (*tupyogu*, 10,461–13,522 units). Klimek et al. [2012] require an average electorate of at most 5,000 per unit. Polling districts meet this requirement (mean electorates of roughly 2,500–3,000), but dong do not (roughly 9,700–11,600). Klimek et al. [2018] also note that their approach is less reliable in smaller datasets. Their Turkish data have 153,701 polling stations averaging 355 voters, so even the Korean polling districts are an order of magnitude fewer and larger. As in both papers, we remove units with fewer than 100 eligible voters. We also remove units where either major party fielded no candidate (in the 2008 and 2012 general elections), under both leader codings. Constituency-level samples of about 250 units are too small for this method and are not used.

Datasets. We fit the model to the real data as reported (“raw fingerprints”) and to data with province means removed (“residualized fingerprints”). Because strong province effects may pull the distributions away from the Normal shape that the model assumes, we create the residualized version not contained in the original fingerprint-model papers. This version is our new contribution. We also fit the joint-null dataset, generated by the same procedure and seed as in the eforensics analysis. The marginal null is not a meaningful fraud-free comparison here. It keeps each unit at its observed turnout and vote share and adds only small independent noise, so its fingerprint is essentially the real one and would inherit any fraud-like cluster in the real data. (The eforensics model is different, because it judges each unit against the fitted fraud-free distribution, so unit-level noise changes what it sees.) The joint null removes this unit-specific structure.

Results at the dong level. Each level gives ten fits: five elections under two leader codings. Table D.1 and the summary figure in the main text compare the real-data fits with the joint-null fits. Extreme fraud is essentially absent: f_e never exceeds 0.07% in any real-data or joint-null fit at either level. The incremental-fraud estimates fail the validation at both levels, but in different ways. On the raw dong-level fingerprints, none of the ten real-data fits falls below 5%. The joint null is similar. None of its ten fits falls below 5% either, the real-data and joint-null estimates correlate at $r = 0.80$ across the ten pairs, and the null’s estimate is at least as large as the real one in 3 of the 10 pairs. With the national winner as leader, the 2007 presidential election gives 75% for the real dataset and 70% for the null, and the 2008 general election gives 92% and 77%. For the 2002 presidential election, the null’s estimate (66%) far exceeds the real one (15%).

Results at the polling-district level. Figures D.1 and D.2 show the polling-district fingerprints with the

national winner as leader. On the raw fingerprints, the joint null falls below 5% in 4 of its 10 fits. With the national winner as leader, the real estimate is much larger than the null's in the 2007 presidential election (43% against 2%) and the 2012 general election (59% against 1%). But the null still returns 85% for the 2008 general election, where the real estimate is 13%. With the losing conservative candidate as leader, it returns 87% for the 2002 presidential election, against 10% for the real data. Across the ten pairs, the real-data and joint-null estimates are weakly negatively correlated ($r = -0.42$).

Province-adjusted fingerprints. Removing province means does not help at either level. At the dong level, only 1 of the 10 estimates falls below 5% for the real data, and likewise for the null. The real-data and joint-null estimates correlate at $r = 0.17$ across the ten pairs. At the polling-district level, 5 of the 10 fall below 5% for each, and the correlation is $r = -0.05$. The null gives the larger estimate in 7 of the 10 dong-level pairs and in 4 of the 10 polling-district pairs, and the province-adjusted polling-district estimate for the 2012 presidential election is 86%.

Stability across levels. Sampling uncertainty is smaller at the polling-district level: the median standard deviation across sweeps in Table D.1 is 3.8 points, against 7.4 at the dong level. Yet the estimates are not stable across levels. On the raw fingerprints, the same election and leader coding give fraud shares that differ between the two levels by a median of 10.5 points and by up to 80 points. The 2008 general election gives 92% at the dong level and 13% at the polling-district level, while the 2012 general election moves the other way, from 22% to 59%. Klimek et al. [2012] present insensitivity to the aggregation level as a key property of their method, but this does not hold for the Korean data.

Excluding regions with strong partisan leanings. Table D.2 and Figure D.3 show that excluding such regions does not resolve these problems and can move estimates in either direction. At the dong level, removing Honam alone raises the 2012 presidential estimate from 18% to 58% and lowers the 2008 general-election estimate from 92% to 33%. Removing Honam and Yeongnam together lowers the 2007 presidential estimate from 75% to 37% but raises the 2012 general-election estimate from 22% to 59% and the 2002 presidential estimate from 15% to 50%. At the polling-district level, the 2012 presidential estimate rises from 15% to 38% without Honam and to 50% without Honam and Gyeongbuk/Daegu. The 2007 presidential estimate falls from 43% to 15% without Honam and Yeongnam, and the 2012 general-election estimate stays near 60% under every exclusion. Figures D.1 and D.3 show part of the reason. With the conservative candidate as leader, as in the 2012 presidential election, the leader-share distribution has two modes, and the Honam cluster lies far below the national mode. This cluster widens the estimated spread of fraud-free units, so the turnout/leader-share pairs of the fitted model spread over nearly the whole leader-share axis.

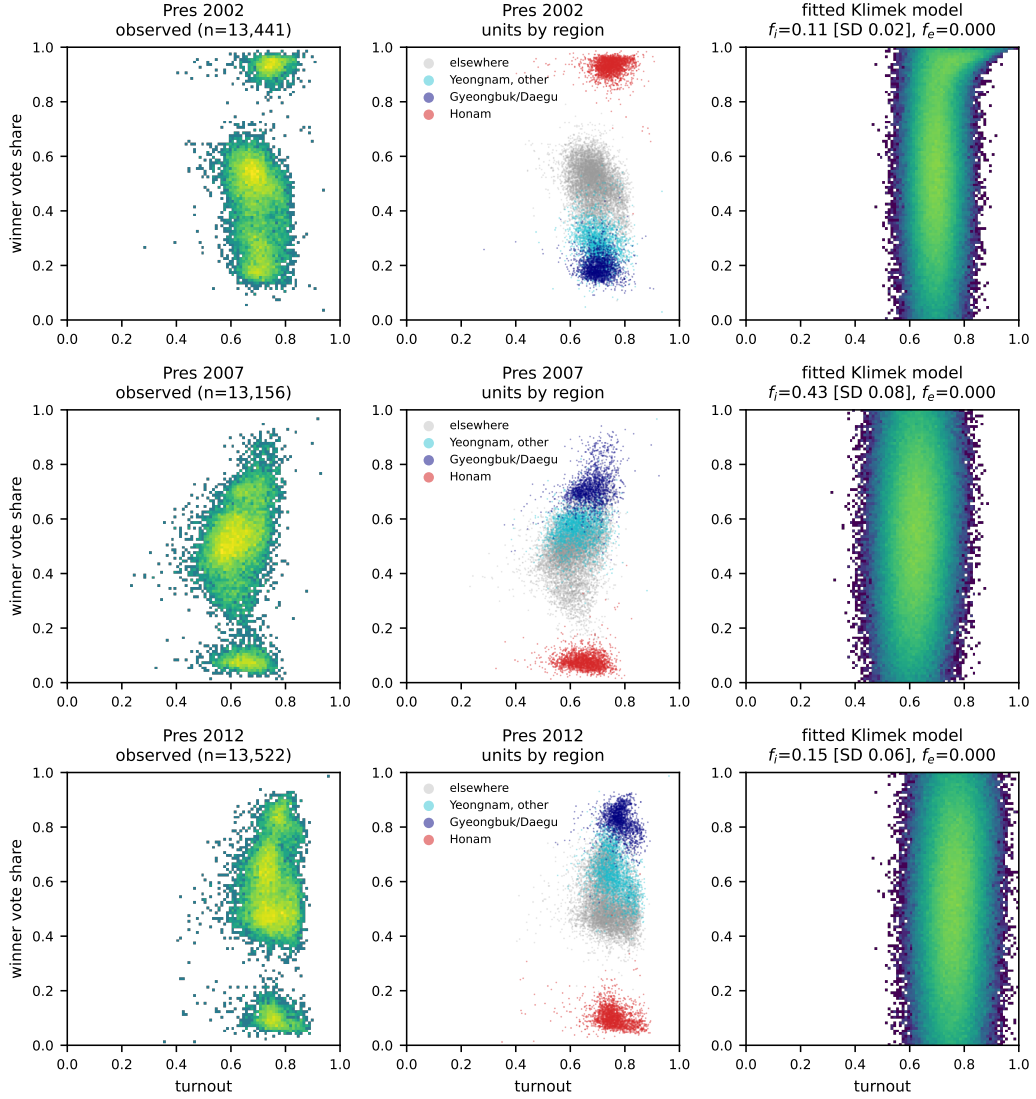


Figure D.1: Election fingerprints of the three presidential elections held before early voting existed (polling-district level), with the national winner as leader: observed two-dimensional histogram of unit turnout against winner vote share at the fit's 1% resolution (left), every unit colored by region (center), with Honam (Gwangju, Jeonbuk, Jeonnam) in red, Gyeongbuk and Daegu in dark blue and the rest of Yeongnam (Busan, Ulsan, Gyeongnam) in light blue, and the fingerprint of the fitted Klimek et al. [2018]-procedure fraud model (right), drawn from a large simulated sample of 200,000 units at the fraud-free distributions computed from the observed data and the mean fitted f_i , f_e and α . Panel titles give the fitted incremental and extreme fraud shares (f_i is the mean across 200 sweeps, with its standard deviation in brackets). Logarithmic color scale shared within each row for the left and right panels. Units far below the national mode of the leader share widen the fraud-free leader-share distribution computed from the observed data: the Yeongnam cluster for the Democratic winner in 2002, and the Honam cluster for the conservative winners in 2007 and 2012. The turnout/leader-share pairs of each fitted model therefore spread over nearly the whole leader-share axis.

Table D.1: Klimek et al. fraud-model estimates of the total fraud share $f_i + f_e$ (%) for the real dataset and the joint-null dataset, with the actual winner as leader (the elected president’s party, or for general elections the party winning the most seats), for the five elections held before early voting, at the dong and polling-district levels. Raw fingerprints and fingerprints after removing province means are listed. Each entry is the mean $\pm$ one standard deviation across 200 sweeps.

Election	Winner	Dong				Polling district			
		Raw		Prov.-adjusted		Raw		Prov.-adjusted	
		Real	Null	Real	Null	Real	Null	Real	Null
Pres 2002	Dem.	15 $\pm$ 3	66 $\pm$ 29	5 $\pm$ 6	6 $\pm$ 7	11 $\pm$ 2	11 $\pm$ 2	6 $\pm$ 6	0 $\pm$ 1
Pres 2007	Con.	75 $\pm$ 2	70 $\pm$ 2	7 $\pm$ 7	9 $\pm$ 10	43 $\pm$ 8	2 $\pm$ 2	1 $\pm$ 2	3 $\pm$ 3
Gen 2008	Con.	92 $\pm$ 1	77 $\pm$ 2	6 $\pm$ 5	20 $\pm$ 8	13 $\pm$ 6	85 $\pm$ 1	34 $\pm$ 4	3 $\pm$ 2
Gen 2012	Con.	22 $\pm$ 11	13 $\pm$ 10	23 $\pm$ 11	20 $\pm$ 9	59 $\pm$ 7	1 $\pm$ 1	4 $\pm$ 3	15 $\pm$ 4
Pres 2012	Con.	18 $\pm$ 7	8 $\pm$ 6	20 $\pm$ 15	31 $\pm$ 17	15 $\pm$ 6	8 $\pm$ 5	86 $\pm$ 11	7 $\pm$ 7
Units		2,656–3,576				10,461–13,522			

Table D.2: Estimated incremental-fraud share f_i (%) of the Klimek et al. [2012] model for the five elections held before early voting existed (national winner as leader, raw fingerprints), at the dong and polling-district levels, on all units and with regional voting blocs excluded: H, Honam (Gwangju, Jeonbuk, Jeonnam); Y, Yeongnam (Busan, Daegu, Ulsan, Gyeongbuk, Gyeongnam); GD, Gyeongbuk and Daegu. Each entry is the mean across 200 sweeps, and brackets give one standard deviation below and above the mean (truncated at zero), omitted when the standard deviation is under one percentage point.

Election	All units	Excl. H	Excl. H, Y	Excl. H, GD
<i>Dong</i>				
Pres 2002	15 [12, 18]	16 [1, 31]	50 [24, 75]	13 [0, 27]
Pres 2007	75 [73, 77]	54 [40, 68]	37 [15, 60]	58 [34, 81]
Gen 2008	92 [91, 93]	33 [22, 45]	38 [24, 53]	35 [24, 47]
Gen 2012	22 [11, 33]	26 [15, 36]	59 [41, 77]	25 [11, 40]
Pres 2012	18 [11, 25]	58 [54, 61]	46 [41, 51]	59 [55, 64]
Units retained	2,656–3,576	2,107–2,946	1,552–1,867	1,902–2,465
<i>Polling district</i>				
Pres 2002	11 [9, 13]	15 [2, 28]	26 [7, 45]	9 [0, 19]
Pres 2007	43 [34, 51]	39 [31, 47]	15 [4, 25]	33 [21, 45]
Gen 2008	13 [7, 19]	10 [5, 15]	9 [3, 16]	9 [4, 15]
Gen 2012	59 [52, 67]	61 [55, 68]	61 [52, 71]	65 [57, 74]
Pres 2012	15 [8, 21]	38 [36, 40]	34 [30, 37]	50 [46, 53]
Units retained	10,461–13,522	8,900–11,695	7,016–8,133	8,313–10,155

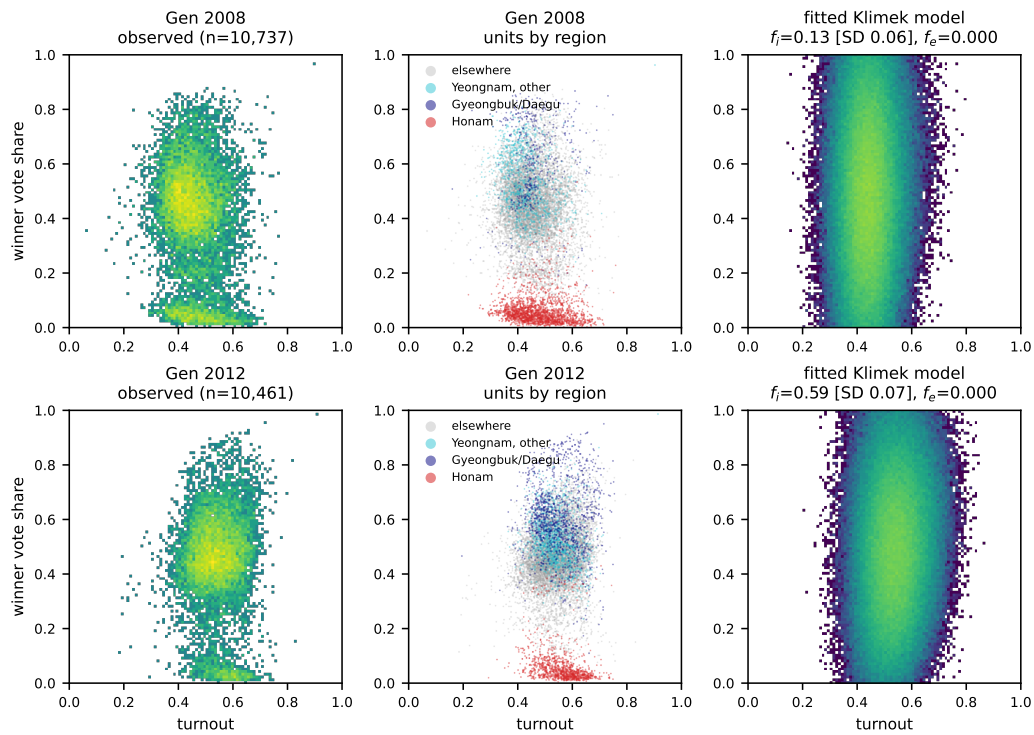


Figure D.2: As Figure D.1, for the two general elections held before early voting existed.

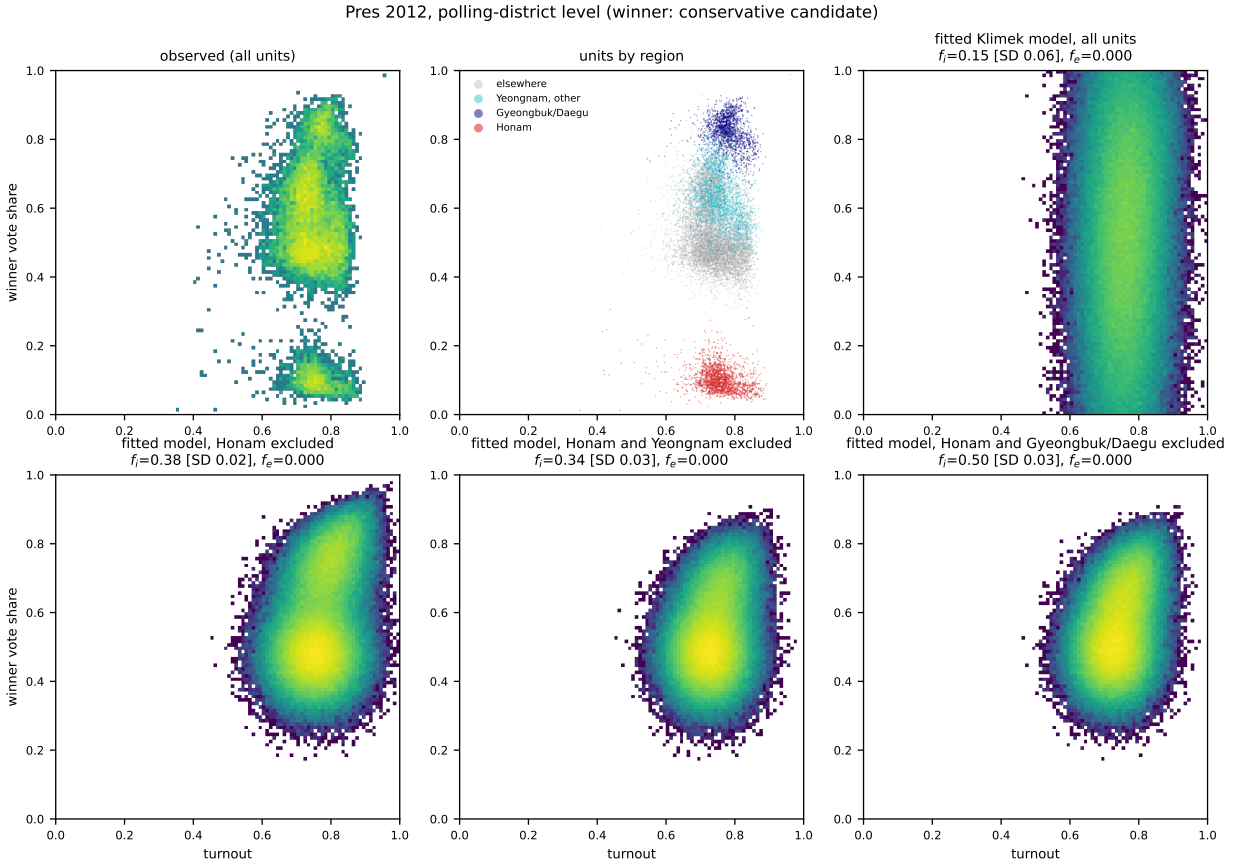


Figure D.3: The 2012 presidential election (polling-district level, conservative winner as leader). Top: observed fingerprint, units colored by region (as in Figure D.1), and the fitted Klimek et al. [2018]-procedure model on all units. Bottom: fitted models with Honam, with Honam and Yeongnam, and with Honam and Gyeongbuk/Daegu excluded. None of the three exclusions brings the estimated incremental fraud to zero, and each raises it. On all units, the Honam cluster, far below the national mode of the conservative leader share, inflates the width of the fraud-free distribution, so the turnout/leader-share pairs of the fitted model spread over nearly the whole leader-share axis.

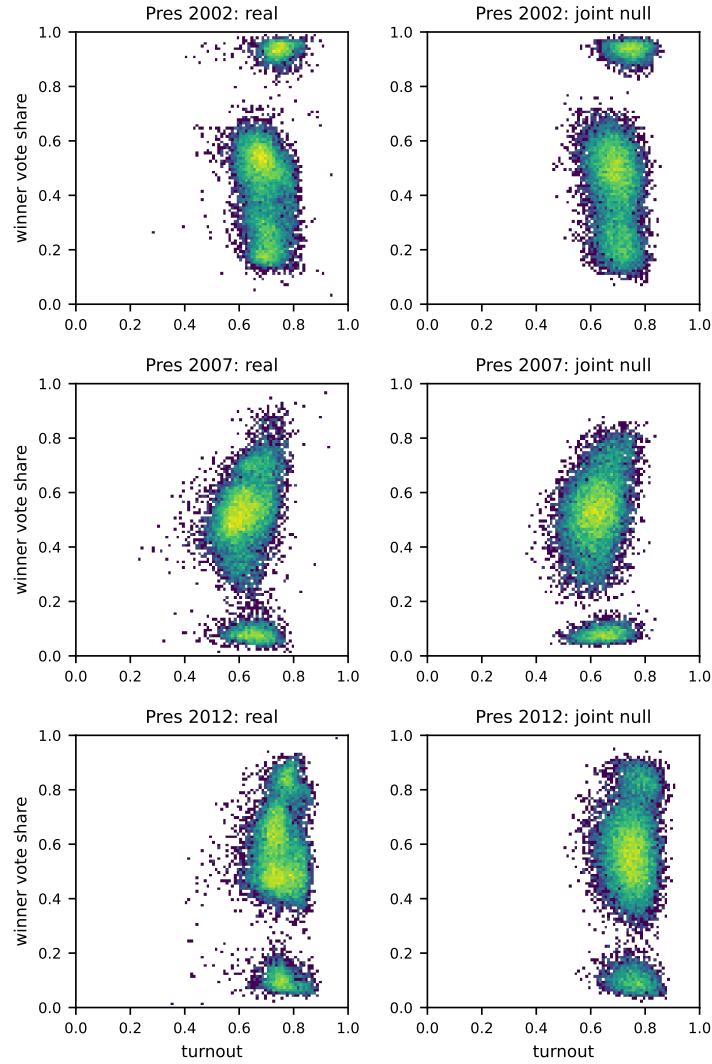


Figure D.4: Election fingerprints of the real dataset (left) and of their joint fraud-free null (right), for the three presidential elections held before early voting existed (polling-district level, national winner as leader). The nulls are generated by the same procedure and seed as in the eforensics analysis, from the Klimek unit tables. Logarithmic color scale shared within each row.

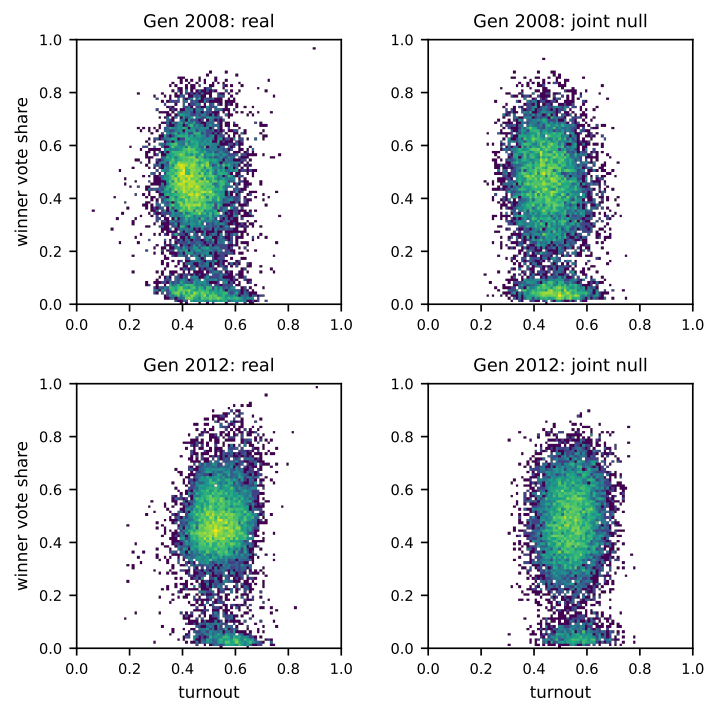


Figure D.5: As Figure D.4, for the two general elections held before early voting existed.

D.2 Estimation

The estimation procedure follows Klimek et al. [2018], which applies the parametric model of Klimek et al. [2012]. On details where Klimek et al. [2018] is silent, we follow the main text and supplementary information of Klimek et al. [2012].

Model [Klimek et al., 2012, SI]. Each model unit’s clean turnout and leader share are drawn from Normal distributions truncated to $(0, 1)$, with means a, v and standard deviations σ_a and $\sqrt{2}\sigma_{vL}$. With probability f_i the unit suffers incremental fraud: a proportion $x_i \sim |N(0, \sigma_{vR})|$ of abstainers and a proportion x_i^α of opposition voters are counted for the leader. With probability f_e it suffers extreme fraud, the same transfer with $y_i = 1 - |N(0, 0.075)|$ in place of x_i . The supplementary information of Klimek et al. [2012] typesets the incremental-fraud draw as $|N(0, \sqrt{\sigma_{vR}})|$. We use σ_{vR} , since it is defined there as the incremental-fraud *width* in vote-share units, and a square root would make the draw’s scale depend on those units.

Fraud-free distributions. Klimek et al. [2018] state only that the means and standard deviations of the fraud-free distributions are estimated from the data, so we use the estimators of Klimek et al. [2012, SI]. To find a and v , we sort the units’ turnout (or leader share) into 1% bins and take the bin with the most units, that is, the mode of the empirical distribution. In their main text, Klimek et al. [2012] describe this value as the point where the empirical distribution is highest, and in their supplementary information as its “(first local) maximum”. We use the most populated bin rather than the first local maximum. σ_{vL} and σ_{vR} are the root-mean-square deviations of leader share from v among units below and above v . σ_a is the corresponding deviation of turnout from a among units below both a and v .

Objective and resolution. Following Klimek et al. [2018], f_i, f_e and α minimize the residual sum of squares between the observed and model two-dimensional turnout/leader-share fingerprints, i.e. the unit counts per cell. This replaces the one-dimensional χ^2 -type statistic on the leader-share distribution used by Klimek et al. [2012]. Klimek et al. [2018] do not state a bin width, so we use the 1% bins of Klimek et al. [2012, SI], a 100×100 fingerprint. The candidate parameter values are those of Klimek et al. [2012, SI]: $f_i, f_e \in \{0, 0.01, \dots, 1\}$ with $f_i + f_e \leq 1$, and $\alpha \in \{0, 0.1, \dots, 5\}$.

Sweeps. As in Klimek et al. [2018], results are taken from 200 sweeps over these candidate values, each with a fresh simulated realization of the model units, and we report the mean and standard deviation of f_i, f_e and α across sweeps. Klimek et al. [2018] note that means and standard deviations basically stop changing after 100 sweeps.

Data rules. As in Klimek et al. [2018] and Klimek et al. [2012], units with fewer than 100 eligible voters are removed, turnout is valid votes over eligible voters, and a unit reporting more valid votes than eligible voters is assigned 100% turnout. No other trimming is applied. In particular, units at 100% turnout or at 0% or 100% leader share are kept, since that corner is where the model’s extreme-fraud mode lies (the eforensics data preparation trims turnout and vote share to $[0.001, 0.999]$). Because both leader codings are fit, units where either major party received no votes are also removed, identically for both codings. In the 2008 and 2012 general elections these are constituencies where that party fielded no candidate, so a zero leader share there reflects the ballot rather than voters’ preferences.

Departure from Klimek et al. [2018]. We make one computational change, which does not affect the results. A direct search simulates the fingerprint for each candidate (f_i, f_e) and scores it separately. We obtain the same best-fitting values faster. In each sweep, every simulated unit receives one random number between 0 and 1, and its fraud class depends only on where that number falls: below f_i (incremental fraud), between f_i and $f_i + f_e$ (extreme fraud) or above $f_i + f_e$ (no fraud). Increasing f_i or f_e by 0.01 therefore changes the class of only the few units whose numbers fall in that step, and the simulated fingerprint changes only in the cells those units occupy. We use these small changes to update the fit criterion for all 101×101 candidate pairs instead of rebuilding the fingerprint for each pair, which gives the same best-fitting values as scoring every pair separately.

Residualized fingerprints. We subtract the province means from logit turnout and logit leader share, add

back the national mean, and then fit the model as above.

D.3 Comparison with alternative estimation procedures

Before settling on the Klimek et al. [2018] procedure, we compared it against two alternatives on data simulated from the model with known parameters: our initial estimator, which fits the fraud-free distributions jointly with the fraud parameters instead of computing them from the observed data before estimation, and a reconstruction of the original Klimek et al. [2012] procedure (fraud-free distributions computed from the observed data, the 2012 paper’s one-dimensional relative statistic on the leader-share distribution, and a search over candidate values of f_i , f_e and α , with the best-fitting values computed directly in the same way as for the 2018 procedure; see Appendix D.2). Table D.3 reports the results.

No procedure recovers well at the smallest sample size (250 units, comparable to a constituency), which is why the Klimek analysis is not run at the constituency level. At dong-like sample sizes (2,700–3,500 units) the three procedures fail differently. Our joint-fit estimator is the most accurate when it works, recovering all three 3,500-unit cases closely, but it is also the least stable: at 2,700 units with true $f_i = 0.15$, it returns $f_i = 0.82$. The errors of the Klimek et al. [2018] procedure run in both directions. On fraud-free data it returns $f_i = 0.30$ at 3,500 units and $f_i = 0.16$ at 2,700, with standard deviations across sweeps of 0.18 and 0.14. Where the true share is 0.30 it returns 0.08, because the width of its fraud-free distribution, computed from the observed data, is itself widened by any fraud in those data and so absorbs part of the fraud signal. It recovers a true 0.15 at 2,700 units (0.16), but with a standard deviation of 0.14. Its extreme-fraud estimate, by contrast, is accurate in every dong-sized case. The reconstructed 2012 procedure returns a large, spurious extreme-fraud share ($f_e = 0.21$ – 0.24) on fraud-free data. The two-dimensional objective of the 2018 procedure avoids this.

We use the Klimek et al. [2018] procedure for the main analysis because it is Klimek and coauthors’ most recent procedure.

Table D.3: Parameter recovery on data simulated from the Klimek model (true clean-component parameters $\tau = 0.70$, $\sigma_\tau = 0.07$, $\nu = 0.50$, $\sigma_\nu = 0.10$, with one simulated dataset per case). “Ours” is a single fit, the 2018 procedure is summarized over 200 sweeps at 1% fingerprint resolution, and the reconstructed 2012 procedure over 100 realizations, as specified in Klimek et al. [2012, SI]. “SD” is the standard deviation of f_i across sweeps or realizations.

n	True		Ours (joint fit)		2018 procedure (2-D)			2012 procedure (1-D)		
	f_i	f_e	f_i	f_e	f_i	f_e	SD	f_i	f_e	SD
3,500	0.00	0.00	0.001	0.000	0.295	0.000	0.181	0.529	0.243	0.210
3,500	0.15	0.03	0.148	0.029	0.103	0.029	0.105	0.037	0.054	0.055
3,500	0.30	0.00	0.331	0.000	0.078	0.000	0.069	0.133	0.232	0.287
2,700	0.00	0.00	0.036	0.000	0.160	0.000	0.138	0.560	0.206	0.210
2,700	0.15	0.03	0.816	0.025	0.157	0.022	0.137	0.106	0.033	0.132
250	0.00	0.00	0.760	0.000	0.780	0.013	0.192	0.549	0.202	0.222
250	0.10	0.02	0.913	0.036	0.581	0.035	0.268	0.676	0.047	0.217
13,500	0.00	0.00	–	–	0.281	0.000	0.108	0.716	0.134	0.170
13,500	0.15	0.03	–	–	0.199	0.031	0.065	0.965	0.006	0.049
13,500	0.30	0.00	–	–	0.345	0.000	0.057	0.982	0.006	0.027

D.4 Complete results

Table D.4 lists every Klimek et al. [2018]-procedure fit at the dong and polling-district levels: five elections, both leader codings, raw and residualized fingerprints, real data and the joint null. The marginal null is not fitted because its fingerprint is essentially the real one, so it is not a fraud-free control for a fingerprint method (Appendix D.1).

Table D.4: All Klimek et al. [2018]-procedure fits: the five elections held before early voting, at the dong and polling-district levels, both leader codings, raw and province-residualized fingerprints, real data and the joint null. Each entry is the mean, with the standard deviation beside it, across 200 sweeps.

Level	Elec.	Leader	Fingerpr.	Source	n	f_i %	SD	f_e %	SD	α	SD
dong	Pres02	Dem.	raw	real	3,514	14.8	2.9	0.00	0.00	0.10	0.00
dong	Pres02	Dem.	raw	joint null	3,514	66.0	29.5	0.06	0.24	0.92	0.65
dong	Pres02	Dem.	resid.	real	3,514	5.3	5.7	0.00	0.00	1.33	0.82
dong	Pres02	Dem.	resid.	joint null	3,514	6.5	6.7	0.01	0.07	1.64	0.91
dong	Pres02	Con.	raw	real	3,514	74.8	2.3	0.00	0.00	0.53	0.06
dong	Pres02	Con.	raw	joint null	3,514	84.0	1.7	0.00	0.00	0.58	0.06
dong	Pres02	Con.	resid.	real	3,514	5.8	5.3	0.00	0.00	1.47	0.81
dong	Pres02	Con.	resid.	joint null	3,514	8.4	8.0	0.01	0.07	1.72	0.87
dong	Pres07	Dem.	raw	real	3,576	11.0	3.1	0.00	0.00	0.19	0.03
dong	Pres07	Dem.	raw	joint null	3,576	9.7	2.7	0.00	0.00	0.15	0.05
dong	Pres07	Dem.	resid.	real	3,576	24.9	13.6	0.01	0.12	2.09	0.86
dong	Pres07	Dem.	resid.	joint null	3,576	3.8	3.9	0.00	0.00	1.63	1.03
dong	Pres07	Con.	raw	real	3,576	74.8	1.8	0.00	0.00	0.40	0.01
dong	Pres07	Con.	raw	joint null	3,576	69.7	2.0	0.00	0.00	0.40	0.01
dong	Pres07	Con.	resid.	real	3,576	6.9	7.4	0.00	0.00	1.75	0.98
dong	Pres07	Con.	resid.	joint null	3,576	9.3	9.9	0.00	0.00	1.84	0.97
dong	Gen08	Dem.	raw	real	2,733	76.3	3.1	0.01	0.12	0.81	0.14
dong	Gen08	Dem.	raw	joint null	2,733	51.1	5.0	0.00	0.00	0.78	0.19
dong	Gen08	Dem.	resid.	real	2,733	4.9	4.0	0.01	0.10	1.27	0.91
dong	Gen08	Dem.	resid.	joint null	2,733	31.9	6.7	0.06	0.26	2.18	1.04
dong	Gen08	Con.	raw	real	2,733	92.2	1.2	0.07	0.26	1.01	0.10
dong	Gen08	Con.	raw	joint null	2,733	77.1	2.3	0.00	0.00	0.70	0.10
dong	Gen08	Con.	resid.	real	2,733	6.2	4.8	0.03	0.16	1.10	0.78
dong	Gen08	Con.	resid.	joint null	2,733	19.6	7.7	0.01	0.12	1.66	0.79
dong	Gen12	Dem.	raw	real	2,656	23.0	11.5	0.02	0.14	2.06	1.00
dong	Gen12	Dem.	raw	joint null	2,656	11.4	7.7	0.01	0.10	1.81	0.92
dong	Gen12	Dem.	resid.	real	2,656	21.5	10.4	0.05	0.22	2.37	0.94
dong	Gen12	Dem.	resid.	joint null	2,656	11.8	6.5	0.01	0.12	1.28	0.77
dong	Gen12	Con.	raw	real	2,656	21.6	11.0	0.01	0.10	1.18	0.59
dong	Gen12	Con.	raw	joint null	2,656	13.4	9.8	0.01	0.07	1.37	0.83
dong	Gen12	Con.	resid.	real	2,656	23.0	11.1	0.01	0.07	2.28	1.03
dong	Gen12	Con.	resid.	joint null	2,656	20.3	9.0	0.01	0.07	1.73	0.98
dong	Pres12	Dem.	raw	real	3,479	5.2	3.0	0.01	0.07	0.49	0.98
dong	Pres12	Dem.	raw	joint null	3,479	10.0	2.3	0.00	0.00	0.10	0.00
dong	Pres12	Dem.	resid.	real	3,479	25.8	19.8	0.01	0.07	2.01	0.89
dong	Pres12	Dem.	resid.	joint null	3,479	30.4	15.8	0.00	0.00	1.94	0.87
dong	Pres12	Con.	raw	real	3,479	18.0	7.3	0.00	0.00	0.44	0.19
dong	Pres12	Con.	raw	joint null	3,479	7.7	5.7	0.01	0.07	0.73	0.60
dong	Pres12	Con.	resid.	real	3,479	20.5	15.4	0.00	0.00	1.98	1.04
dong	Pres12	Con.	resid.	joint null	3,479	30.8	16.6	0.00	0.00	2.08	0.86
p. district	Pres02	Dem.	raw	real	13,441	11.1	1.6	0.00	0.00	0.10	0.00
p. district	Pres02	Dem.	raw	joint null	13,441	10.7	1.5	0.00	0.00	0.10	0.00
p. district	Pres02	Dem.	resid.	real	13,441	6.4	6.0	0.00	0.00	1.95	1.17
p. district	Pres02	Dem.	resid.	joint null	13,441	0.5	0.7	0.00	0.00	0.64	0.99

Table D.4 – continued

Level	Elec.	Leader	Fingerpr.	Source	n	f_i %	SD	f_e %	SD	α	SD
p. district	Pres02	Con.	raw	real	13,441	9.5	5.3	0.00	0.00	0.58	0.14
p. district	Pres02	Con.	raw	joint null	13,441	86.9	0.7	0.00	0.00	0.60	0.02
p. district	Pres02	Con.	resid.	real	13,441	3.1	3.5	0.00	0.00	1.83	1.28
p. district	Pres02	Con.	resid.	joint null	13,441	2.6	3.1	0.00	0.00	1.57	1.15
p. district	Pres07	Dem.	raw	real	13,156	10.6	1.7	0.00	0.00	0.20	0.00
p. district	Pres07	Dem.	raw	joint null	13,156	10.5	1.4	0.00	0.00	0.20	0.00
p. district	Pres07	Dem.	resid.	real	13,156	0.8	1.3	0.00	0.00	0.91	1.17
p. district	Pres07	Dem.	resid.	joint null	13,156	39.7	7.9	0.01	0.07	3.79	0.71
p. district	Pres07	Con.	raw	real	13,156	42.6	8.5	0.00	0.00	0.99	0.42
p. district	Pres07	Con.	raw	joint null	13,156	1.6	1.9	0.00	0.00	0.73	0.76
p. district	Pres07	Con.	resid.	real	13,156	1.5	2.0	0.00	0.00	1.00	0.97
p. district	Pres07	Con.	resid.	joint null	13,156	2.5	2.7	0.00	0.00	1.54	1.09
p. district	Gen08	Dem.	raw	real	10,737	7.7	4.5	0.01	0.07	2.69	1.09
p. district	Gen08	Dem.	raw	joint null	10,737	33.2	2.5	0.00	0.00	0.58	0.07
p. district	Gen08	Dem.	resid.	real	10,737	29.6	4.5	0.00	0.00	3.48	0.95
p. district	Gen08	Dem.	resid.	joint null	10,737	37.5	3.0	0.01	0.07	3.22	0.92
p. district	Gen08	Con.	raw	real	10,737	12.7	5.8	0.00	0.00	2.10	0.89
p. district	Gen08	Con.	raw	joint null	10,737	84.7	1.0	0.00	0.00	0.78	0.06
p. district	Gen08	Con.	resid.	real	10,737	34.4	4.3	0.00	0.00	3.42	0.92
p. district	Gen08	Con.	resid.	joint null	10,737	2.5	1.7	0.00	0.00	0.74	0.59
p. district	Gen12	Dem.	raw	real	10,461	65.0	8.4	0.00	0.00	3.24	0.97
p. district	Gen12	Dem.	raw	joint null	10,461	2.7	2.6	0.00	0.00	1.61	1.15
p. district	Gen12	Dem.	resid.	real	10,461	3.2	3.2	0.00	0.00	2.05	1.30
p. district	Gen12	Dem.	resid.	joint null	10,461	2.2	2.1	0.00	0.00	1.64	1.23
p. district	Gen12	Con.	raw	real	10,461	59.5	7.2	0.00	0.00	1.87	0.73
p. district	Gen12	Con.	raw	joint null	10,461	1.3	1.5	0.00	0.00	0.94	0.95
p. district	Gen12	Con.	resid.	real	10,461	4.5	3.4	0.00	0.00	2.17	1.06
p. district	Gen12	Con.	resid.	joint null	10,461	15.0	4.3	0.00	0.00	0.96	0.25
p. district	Pres12	Dem.	raw	real	13,522	2.1	1.1	0.00	0.00	0.16	0.31
p. district	Pres12	Dem.	raw	joint null	13,522	4.2	1.0	0.00	0.00	0.10	0.00
p. district	Pres12	Dem.	resid.	real	13,522	92.3	7.7	0.00	0.00	3.07	0.96
p. district	Pres12	Dem.	resid.	joint null	13,522	10.7	8.9	0.00	0.00	2.11	1.00
p. district	Pres12	Con.	raw	real	13,522	14.5	6.0	0.00	0.00	0.59	0.14
p. district	Pres12	Con.	raw	joint null	13,522	8.5	5.4	0.00	0.00	0.89	0.51
p. district	Pres12	Con.	resid.	real	13,522	85.5	11.4	0.00	0.00	3.67	0.84
p. district	Pres12	Con.	resid.	joint null	13,522	7.0	7.3	0.00	0.00	1.99	1.14

E Code and data-processing documentation

This appendix documents the code and procedures referenced throughout the paper: what each script does, the software it requires, and the commands needed to reproduce a model run from a raw election-result CSV through to a posterior summary. The full codebase is not reproduced inline because, in total, it comprises several thousand lines of R and Python. The complete, version-controlled source is provided as part of the submission.

E.1 Overview

The code has two layers. A Python layer (i) loads and cleans a raw National Election Commission result file into the unit-level (w, a, N, prov) table required by `eforensics` (leader votes, abstentions, eligible voters, and a province grouping code), (ii) optionally replaces that table with one of the three fraud-free null constructions, and (iii) invokes an R layer that calls the `eforensics` package’s `eforensics()` function

via JAGS and writes the posterior summary back out as CSV. Two supplementary analyses reuse the Python layer: the PSRF stopping-rule check, which reruns all constituency-level cells through the same R layer with a different stopping rule, and the Klimek-type fingerprint analysis, which is pure Python and does not use R or JAGS. Orchestration across all combinations of elections, levels, channels, sources, and leader codings is handled by shell scripts that launch many single-cell Python invocations concurrently and log each cell's output separately, so that a long-running batch can be interrupted and resumed without repeating completed cells.

E.2 Software requirements

R (≥ 4.0 recommended), with the packages `eforensics` [Ferrari et al., 2019] (installed from Walter Mebane's fork, `UMeforensics/eforensics_public` on GitHub, rather than any CRAN mirror, to guarantee the exact model implementation referenced throughout this paper), `coda` (posterior summary and Gelman–Rubin diagnostics, Gelman and Rubin, 1992), and, optionally, `diptest` (Hartigan dip test for the Mebane multimodality diagnostic, which is reported as unavailable if this package is absent). **JAGS** ≥ 4.3 must be installed separately and on the system path, since `eforensics` calls it as an external MCMC sampler rather than bundling it. **Python** ≥ 3.8 , with `numpy` and `pandas` for the data-preparation and null-generation layer, and, for the post-processing scripts that build this paper's tables and figures, `matplotlib`. The Klimek-type analysis additionally requires `scipy`. No Python package is required to have any dependency on R, and the two communicate only through CSV files written to a temporary directory and an `Rscript` subprocess call.

E.3 Script reference

mebane_crosscheck.R. Fits the `qbl` model to a single election/level/channel/source cell. Takes an input CSV with columns `leader_votes`, `total_votes`, `eligible`, `prov_code`, and writes a one-row summary CSV (mixture weights, credible intervals, $\hat{R}$, the Mebane multimodality diagnostics) and a per-unit MAP-classification CSV. Usage: `Rscript mebane_crosscheck.R <input.csv> <output.csv> [n_iter] [n_chains] [burn_in] [use_parcomp] [n_adapt]`.

mebane_crosscheck_selfnull_fit.R. Identical to `mebane_crosscheck.R`, but additionally monitors and writes out the posterior-median class-1 (no-fraud) hyperparameters needed to construct the model-consistent self-null: the raw province design-matrix coefficients on the turnout and vote-share equations, the corresponding random-intercept means, and the random-intercept variances. Writes two additional output files, `*_hyperparams.csv` and `*_provlevels.csv` (the latter records the province factor-level ordering needed to map the coefficients back onto the original province codes). Usage: `Rscript mebane_crosscheck_selfnull_fit.R <input.csv> <output.csv> [n_iter] [n_chains] [burn_in] [use_parcomp]`.

mebane_r_driver_bidirectional.py. The main Python data layer. Contains the per-election configuration (candidate name patterns, result-file names, early-voting status), the Korean-language CSV parser that builds dong- and constituency-level unit tables from a raw result file, the marginal- and joint-null generators, and the driver function that writes a unit table to a temporary CSV, calls `mebane_crosscheck.R` via `Rscript`, and reads the result back in. `leader_side` ('dem' or 'con') controls which party or candidate is coded as the model's leader. Usage: `python3 mebane_r_driver_bidirectional.py <elec_id> <level> <channel> <source> [leader_side] [r_n_iter] [r_n_chains] [r_burn_in] [use_parcomp] [r_n_adapt]`, where `source` is one of `real`, `marginal_null`, `joint_null`.

mebane_self_null_driver.py. Implements the two-stage self-null procedure: it calls `mebane_crosscheck_selfnull_fit.R` on the real data to harvest class-1 hyperparameters, simu-

lates a synthetic dataset from those hyperparameters with every unit forced into the no-fraud class, and refits `mebane_crosscheck.R` to that synthetic dataset. The form of the leader-vote probability (observed- or latent-turnout) is read from the `EFORENSICS_WFORM` environment variable by the simulation step as well as by both fitting stages, so under the latent-turnout form the synthetic class-1 data are drawn from $W_i \sim \text{Binomial}(N_i, \tau_i \nu_i)$ rather than from the package’s $W_i \sim \text{Binomial}(N_i, \nu_i(1 - A_i/N_i))$, and the null stays consistent with the model it is refit under. Under the latent-turnout form, a simulated unit whose leader votes exceed its votes cast has its abstentions and leader votes redrawn together until feasible, since the package rejects such data (Appendix C.4). Depends on `mebane_r_driver_bidirectional.py` for data loading and on `mebane_crosscheck_selfnull_fit.R/mebane_crosscheck.R` for the two model fits. Usage: `python3 mebane_self_null_driver.py <elec_id> <level> <channel> <leader_side> [r_n_iter] [r_n_chains] [r_burn_in] [use_parcomp] [seed]`.

`run_mebane_r_all_parallel_bidirectional.sh` and `run_mebane_self_null_parallel.sh`. Bash drivers that loop over every (election, level, channel, source, leader_side) combination in the study, launching `mebane_r_driver_bidirectional.py` or `mebane_self_null_driver.py` as concurrent background subprocesses (bounded by a configurable `MAX_PARALLEL`), and writing each cell’s console output to its own log file plus a short summary snippet. A cell with an existing non-empty log is skipped unless `FORCE_RERUN=1` is set, making both scripts safe to interrupt and resume. Environment variables documented in each script’s header (`R_N_ITER`, `R_N_CHAINS`, `R_BURN_IN`, `LEADER_SIDES`, `MAX_PARALLEL`, `SEED_BASE`, `MEBANE_PROVINCE_GROUPING`, among others) control the MCMC settings and scope of a given run without editing the script itself.

`run_all_eforensics_dem.sh`, `run_all_eforensics_con.sh`. Run every eforensics estimation in the paper for one leader coding, eight cells at a time: the short-chain configuration, the latent-turnout form, the constituency-level PSRF check and both model-consistent self-nulls. `run_selfnull_con.sh` runs only the two conservative-coded self-nulls, and `run_psrfdong.sh` runs the PSRF check at the dong level. Each step writes to its own output directory and fixes its settings explicitly, and a check at start-up refuses to run outdated code.

`collect_eforensics_results.py`. Rebuilds the aggregated result files for the real data and both nulls (fraud shares, mixture components and multimodality diagnostics) directly from the run logs. It accepts only logs written by the current code.

`make_chainlength_check.py`, `make_eforensics_stats.py`. The chain-length check of the main text, comparing each constituency-level PSRF run with its short-chain counterpart (figure and statistics), and the script that computes every eforensics number cited in the main text, together with the bodies of its eforensics tables.

`parse_selfnull_logs.py`. Parses the self-null sweep’s per-cell logs into the aggregated CSV that the self-null table and figure scripts read, recording which form each cell used. It is the counterpart of `parse_mebane_logs.py` for that sweep. `make_appendix_table_selfnull.py` and `make_fig7_selfnull.py` take optional input and output paths, so the same table and figure can be built for a latent-turnout sweep without overwriting the observed-turnout versions.

`make_appendix_table.py`, `make_appendix_table_components.py`, `make_appendix_table_mebane_md.py`, and `make_appendix_table_selfnull.py`. Each reads an aggregated results CSV (`appendix_full_results.csv`, `appendix_components_results.csv`, `appendix_mebane_md_results.csv`, or `appendix_selfnull_results.csv`, one row per completed cell) and writes the corresponding LaTeX longtable included in Appendix C.

`make_figures.py`. Reads the same aggregated results CSVs and generates the eforensics figures in the main text as PDF files, and `make_fig7_selfnull.py` regenerates the self-null scatter plot from `appendix_selfnull_results.csv`.

`parse_mebane_logs.py`. Parses the `RESULT` blocks of the per-run logs written by the drivers into

one CSV per sweep.

run_psrfs_constituency.sh and **parse_psrfs_logs.py**. The PSRF stopping-rule check (Appendix C.5). The shell script reruns all constituency-level cells with `EFORENSICS_CONV=psrf`, which makes `mebane_crosscheck.R` pass `mcmc.conv.diagnostic = "PSRF"` to `eforensics()` (at most `EFORENSICS_MAX_AUTO` extensions, default 10). It refuses to start if the R script lacks this switch, and every log records the rule in force. Logs go to `output_data_psrfs/`, and the parser writes `psrf_constituency_results.csv`, one row per run.

klimek_units.py. Builds the Klimek-analysis unit tables for the five elections held before early voting, at the dong and polling-district (*tupyogu*) levels, from the same raw result files, applying the data rules of Appendix D.2.

klimek_2018.py, **klimek_2018_verify.py**. The Klimek et al. [2018]-procedure estimator used for all Klimek results in the paper (Appendix D.2), and the check that its exact solve returns the same minimizer as scoring every candidate pair directly.

klimek_2018_all.py. Runs the estimator over all 80 combinations (5 elections $\times$ 2 levels $\times$ 2 leader codings $\times$ raw/residualized $\times$ real/joint null, 80 fits) and the regional-exclusion refits (30 fits). It is resumable and writes `klimek_2018_all_results.csv`, `klimek_2018_exclusion_results.csv` and the per-sweep fits.

klimek_2018_from_sweeps.py. Rebuilds results rows from the per-sweep files when only those are copied back from another machine. The sample size and the fraud-free distribution parameters are recomputed deterministically. With `--check` it rebuilds the existing rows and compares them with the stored ones.

klimek_2018_winner_table.py, **klimek_2018_exclusion_table.py**, **klimek_2018_summary_figure.py**, **make_klimek_preearly_figures.py** (with **klimek_2018_viz.py**), **klimek_2018_stats.py**. Build the main text's Klimek tables, summary figure and fingerprint figures from those CSVs without refitting, and print every Klimek number cited in the text.

klimek_analysis.py, **klimek_recovery_test.py**. Our initial joint-fit estimator and its parameter-recovery test. Not used for any Klimek result in the paper. Its recovery results supply the "ours" column of Table D.3.

klimek_original.py, **klimek_original_recovery_test.py**, **klimek_2018_recovery_test.py**. A reconstruction of the original Klimek et al. [2012] one-dimensional procedure, and the recovery tests of the reconstructed 2012 procedure and of `klimek_2018.py`, used in the comparison of Appendix D.3. They write to `klimek_original_output/`.

ef_qbl_tau.R, **make_tau_comparison.py**. The latent-turnout fork of the package's `qbl` model, activated by setting `EFORENSICS_WFORM=tau` (it replaces only the leader-vote probability statement and stops if that statement is not found in the expected form), and the script that parses the latent-turnout logs, pairs each run with its observed-turnout counterpart, and writes the latent-turnout summary table, Table C.8 and the statistics cited in the text (`tau_stats.txt`).

make_appendix_table_extra.py. Generates Table C.10 from the CSVs above.

make_appendix_table_klimek2018.py. Generates Tables D.3 and D.4 (Appendices D.3–D.4).

E.4 Reproducing a single cell end to end

1. Install R, JAGS, and the R packages listed above, and install `eforensics` from `UMeforensics/eforensics_public` rather than CRAN.
2. Download the relevant election-result CSV file(s) from the `data.go.kr` pages listed under Data availability, and place them in the working directory under the filenames referenced in the per-election

configuration (e.g. `21st_election_result.csv` for the 2020 general election).

3. Run, e.g., `python3 mebane_r_driver_bidirectional.py 21 dong early real dem 1000 4 300 FALSE 1000` to fit the real-data cell for the 21st general election, dong level, in-precinct early-voting channel, Democratic-coded leader, at the short-chain configuration's settings. Substitute `marginal_null` or `joint_null` for `real` to fit the corresponding null instead, or use `mebane_self_null_driver.py` for the model-consistent self-null.
4. For all the eforensics runs used in this paper, run `./run_all_eforensics_dem.sh` and `./run_all_eforensics_con.sh`. Each runs, for one leader coding, the short-chain configuration for the real data and both nulls, both model-consistent self-nulls, the latent-turnout form and the constituency-level PSRF check, writing to `output_data_v2/`, `output_data_v2_tau/` and `output_data_v2_psr/`. The underlying launchers, `run_mebane_r_all_parallel_bidirectional.sh` and `run_mebane_self_null_parallel.sh`, accept an optional list of election IDs and a level as positional arguments to restrict scope, and take MCMC settings, concurrency, the form of the leader-vote probability and leader coding from the environment variables documented in their headers.
5. Collect the results with `collect_eforensics_results.py` (real data and both nulls), `parse_selfnull_logs.py` (both self-nulls), `parse_psr_logs.py` and `make_tau_comparison.py`. Then run `make_appendix_table.py`, `make_appendix_table_components.py`, `make_appendix_table_mebane_md.py`, `make_appendix_table_selfnull.py`, `make_appendix_table_extra.py`, `make_figures.py`, `make_fig7_selfnull.py`, `make_chainlength_check.py` and `make_eforensics_stats.py`, which prints every eforensics number cited in the main text.
6. For the Klimek-type analysis, which needs no R or JAGS, run `python3 klimek_2018_all.py`, then `klimek_2018_winner_table.py`, `klimek_2018_exclusion_table.py`, `klimek_2018_summary_figure.py`, `make_klimek_preearly_figures.py` and `klimek_2018_stats.py`. For the appendix, run `klimek_2018_recovery_test.py`, `klimek_original_recovery_test.py` and `make_appendix_table_klimek2018.py`.

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